

# Clinical experiment report

Dataset overview | synthetic non-small cell lung cancer (NSCLC)

**N = 50,000 simulated patients per dataset version.** There are **43 covariate columns**, including derived indicators, plus a patient identifier and one outcome: natural-log progression-free survival (PFS) in months, fully observed without censoring.

**Key finding deliberately flipped:** NLR  $\geq 3$  versus  $< 3$  is associated with **shorter PFS in the expected version** and **longer PFS in the surprising version**. Its injected log-PFS coefficient changes from **-0.45 to +0.45**. The paired versions share covariates and random residuals; only this outcome term changes.

| Covariate group                          | Included variables                                                                                                                                                    |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Demographics and disease (7)             | age; sex; smoking status; ECOG performance status; histology; stage IV indicator; brain metastases                                                                    |
| Molecular and immune markers (7)         | EGFR mutation; KRAS G12C; ALK fusion; STK11 mutation; BRCA2 mutation; PD-L1 tumor proportion score; high tumor mutational burden                                      |
| Laboratory and nutritional measures (15) | albumin; LDH; 6-month weight loss; CRP; NLR; hemoglobin; alkaline phosphatase; AST; ALT; total bilirubin; creatinine; blood urea nitrogen; sodium; potassium; calcium |
| Treatment indicators (4)                 | pembrolizumab; sotorasib; olaparib; osimertinib                                                                                                                       |
| Constructed research variables (6)       | signature A; signature B; signature C; marker D; marker E; marker F                                                                                                   |
| Derived threshold indicators (4)         | ECOG $\geq 2$ ; CRP $\geq 10$ mg/L; albumin $\leq 3.5$ g/dL; NLR $\geq 3$                                                                                             |

| Planted finding category | Associations encoded in the synthetic data                                                                                                                                                                                                |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected                 | Stage IV and CRP $\geq 10$ mg/L each predict shorter PFS in both versions. NLR $\geq 3$ also predicts shorter PFS in the expected version.                                                                                                |
| Surprising               | ECOG $\geq 2$ predicts longer PFS in both versions. In the surprising version, NLR $\geq 3$ also predicts longer PFS.                                                                                                                     |
| Neutral                  | BRCA2 mutation predicts longer PFS within A $\geq 0$ and B $\geq 0.5$ ; marker D positivity predicts longer PFS within A $\geq 0$ and C $\leq 0.5$ , in both versions. Neutral describes the prior-knowledge category, not a null effect. |

**Four versions:** expected-unmasked, expected-masked, surprising-unmasked, surprising-masked. Masking changes names and categorical text labels while preserving the numerical data. Each version has six planted findings: **3 expected / 2 neutral / 1 surprising** before the focal flip, versus **2 / 2 / 2** after it. These categories follow the frozen benchmark specification. Full comparisons are on page 3.

# Results snapshot and contents

NSCLC discovery benchmark | tables and figures with 95% confidence intervals

**Results snapshot:** 2026-09-12T20:27:30.468069+00:00  
**820 selected run slots;** 787 successfully completed; 33 ended with errors; 0 unfinished or queued.

**The manipulated discovery:** Neutrophil-to-lymphocyte ratio (NLR)  $\geq 3$  versus  $< 3$ . Its simulated relationship with progression-free survival (PFS) changes from **shorter PFS (expected)** to **longer PFS (surprising)**. The other five discovery definitions remain fixed.

**Four experimental conditions:** expected-unmasked, expected-masked, surprising-unmasked, and surprising-masked. Expected/surprising describes the focal finding's direction. Masked/unmasked describes the variable labels visible to the agent; masking preserves the numerical data.

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All findings described here are deliberately embedded in synthetic data. The categories reflect the benchmark's frozen specification.

# Clinical questions and planted discoveries

50,000 simulated NSCLC patients per version | six planted comparisons

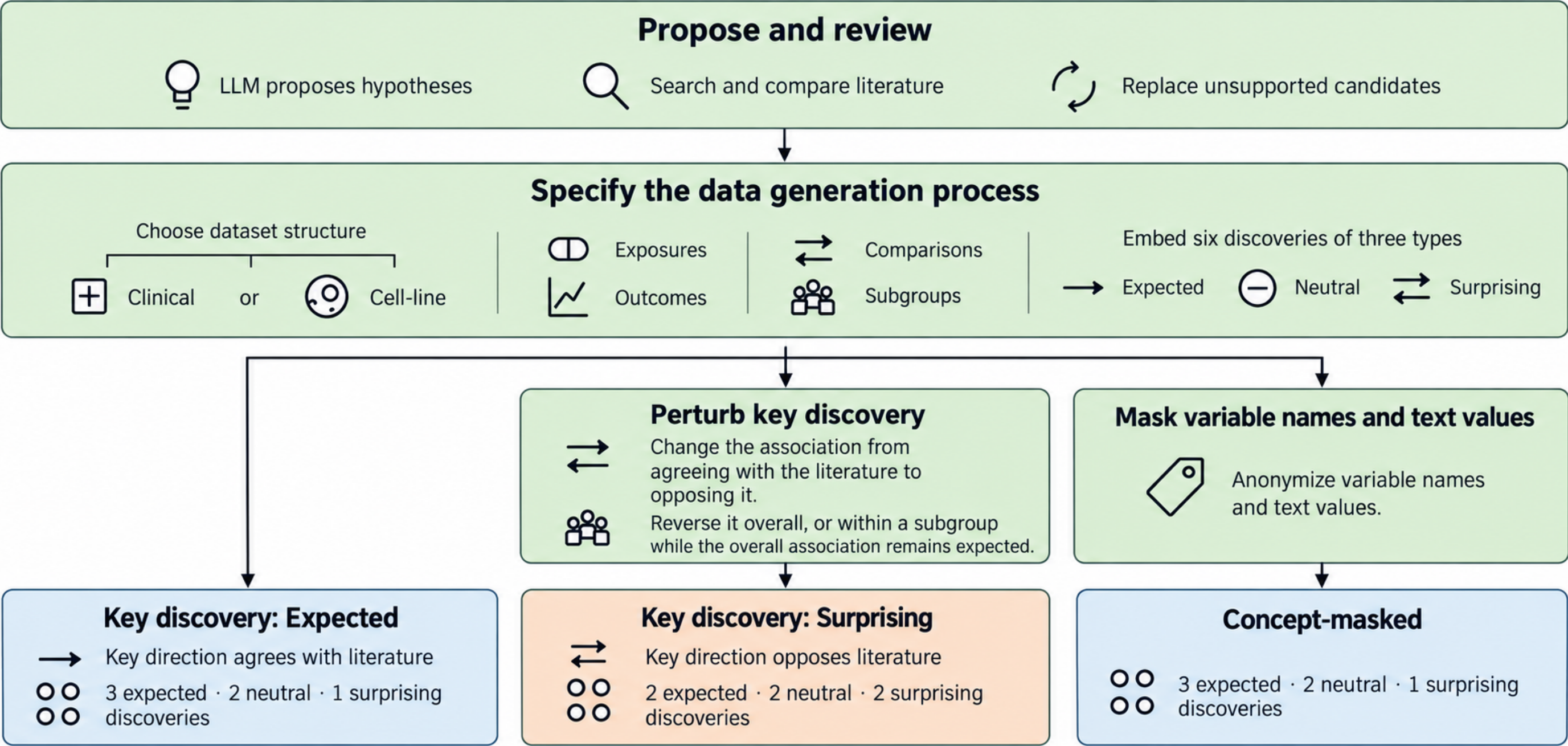
The question for each comparison is whether progression-free survival differs between the exposed and comparator groups. The endpoint is mean natural-log PFS in months, fully observed without censoring. A, B, and C are constructed research signatures in arbitrary units; marker D is a constructed binary assay.

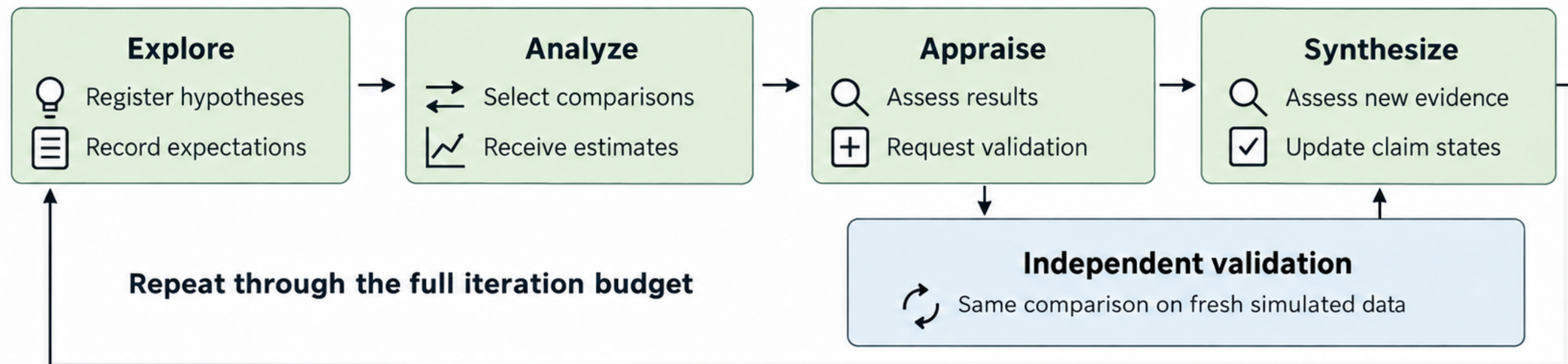
| Underlying comparison                                                                               | Masked exposure (main grid) | Expected-focal dataset                                | Surprising-focal dataset                              |
|-----------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------|-------------------------------------------------------|
| <b>FLIPPED: Neutrophil-to-lymphocyte ratio (NLR) <math>\geq 3</math> versus <math>&lt; 3</math></b> | feature_006                 | Shorter PFS; expected<br>Injected coefficient: -0.45  | Longer PFS; surprising<br>Injected coefficient: +0.45 |
| Stage IV versus other stages                                                                        | feature_025                 | Shorter PFS; expected<br>Injected coefficient: -0.45  | Shorter PFS; expected<br>Injected coefficient: -0.45  |
| C-reactive protein (CRP) $\geq 10$ versus $< 10$ mg/L                                               | feature_028                 | Shorter PFS; expected<br>Injected coefficient: -0.45  | Shorter PFS; expected<br>Injected coefficient: -0.45  |
| ECOG performance status $\geq 2$ versus $< 2$                                                       | feature_041                 | Longer PFS; surprising<br>Injected coefficient: +0.45 | Longer PFS; surprising<br>Injected coefficient: +0.45 |
| BRCA2 mutation versus no mutation, within A $\geq 0$ and B $\geq 0.5$                               | feature_024                 | Longer PFS; neutral<br>Injected coefficient: +0.45    | Longer PFS; neutral<br>Injected coefficient: +0.45    |
| Research marker D positive versus negative, within A $\geq 0$ and C $\leq 0.5$                      | feature_036                 | Longer PFS; neutral<br>Injected coefficient: +0.45    | Longer PFS; neutral<br>Injected coefficient: +0.45    |

**What changes:** the NLR  $\geq 3$  term changes from -0.45 to +0.45 log-PFS units. The paired datasets share covariates and random residuals; only this outcome contribution changes. These coefficients are injected model terms, not hazard ratios or estimated marginal contrasts.

**What stays surprising:** ECOG  $\geq 2$  predicts longer PFS in both versions. The inventory therefore changes from 3 expected / 2 neutral / 1 surprising to 2 expected / 2 neutral / 2 surprising findings. Both dataset versions contain a mixture of discovery categories.

**Main-grid masked identities:** NLR  $\geq 3$  = feature\_006; A = feature\_013; B = feature\_042; C = feature\_002. Numerical subgroup thresholds and outcomes are preserved. Source: the frozen pair, equations, and masking map listed in the source appendix and clinical\_context.json.





**Shared claim ledger - available at every stage**

| Claim judgment: what does the agent believe?                                             | Investigation status: is more work planned?                                                    |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> <b>Accepted</b> - agent currently endorses the claim | <input checked="" type="checkbox"/> <b>Active</b> - pursuing further investigation             |
| <input checked="" type="checkbox"/> <b>Rejected</b> - agent currently rejects the claim  | <input checked="" type="checkbox"/> <b>Deferred</b> - investigation paused                     |
| <input checked="" type="checkbox"/> <b>Unresolved</b> - agent remains undecided          | <input checked="" type="checkbox"/> <b>Closed</b> - no further investigation currently planned |

These fields are independent: a claim can be Accepted + Active or Rejected + Closed.  
Field statuses can change during each iteration.

**Outcomes**

- Discovery recall
- Precision
- F1\*
- Exploration coverage
- Evidence responsiveness

# Reading the results

What each metric measures and how it is calculated

**Multiple findings per dataset:** each version contains six planted findings (listed on page 3), and a run can propose additional findings. Precision and recall count findings within each run. The reported values then average run scores within each model/harness/workflow and experimental condition; individual patients are not the scoring units.

| Metric (all on a 0-100 scale) | Meaning and calculation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Precision (%)                 | <b>Of the findings the agent accepts at the end, how many hold up?</b> $100 \times (\text{distinct accepted claims that were tested and supported on fresh confirmation data}) / (\text{all distinct accepted claims})$ . Example: 3 confirmed out of 4 accepted = 75%. Additional findings outside the six planted targets can count. Duplicate claims count once; no accepted claims makes precision unavailable.                                                                                                                                                                                                                  |
| Recall (%)                    | <b>How many of the planted findings did the agent recover?</b> Within each expected, neutral, and surprising finding category, divide recovered targets by available targets; average those three fractions and multiply by 100. Recovery requires testing, final acceptance, fresh-data confirmation, and an exact match including subgroup and direction. Each target counts once. In the expected-focal dataset, recovering only its 3 expected findings gives $(3/3 + 0/2 + 0/1) / 3 \times 100 = 33.3\%$ .                                                                                                                      |
| Scientific F1*                | <b>Combines precision and recall:</b> $2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$ , calculated per run and then averaged. Undefined precision or a zero denominator gives $F1 = 0$ under the evaluator's convention.                                                                                                                                                                                                                                                                                                                                                                       |
| Exploration E                 | <b>How broadly and how early did the agent investigate the planted comparisons?</b> At the end of each of 25 rounds, calculate the fraction of each finding category tested so far. Average across the three categories and all 25 rounds, then multiply by 100. A valid test of the exact comparison counts regardless of the proposed direction, test result, or later acceptance. Repeating a test adds no coverage. Earlier tests contribute to more rounds and therefore earn more credit. Biomni uses 25 reporting rounds.                                                                                                     |
| Evidence responsiveness B     | <b>Does the agent's judgment follow the validation evidence?</b> Score its recorded decision at the scheduled check two rounds after each eligible validation result: accept when the result's whole interval is above the minimum-effect cutoff; reject when wholly below; leave unresolved when the interval spans or touches the cutoff. These are the supported, excluded, and ambiguous evidence classes. Within a run and class, $\text{accuracy} = \text{correct decisions} / \text{eligible results}$ . Average each class's accuracy across available runs, then average the three class means equally and multiply by 100. |
| Focal recovery (%)            | <b>Was the flipped NLR/PFS finding recovered?</b> Each run scores 100 for an exact, tested, accepted, independently confirmed recovery, or 0 otherwise. The displayed mean is the percentage of runs recovering this one finding.                                                                                                                                                                                                                                                                                                                                                                                                    |

Finding categories describe the six targets. Expected/surprising column labels describe which direction was assigned to the focal NLR finding. These are different uses of the same terms. Scoring and aggregation details continue on page 7.

# Reading the results: scoring details

Eligible observations, uncertainty, and execution accounting

**Scientific cohort:** all ended runs with reports, including runs that ended with errors. Unfinished and queued runs never receive zero scores. Means are calculated separately for each of the four conditions. Runs within a base dataset receive equal weight; base datasets then receive equal weight. This report has one base dataset pair. Undefined observations are omitted. Scientific F1 and focal recovery use the committed scientific results without the legacy whole-run zero penalty for execution errors.

**Confirmation and effect thresholds:** after a run, the evaluator tests its final accepted claims on fresh simulated data. Confirmation requires the interval's lower bound to exceed the benchmark's minimum effect in the claimed direction, with a multiplicity adjustment across final accepted claims. The NSCLC cutoff is 0.10 natural-log PFS units. An excluded result rules out reaching this cutoff; it need not imply an exactly zero or opposite effect. F1\* combines category-balanced target recall with a precision measure that can credit additional findings.

**Evidence responsiveness eligibility:** use voluntary and automatically delivered validation results. Invalid results and results whose two-round follow-up falls beyond the budget or reached run history are excluded. A missing decision at an eligible follow-up scores zero. If a run lacks an evidence class, its accuracy for that class is unavailable. Other runs can supply that class; the aggregate B is unavailable only when a class is absent across all eligible runs. Each class receives equal weight even if the numbers of results differ.

**95% CIs:** 10,000 seeded whole-run bootstrap draws; recompute the class-balanced B statistic in each draw. Focal recovery uses Wilson intervals. CIs are unavailable with too few eligible runs or evidence classes. Constant continuous scores can have zero-width bootstrap intervals. These are marginal intervals, not paired effect tests or multiplicity-adjusted intervals. They describe repeated-run uncertainty on this fixed dataset, not uncertainty across clinical datasets, and are distinct from the validation-result intervals used to score B.

**Counts and tokens:** attempted counts selected run identities that started; successfully completed means all stages succeeded; errors means status failed. Counts span all four conditions. Median output tokens use successful runs only, include returned retries and peer/chair calls, and can be lower bounds when usage is missing. Per-condition denominators and usage coverage are in the companion CSVs.

**Workflows:** persistent retains bounded conversation and the ledger; sequential starts fresh at each stage with the ledger; deliberative uses peer drafts and a chair. Codex denotes CLI transport within this project's scientific runner. Biomni uses its own workflow with full resources and xhigh reasoning; other runs use medium. Mixed code generations and budgets remain important comparison limitations.

# Reading the results: summary and interaction

Overall performance and the label-by-surprise contrast answer different questions

**Weighted condition score C (0-100):** within each model/harness/workflow and condition, combine the aggregate metric values:  **$C = 0.35 \times F1 + 0.25 \times \text{focal recovery} + 0.20 \times \text{exploration} + 0.20 \times \text{responsiveness}$** . Use the reported mean of per-run F1, not F1 recalculated from mean precision and recall. B retains its evidence-class balancing. Precision and recall enter through F1; extra focal weight deliberately emphasizes the flipped finding.

**Grounded discovery and adaptation S (0-100; higher is better):**  **$S = (C_{EU} \times C_{EM} \times C_{SU} \times C_{SM})^{1/4}$** . EU = expected-unmasked; EM = expected-masked; SU = surprising-unmasked; SM = surprising-masked. All four conditions have equal weight. This geometric mean makes weakness in one condition harder to offset elsewhere. A true zero C gives  $S = 0$ ; a missing required component or condition makes S unavailable, without reweighting.

**Separate label-by-surprise interaction I (percentage points):**  **$I = (\text{focal}_{EU} - \text{focal}_{SU}) - (\text{focal}_{EM} - \text{focal}_{SM})$** . Positive means the expected-to-surprising recovery drop is larger with recognizable labels; negative means it is larger with masked labels. Example: an unmasked drop of 40 points minus a masked drop of 10 points gives  $I = +30$ . Zero can reflect uniformly good or uniformly poor recovery, so read I alongside S and the focal condition table. I is not subtracted from S and is not a higher-is-better score.

**95% CIs for C and S:** 10,000 seeded bootstrap samples of whole runs, drawn independently within each condition. Recompute all component means, B's class balance, C, and then S in each draw, preserving within-run metric relationships. Do not average CI endpoints or pair runs merely by replicate number. At least two observations per required component/class and 95% valid draws are required. Sparse evidence can leave a point estimate available but its CI unavailable; constant samples can yield zero-width bootstrap CIs.

**95% CI for I:** Wilson-based method of variance estimates recovery (MOVER), combining the four independent focal-recovery proportions. This avoids a collapsed interval when a condition has all successes or all failures. The intervals are approximate, conditional on this fixed clinical dataset pair, and not adjusted for multiple comparisons. Method: [Zou, Huang and Zhang \(2009\), Computational Statistics & Data Analysis 53:1080-1085](#).

**Interpretation limits:** S is a provisional descriptive index with weights chosen after these experiments, not a validated or preregistered endpoint. I does not establish pretraining bias: masking also changes interpretability. Focal recovery avoids changing category weights when the target flips. Exploration covers the six planted comparisons; B includes automatic validation and already-correct judgments. Neither directly measures keeping a claim open or appropriately choosing validation.

# Precision (%)

Persistent + Biomni: first 8 rows | Other workflows below the divider | Means and 95% CIs

| LLM           | Harness       | Workflow     | Effort | Attempted runs | Completed runs | Failed runs | Median output tokens / success | Expected unmasked |             | Expected masked |             | Surprising unmasked |             | Surprising masked |             |
|---------------|---------------|--------------|--------|----------------|----------------|-------------|--------------------------------|-------------------|-------------|-----------------|-------------|---------------------|-------------|-------------------|-------------|
|               |               |              |        |                |                |             |                                | Mean              | 95% CI      | Mean            | 95% CI      | Mean                | 95% CI      | Mean              | 95% CI      |
| Astra         | Codex         | persistent   | medium | 40             | 40             | 0           | 85,653                         | 97.7              | 95.1-99.7   | 93.1            | 90.0-96.2   | 93.5                | 89.5-97.1   | 93.5              | 89.7-97.2   |
| Sol           | Codex         | persistent   | medium | 40             | 40             | 0           | 172,152                        | 95.7              | 94.6-96.8   | 96.0            | 94.4-97.7   | 91.4                | 87.1-95.3   | 93.2              | 87.9-97.0   |
| Terra         | Codex         | persistent   | medium | 40             | 40             | 0           | 56,127.5                       | 97.6              | 95.6-99.4   | 98.4            | 97.5-99.3   | 97.0                | 94.8-98.8   | 97.8              | 93.7-100.0  |
| Luna          | Codex         | persistent   | medium | 40             | 36             | 4           | 77,247.5                       | 59.5              | 36.9-78.5   | 91.9            | 81.9-98.2   | 72.2                | 64.1-80.2   | 85.2              | 73.7-95.8   |
| Claude Opus 5 | Custom runner | persistent   | medium | 20             | 20             | 0           | 321,510.5                      | 92.5              | 86.2-98.4   | 93.9            | 89.4-98.4   | 93.5                | 89.8-97.2   | 91.9              | 82.4-98.3   |
| Qwen 3.8 27B  | Biomni        | Biomni       | xhigh  | 40             | 34             | 6           | 81,632.5                       | 97.6              | 92.9-100.0  | 97.4            | 93.5-100.0  | 95.5                | 89.0-100.0  | 99.5              | 98.5-100.0  |
| Qwen 3.8 27B  | Custom runner | persistent   | medium | 40             | 38             | 2           | 183,076                        | 100.0             | 100.0-100.0 | 96.0            | 90.4-100.0  | 95.0                | 85.0-100.0  | 89.7              | 79.2-98.3   |
| Gemma 4 31B   | Custom runner | persistent   | medium | 40             | 37             | 3           | 199,780                        | 99.2              | 97.7-100.0  | 100.0           | 100.0-100.0 | 82.4                | 68.9-93.8   | 99.3              | 98.0-100.0  |
| Astra         | Codex         | sequential   | medium | 40             | 39             | 1           | 80,126                         | 97.5              | 95.6-99.0   | 94.8            | 92.2-96.9   | 95.9                | 93.5-98.2   | 88.7              | 83.1-94.1   |
| Astra         | Codex         | deliberative | medium | 40             | 40             | 0           | 255,276                        | 99.0              | 98.1-99.8   | 94.8            | 92.0-97.4   | 99.3                | 98.6-100.0  | 93.8              | 88.7-98.3   |
| Sol           | Codex         | sequential   | medium | 40             | 40             | 0           | 176,867.5                      | 93.4              | 91.6-95.0   | 94.9            | 92.1-97.0   | 90.2                | 86.5-93.5   | 88.7              | 82.6-93.8   |
| Sol           | Codex         | deliberative | medium | 40             | 38             | 2           | 547,345.5                      | 93.1              | 89.5-96.2   | 93.8            | 90.5-96.5   | 89.1                | 85.1-92.7   | 92.2              | 86.0-97.2   |
| Terra         | Codex         | sequential   | medium | 40             | 39             | 1           | 58,512                         | 93.5              | 90.8-96.4   | 99.1            | 97.2-100.0  | 94.4                | 91.2-97.1   | 97.7              | 94.5-100.0  |
| Terra         | Codex         | deliberative | medium | 40             | 39             | 1           | 160,931                        | 95.7              | 92.8-98.3   | 99.8            | 99.3-100.0  | 91.1                | 78.8-98.6   | 91.6              | 80.0-100.0  |
| Luna          | Codex         | sequential   | medium | 40             | 39             | 1           | 79,146                         | 76.2              | 62.9-86.8   | 89.3            | 83.1-95.1   | 58.2                | 40.9-72.5   | 80.8              | 61.8-95.9   |
| Luna          | Codex         | deliberative | medium | 40             | 38             | 2           | 237,172.5                      | 60.9              | 41.2-78.8   | 90.5            | 84.0-96.1   | 67.7                | 56.2-78.1   | 90.0              | 82.4-96.7   |
| Claude Opus 5 | Custom runner | sequential   | medium | 20             | 19             | 1           | 309,733                        | 98.0              | 95.8-99.8   | 93.8            | 91.1-96.5   | 94.4                | 90.6-97.8   | 92.9              | 86.4-96.8   |
| Claude Opus 5 | Custom runner | deliberative | medium | 20             | 18             | 2           | 1,222,954                      | 97.2              | 95.4-98.9   | 96.5            | 94.6-98.5   | 93.3                | 86.5-97.7   | 94.6              | 90.0-98.4   |
| Qwen 3.8 27B  | Custom runner | sequential   | medium | 40             | 40             | 0           | 265,093.5                      | 100.0             | 100.0-100.0 | 91.8            | 76.8-100.0  | 86.0                | 73.0-98.0   | 99.3              | 97.9-100.0  |
| Qwen 3.8 27B  | Custom runner | deliberative | medium | 40             | 40             | 0           | 822,696.5                      | 100.0             | 100.0-100.0 | 94.9            | 86.9-100.0  | 100.0               | 100.0-100.0 | 99.5              | 98.5-100.0  |
| Gemma 4 31B   | Custom runner | sequential   | medium | 40             | 40             | 0           | 310,695.5                      | 99.1              | 97.3-100.0  | 100.0           | 100.0-100.0 | 97.9                | 94.3-100.0  | 97.4              | 92.1-100.0  |
| Gemma 4 31B   | Custom runner | deliberative | medium | 40             | 33             | 7           | 810,756                        | 99.4              | 98.2-100.0  | 97.9            | 93.7-100.0  | 96.3                | 90.9-100.0  | 100.0             | 100.0-100.0 |

Cohort: all ended runs with reports, including runs that ended with errors. Counts and token medians span all four conditions; condition means use available scored runs. NA = unavailable, not zero. 95% CIs: whole-run bootstrap; Wilson for focal recovery. NA in a CI column means insufficient eligible runs or evidence. Exact model identifiers are in the appendix. Full-precision values and per-condition counts: results.csv and condition\_metrics.csv.

# Recall (%)

Persistent + Biomni: first 8 rows | Other workflows below the divider | Means and 95% CIs

| LLM           | Harness       | Workflow     | Effort | Attempted runs | Completed runs | Failed runs | Median output tokens / success | Expected unmasked |           | Expected masked |           | Surprising unmasked |           | Surprising masked |           |
|---------------|---------------|--------------|--------|----------------|----------------|-------------|--------------------------------|-------------------|-----------|-----------------|-----------|---------------------|-----------|-------------------|-----------|
|               |               |              |        |                |                |             |                                | Mean              | 95% CI    | Mean            | 95% CI    | Mean                | 95% CI    | Mean              | 95% CI    |
| Astra         | Codex         | persistent   | medium | 40             | 40             | 0           | 85,653                         | 68.3              | 66.7-71.7 | 66.7            | 66.7-66.7 | 66.7                | 66.7-66.7 | 68.3              | 66.7-71.7 |
| Sol           | Codex         | persistent   | medium | 40             | 40             | 0           | 172,152                        | 63.3              | 56.7-66.7 | 66.7            | 66.7-66.7 | 61.7                | 51.7-66.7 | 66.7              | 66.7-66.7 |
| Terra         | Codex         | persistent   | medium | 40             | 40             | 0           | 56,127.5                       | 51.1              | 37.8-63.3 | 66.7            | 66.7-66.7 | 60.0                | 51.7-65.0 | 61.7              | 53.3-66.7 |
| Luna          | Codex         | persistent   | medium | 40             | 36             | 4           | 77,247.5                       | 13.3              | 4.4-22.2  | 37.8            | 23.3-52.2 | 21.7                | 11.7-31.7 | 41.7              | 30.0-53.3 |
| Claude Opus 5 | Custom runner | persistent   | medium | 20             | 20             | 0           | 321,510.5                      | 66.7              | 66.7-66.7 | 66.7            | 66.7-66.7 | 66.7                | 66.7-66.7 | 66.7              | 66.7-66.7 |
| Qwen 3.8 27B  | Biomni        | Biomni       | xhigh  | 40             | 34             | 6           | 81,632.5                       | 63.3              | 56.7-66.7 | 66.7            | 66.7-66.7 | 56.7                | 41.7-66.7 | 66.7              | 66.7-66.7 |
| Qwen 3.8 27B  | Custom runner | persistent   | medium | 40             | 38             | 2           | 183,076                        | 60.0              | 50.0-66.7 | 58.9            | 47.8-66.7 | 60.0                | 50.0-66.7 | 58.3              | 50.0-66.7 |
| Gemma 4 31B   | Custom runner | persistent   | medium | 40             | 37             | 3           | 199,780                        | 38.9              | 31.1-48.9 | 65.6            | 63.3-66.7 | 50.0                | 40.0-60.0 | 66.7              | 66.7-66.7 |
| Astra         | Codex         | sequential   | medium | 40             | 39             | 1           | 80,126                         | 63.3              | 56.7-66.7 | 73.3            | 68.3-78.3 | 70.0                | 66.7-76.7 | 68.3              | 66.7-71.7 |
| Astra         | Codex         | deliberative | medium | 40             | 40             | 0           | 255,276                        | 68.3              | 66.7-71.7 | 66.7            | 66.7-66.7 | 66.7                | 66.7-66.7 | 66.7              | 66.7-66.7 |
| Sol           | Codex         | sequential   | medium | 40             | 40             | 0           | 176,867.5                      | 54.4              | 41.1-66.7 | 66.7            | 66.7-66.7 | 58.3                | 46.7-66.7 | 66.7              | 66.7-66.7 |
| Sol           | Codex         | deliberative | medium | 40             | 38             | 2           | 547,345.5                      | 48.9              | 34.4-63.3 | 66.7            | 66.7-66.7 | 38.3                | 23.3-53.3 | 66.7              | 66.7-66.7 |
| Terra         | Codex         | sequential   | medium | 40             | 39             | 1           | 58,512                         | 52.2              | 41.1-63.3 | 66.7            | 66.7-66.7 | 55.0                | 45.0-65.0 | 65.0              | 61.7-66.7 |
| Terra         | Codex         | deliberative | medium | 40             | 39             | 1           | 160,931                        | 54.4              | 42.2-66.7 | 64.4            | 60.0-66.7 | 35.0                | 23.3-46.7 | 66.7              | 66.7-66.7 |
| Luna          | Codex         | sequential   | medium | 40             | 39             | 1           | 79,146                         | 4.4               | 0.0-8.9   | 51.1            | 41.1-60.0 | 6.7                 | 0.0-13.3  | 36.7              | 23.3-50.0 |
| Luna          | Codex         | deliberative | medium | 40             | 38             | 2           | 237,172.5                      | 4.4               | 0.0-10.0  | 41.1            | 27.8-54.4 | 10.0                | 5.0-15.0  | 40.0              | 26.7-53.3 |
| Claude Opus 5 | Custom runner | sequential   | medium | 20             | 19             | 1           | 309,733                        | 66.7              | 66.7-66.7 | 66.7            | 66.7-66.7 | 66.7                | 66.7-66.7 | 63.3              | 56.7-66.7 |
| Claude Opus 5 | Custom runner | deliberative | medium | 20             | 18             | 2           | 1,222,954                      | 66.7              | 66.7-66.7 | 66.7            | 66.7-66.7 | 66.7                | 66.7-66.7 | 66.7              | 66.7-66.7 |
| Qwen 3.8 27B  | Custom runner | sequential   | medium | 40             | 40             | 0           | 265,093.5                      | 46.7              | 36.7-56.7 | 60.0            | 50.0-66.7 | 53.3                | 45.0-60.0 | 61.7              | 53.3-66.7 |
| Qwen 3.8 27B  | Custom runner | deliberative | medium | 40             | 40             | 0           | 822,696.5                      | 53.3              | 43.3-63.3 | 50.0            | 35.6-63.3 | 56.7                | 46.7-66.7 | 51.7              | 40.0-63.3 |
| Gemma 4 31B   | Custom runner | sequential   | medium | 40             | 40             | 0           | 310,695.5                      | 42.2              | 30.0-54.4 | 66.7            | 66.7-66.7 | 53.3                | 41.7-63.3 | 66.7              | 66.7-66.7 |
| Gemma 4 31B   | Custom runner | deliberative | medium | 40             | 33             | 7           | 810,756                        | 56.7              | 46.7-66.7 | 66.7            | 66.7-66.7 | 56.7                | 48.3-65.0 | 66.7              | 66.7-66.7 |

Cohort: all ended runs with reports, including runs that ended with errors. Counts and token medians span all four conditions; condition means use available scored runs. NA = unavailable, not zero. 95% CIs: whole-run bootstrap; Wilson for focal recovery. NA in a CI column means insufficient eligible runs or evidence. Exact model identifiers are in the appendix. Full-precision values and per-condition counts: results.csv and condition\_metrics.csv.

# Scientific F1\*

Persistent + Biomni: first 8 rows | Other workflows below the divider | Means and 95% CIs

| LLM           | Harness       | Workflow     | Effort | Attempted runs | Completed runs | Failed runs | Median output tokens / success | Expected unmasked |           | Expected masked |           | Surprising unmasked |           | Surprising masked |           |
|---------------|---------------|--------------|--------|----------------|----------------|-------------|--------------------------------|-------------------|-----------|-----------------|-----------|---------------------|-----------|-------------------|-----------|
|               |               |              |        |                |                |             |                                | Mean              | 95% CI    | Mean            | 95% CI    | Mean                | 95% CI    | Mean              | 95% CI    |
| Astra         | Codex         | persistent   | medium | 40             | 40             | 0           | 85,653                         | 80.3              | 78.5-83.0 | 77.6            | 76.5-78.7 | 77.8                | 76.4-79.0 | 78.8              | 76.5-81.8 |
| Sol           | Codex         | persistent   | medium | 40             | 40             | 0           | 172,152                        | 75.6              | 69.7-78.8 | 78.7            | 78.1-79.2 | 72.4                | 62.0-78.4 | 77.6              | 75.6-79.0 |
| Terra         | Codex         | persistent   | medium | 40             | 40             | 0           | 56,127.5                       | 64.3              | 49.9-76.0 | 79.5            | 79.2-79.8 | 73.4                | 66.8-78.4 | 75.1              | 68.4-80.0 |
| Luna          | Codex         | persistent   | medium | 40             | 36             | 4           | 77,247.5                       | 18.8              | 6.7-31.1  | 48.9            | 32.0-64.7 | 30.0                | 16.2-43.0 | 53.3              | 41.3-65.0 |
| Claude Opus 5 | Custom runner | persistent   | medium | 20             | 20             | 0           | 321,510.5                      | 77.4              | 75.1-79.5 | 77.9            | 76.3-79.5 | 77.8                | 76.5-79.1 | 77.1              | 73.4-79.4 |
| Qwen 3.8 27B  | Biomni        | Biomni       | xhigh  | 40             | 34             | 6           | 81,632.5                       | 76.6              | 69.9-80.0 | 79.1            | 77.6-80.0 | 67.9                | 51.3-78.7 | 79.8              | 79.5-80.0 |
| Qwen 3.8 27B  | Custom runner | persistent   | medium | 40             | 38             | 2           | 183,076                        | 74.0              | 65.0-80.0 | 71.6            | 58.5-80.0 | 73.0                | 63.0-80.0 | 70.6              | 61.1-79.2 |
| Gemma 4 31B   | Custom runner | persistent   | medium | 40             | 37             | 3           | 199,780                        | 54.5              | 47.2-63.6 | 79.1            | 77.4-80.0 | 61.1                | 50.3-71.9 | 79.8              | 79.3-80.0 |
| Astra         | Codex         | sequential   | medium | 40             | 39             | 1           | 80,126                         | 76.2              | 70.2-79.5 | 82.4            | 79.1-85.9 | 80.5                | 78.1-84.4 | 77.0              | 74.0-80.3 |
| Astra         | Codex         | deliberative | medium | 40             | 40             | 0           | 255,276                        | 80.8              | 79.4-83.1 | 78.2            | 77.3-79.1 | 79.8                | 79.5-80.0 | 77.8              | 76.0-79.4 |
| Sol           | Codex         | sequential   | medium | 40             | 40             | 0           | 176,867.5                      | 66.3              | 54.3-77.4 | 78.3            | 77.3-79.0 | 69.0                | 57.6-77.1 | 75.9              | 73.5-77.9 |
| Sol           | Codex         | deliberative | medium | 40             | 38             | 2           | 547,345.5                      | 60.2              | 45.4-73.2 | 77.9            | 76.7-78.9 | 49.6                | 35.1-64.4 | 77.2              | 74.8-79.1 |
| Terra         | Codex         | sequential   | medium | 40             | 39             | 1           | 58,512                         | 65.5              | 54.9-75.8 | 79.7            | 79.0-80.0 | 67.6                | 56.5-76.3 | 77.9              | 74.9-80.0 |
| Terra         | Codex         | deliberative | medium | 40             | 39             | 1           | 160,931                        | 66.9              | 52.9-78.0 | 78.1            | 74.4-80.0 | 48.1                | 33.7-61.2 | 76.4              | 71.5-80.0 |
| Luna          | Codex         | sequential   | medium | 40             | 39             | 1           | 79,146                         | 6.9               | 0.0-13.8  | 64.0            | 54.1-73.3 | 9.1                 | 0.0-18.3  | 48.3              | 32.2-63.7 |
| Luna          | Codex         | deliberative | medium | 40             | 38             | 2           | 237,172.5                      | 6.8               | 0.0-14.3  | 53.5            | 38.7-67.7 | 16.2                | 8.0-24.3  | 51.8              | 36.2-66.0 |
| Claude Opus 5 | Custom runner | sequential   | medium | 20             | 19             | 1           | 309,733                        | 79.3              | 78.6-79.9 | 77.9            | 77.0-78.8 | 78.1                | 76.8-79.3 | 74.9              | 69.9-78.7 |
| Claude Opus 5 | Custom runner | deliberative | medium | 20             | 18             | 2           | 1,222,954                      | 79.1              | 78.5-79.6 | 78.8            | 78.2-79.5 | 77.7                | 75.2-79.2 | 78.2              | 76.5-79.5 |
| Qwen 3.8 27B  | Custom runner | sequential   | medium | 40             | 40             | 0           | 265,093.5                      | 62.0              | 53.0-71.0 | 71.6            | 59.2-80.0 | 65.4                | 56.2-73.3 | 75.5              | 69.5-80.0 |
| Qwen 3.8 27B  | Custom runner | deliberative | medium | 40             | 40             | 0           | 822,696.5                      | 68.0              | 59.0-77.0 | 62.4            | 44.9-76.2 | 71.0                | 62.0-80.0 | 65.7              | 54.5-76.7 |
| Gemma 4 31B   | Custom runner | sequential   | medium | 40             | 40             | 0           | 310,695.5                      | 56.0              | 40.9-69.2 | 80.0            | 80.0-80.0 | 66.8                | 55.5-76.5 | 79.0              | 77.0-80.0 |
| Gemma 4 31B   | Custom runner | deliberative | medium | 40             | 33             | 7           | 810,756                        | 70.8              | 61.8-79.4 | 79.2            | 77.7-80.0 | 70.7                | 62.4-78.4 | 80.0              | 80.0-80.0 |

Cohort: all ended runs with reports, including runs that ended with errors. Counts and token medians span all four conditions; condition means use available scored runs. NA = unavailable, not zero. 95% CIs: whole-run bootstrap; Wilson for focal recovery. NA in a CI column means insufficient eligible runs or evidence. Exact model identifiers are in the appendix. Full-precision values and per-condition counts: results.csv and condition\_metrics.csv.

# Exploration E

Persistent + Biomni: first 8 rows | Other workflows below the divider | Means and 95% CIs

| LLM           | Harness       | Workflow     | Effort | Attempted runs | Completed runs | Failed runs | Median output tokens / success | Expected unmasked |           | Expected masked |           | Surprising unmasked |           | Surprising masked |           |
|---------------|---------------|--------------|--------|----------------|----------------|-------------|--------------------------------|-------------------|-----------|-----------------|-----------|---------------------|-----------|-------------------|-----------|
|               |               |              |        |                |                |             |                                | Mean              | 95% CI    | Mean            | 95% CI    | Mean                | 95% CI    | Mean              | 95% CI    |
| Astra         | Codex         | persistent   | medium | 40             | 40             | 0           | 85,653                         | 65.8              | 65.6-66.0 | 64.3            | 64.1-64.4 | 65.2                | 65.0-65.3 | 64.5              | 64.1-64.9 |
| Sol           | Codex         | persistent   | medium | 40             | 40             | 0           | 172,152                        | 66.0              | 65.7-66.3 | 61.7            | 59.9-63.2 | 60.5                | 50.3-65.9 | 62.3              | 61.3-63.1 |
| Terra         | Codex         | persistent   | medium | 40             | 40             | 0           | 56,127.5                       | 59.2              | 46.5-66.0 | 63.1            | 61.5-64.4 | 65.6                | 64.8-66.2 | 64.0              | 62.1-65.1 |
| Luna          | Codex         | persistent   | medium | 40             | 36             | 4           | 77,247.5                       | 25.3              | 7.1-44.0  | 47.1            | 34.1-57.7 | 38.1                | 21.3-52.9 | 43.7              | 31.7-54.9 |
| Claude Opus 5 | Custom runner | persistent   | medium | 20             | 20             | 0           | 321,510.5                      | 64.0              | 61.5-65.3 | 61.3            | 59.6-62.8 | 63.9                | 61.9-65.5 | 63.9              | 63.6-64.0 |
| Qwen 3.8 27B  | Biomni        | Biomni       | xhigh  | 40             | 34             | 6           | 81,632.5                       | 65.9              | 64.7-66.7 | 66.7            | 66.7-66.7 | 59.4                | 46.1-66.5 | 66.7              | 66.7-66.7 |
| Qwen 3.8 27B  | Custom runner | persistent   | medium | 40             | 38             | 2           | 183,076                        | 64.8              | 64.4-65.2 | 59.4            | 56.0-62.1 | 64.7                | 64.1-65.1 | 61.1              | 60.3-62.0 |
| Gemma 4 31B   | Custom runner | persistent   | medium | 40             | 37             | 3           | 199,780                        | 62.5              | 60.1-64.4 | 62.3            | 59.1-64.2 | 63.5                | 62.5-64.3 | 63.9              | 62.7-64.9 |
| Astra         | Codex         | sequential   | medium | 40             | 39             | 1           | 80,126                         | 65.6              | 65.5-65.8 | 66.0            | 64.2-68.4 | 66.2                | 64.9-68.5 | 64.8              | 64.2-65.6 |
| Astra         | Codex         | deliberative | medium | 40             | 40             | 0           | 255,276                        | 66.3              | 65.2-68.2 | 64.1            | 63.9-64.3 | 64.8                | 64.7-65.0 | 64.3              | 63.9-64.6 |
| Sol           | Codex         | sequential   | medium | 40             | 40             | 0           | 176,867.5                      | 60.4              | 49.1-66.2 | 61.4            | 60.4-62.2 | 60.5                | 50.0-66.0 | 62.1              | 60.7-63.3 |
| Sol           | Codex         | deliberative | medium | 40             | 38             | 2           | 547,345.5                      | 54.0              | 37.9-65.8 | 61.4            | 60.6-62.3 | 40.5                | 25.6-55.5 | 61.4              | 60.5-62.3 |
| Terra         | Codex         | sequential   | medium | 40             | 39             | 1           | 58,512                         | 64.9              | 63.9-65.9 | 61.2            | 59.1-63.1 | 61.5                | 57.8-64.7 | 62.8              | 61.6-63.8 |
| Terra         | Codex         | deliberative | medium | 40             | 39             | 1           | 160,931                        | 59.7              | 48.5-65.8 | 57.4            | 51.1-62.1 | 59.6                | 49.4-66.5 | 60.9              | 57.1-64.1 |
| Luna          | Codex         | sequential   | medium | 40             | 39             | 1           | 79,146                         | 18.7              | 5.6-32.0  | 52.4            | 47.3-56.6 | 29.5                | 14.1-44.6 | 44.1              | 35.5-52.1 |
| Luna          | Codex         | deliberative | medium | 40             | 38             | 2           | 237,172.5                      | 17.8              | 4.4-32.2  | 36.0            | 24.2-47.6 | 18.8                | 7.1-31.7  | 48.7              | 41.3-55.5 |
| Claude Opus 5 | Custom runner | sequential   | medium | 20             | 19             | 1           | 309,733                        | 65.4              | 65.2-65.7 | 61.9            | 61.3-62.8 | 64.8                | 64.0-65.5 | 63.9              | 63.1-64.7 |
| Claude Opus 5 | Custom runner | deliberative | medium | 20             | 18             | 2           | 1,222,954                      | 64.2              | 63.2-65.2 | 58.0            | 54.8-61.3 | 64.7                | 64.7-64.7 | 62.0              | 61.1-63.2 |
| Qwen 3.8 27B  | Custom runner | sequential   | medium | 40             | 40             | 0           | 265,093.5                      | 63.5              | 62.0-64.8 | 58.7            | 55.5-60.9 | 63.6                | 62.1-64.9 | 57.5              | 51.1-61.4 |
| Qwen 3.8 27B  | Custom runner | deliberative | medium | 40             | 40             | 0           | 822,696.5                      | 65.0              | 64.2-65.8 | 51.3            | 40.8-60.7 | 64.9                | 64.4-65.5 | 48.0              | 37.3-57.6 |
| Gemma 4 31B   | Custom runner | sequential   | medium | 40             | 40             | 0           | 310,695.5                      | 63.0              | 60.4-64.6 | 63.8            | 62.8-64.6 | 56.9                | 50.3-62.1 | 64.3              | 63.6-65.0 |
| Gemma 4 31B   | Custom runner | deliberative | medium | 40             | 33             | 7           | 810,756                        | 64.9              | 63.5-65.8 | 62.8            | 60.2-64.4 | 62.2                | 58.8-64.7 | 64.5              | 63.5-65.1 |

Cohort: all ended runs with reports, including runs that ended with errors. Counts and token medians span all four conditions; condition means use available scored runs. NA = unavailable, not zero. 95% CIs: whole-run bootstrap; Wilson for focal recovery. NA in a CI column means insufficient eligible runs or evidence. Exact model identifiers are in the appendix. Full-precision values and per-condition counts: results.csv and condition\_metrics.csv.

# Evidence responsiveness B

Persistent + Biomni: first 8 rows | Other workflows below the divider | Means and 95% CIs

| LLM           | Harness       | Workflow     | Effort | Attempted runs | Completed runs | Failed runs | Median output tokens / success | Expected unmasked |             | Expected masked |            | Surprising unmasked |            | Surprising masked |            |
|---------------|---------------|--------------|--------|----------------|----------------|-------------|--------------------------------|-------------------|-------------|-----------------|------------|---------------------|------------|-------------------|------------|
|               |               |              |        |                |                |             |                                | Mean              | 95% CI      | Mean            | 95% CI     | Mean                | 95% CI     | Mean              | 95% CI     |
| Astra         | Codex         | persistent   | medium | 40             | 40             | 0           | 85,653                         | 100.0             | 100.0-100.0 | 94.2            | 89.2-98.3  | 87.5                | 80.0-94.6  | 95.4              | 91.7-98.5  |
| Sol           | Codex         | persistent   | medium | 40             | 40             | 0           | 172,152                        | 87.0              | 77.8-95.2   | 93.1            | 86.4-100.0 | 79.8                | 72.6-87.5  | 89.2              | 81.7-96.7  |
| Terra         | Codex         | persistent   | medium | 40             | 40             | 0           | 56,127.5                       | 86.8              | 78.1-95.6   | 87.8            | 68.9-100.0 | 83.4                | 71.1-95.8  | 92.2              | 85.0-98.3  |
| Luna          | Codex         | persistent   | medium | 40             | 36             | 4           | 77,247.5                       | 71.0              | 57.9-82.7   | 79.0            | 66.5-90.6  | 93.0                | 85.5-98.3  | 80.1              | 67.1-90.4  |
| Claude Opus 5 | Custom runner | persistent   | medium | 20             | 20             | 0           | 321,510.5                      | 87.5              | 72.2-100.0  | 75.0            | 66.7-91.7  | 69.4                | 66.7-75.0  | 68.7              | 57.3-80.0  |
| Qwen 3.8 27B  | Biomni        | Biomni       | xhigh  | 40             | 34             | 6           | 81,632.5                       | 61.7              | 54.9-66.2   | 63.5            | 60.6-66.0  | 57.5                | 49.3-63.0  | 64.3              | 60.0-66.7  |
| Qwen 3.8 27B  | Custom runner | persistent   | medium | 40             | 38             | 2           | 183,076                        | 63.3              | NA          | 59.1            | 52.8-64.4  | 60.4                | 54.2-65.6  | 57.4              | 49.4-64.2  |
| Gemma 4 31B   | Custom runner | persistent   | medium | 40             | 37             | 3           | 199,780                        | 60.0              | 52.5-65.8   | 62.3            | 59.8-64.7  | 65.0                | 61.7-66.7  | 63.8              | 61.1-66.1  |
| Astra         | Codex         | sequential   | medium | 40             | 39             | 1           | 80,126                         | 87.5              | 76.2-100.0  | 96.9            | 92.5-100.0 | 96.2                | 92.3-100.0 | 93.7              | 87.8-98.7  |
| Astra         | Codex         | deliberative | medium | 40             | 40             | 0           | 255,276                        | 97.6              | 92.6-100.0  | 97.2            | 93.3-100.0 | 97.9                | 94.5-100.0 | 97.8              | 94.4-100.0 |
| Sol           | Codex         | sequential   | medium | 40             | 40             | 0           | 176,867.5                      | 91.6              | 85.0-97.2   | 87.2            | 80.6-93.3  | 88.8                | 80.5-95.8  | 84.7              | 74.4-93.9  |
| Sol           | Codex         | deliberative | medium | 40             | 38             | 2           | 547,345.5                      | 91.7              | 84.8-97.5   | 89.4            | 76.7-100.0 | 91.4                | 84.4-97.2  | 90.9              | 84.5-97.2  |
| Terra         | Codex         | sequential   | medium | 40             | 39             | 1           | 58,512                         | 90.6              | 79.8-100.0  | 91.1            | 76.7-100.0 | 84.0                | 73.6-94.4  | 94.4              | 87.9-100.0 |
| Terra         | Codex         | deliberative | medium | 40             | 39             | 1           | 160,931                        | 80.0              | 66.1-93.3   | 92.8            | 80.0-100.0 | 89.6                | 79.4-97.6  | 91.1              | 85.4-96.7  |
| Luna          | Codex         | sequential   | medium | 40             | 39             | 1           | 79,146                         | 78.4              | 65.0-89.3   | 76.4            | 60.9-91.1  | 78.4                | 70.5-86.0  | 75.0              | 63.6-84.9  |
| Luna          | Codex         | deliberative | medium | 40             | 38             | 2           | 237,172.5                      | 74.0              | 63.8-83.3   | 86.8            | 73.7-96.5  | 87.3                | 81.0-93.1  | 91.8              | 85.6-98.3  |
| Claude Opus 5 | Custom runner | sequential   | medium | 20             | 19             | 1           | 309,733                        | 70.0              | 66.7-76.7   | 68.6            | 66.7-73.3  | 75.0                | 66.7-91.7  | 76.7              | 66.7-90.0  |
| Claude Opus 5 | Custom runner | deliberative | medium | 20             | 18             | 2           | 1,222,954                      | 74.1              | 66.7-88.9   | 73.9            | 66.7-82.2  | 73.6                | 68.0-79.1  | 81.8              | 69.6-93.3  |
| Qwen 3.8 27B  | Custom runner | sequential   | medium | 40             | 40             | 0           | 265,093.5                      | 63.6              | NA          | 60.2            | 53.0-65.8  | 55.7                | 45.0-64.1  | 64.3              | 61.9-66.7  |
| Qwen 3.8 27B  | Custom runner | deliberative | medium | 40             | 40             | 0           | 822,696.5                      | 62.4              | NA          | 61.4            | 56.9-65.0  | 57.8                | NA         | 64.3              | 62.0-66.7  |
| Gemma 4 31B   | Custom runner | sequential   | medium | 40             | 40             | 0           | 310,695.5                      | 64.2              | 61.3-66.7   | 61.4            | 58.0-64.4  | 63.7                | 60.5-66.7  | 65.1              | 63.5-66.7  |
| Gemma 4 31B   | Custom runner | deliberative | medium | 40             | 33             | 7           | 810,756                        | 64.9              | 62.4-66.7   | 60.9            | 57.9-63.4  | 63.2                | 59.3-66.7  | 65.6              | 63.9-66.7  |

Cohort: all ended runs with reports, including runs that ended with errors. Counts and token medians span all four conditions; condition means use available scored runs. NA = unavailable, not zero. 95% CIs: whole-run bootstrap; Wilson for focal recovery. NA in a CI column means insufficient eligible runs or evidence. Exact model identifiers are in the appendix. Full-precision values and per-condition counts: results.csv and condition\_metrics.csv.

# Focal recovery (%)

Persistent + Biomni: first 8 rows | Other workflows below the divider | Means and 95% CIs

| LLM           | Harness       | Workflow     | Effort | Attempted runs | Completed runs | Failed runs | Median output tokens / success | Expected unmasked |            | Expected masked |            | Surprising unmasked |            | Surprising masked |            |
|---------------|---------------|--------------|--------|----------------|----------------|-------------|--------------------------------|-------------------|------------|-----------------|------------|---------------------|------------|-------------------|------------|
|               |               |              |        |                |                |             |                                | Mean              | 95% CI     | Mean            | 95% CI     | Mean                | 95% CI     | Mean              | 95% CI     |
| Astra         | Codex         | persistent   | medium | 40             | 40             | 0           | 85,653                         | 100.0             | 72.2-100.0 | 100.0           | 72.2-100.0 | 100.0               | 72.2-100.0 | 100.0             | 72.2-100.0 |
| Sol           | Codex         | persistent   | medium | 40             | 40             | 0           | 172,152                        | 100.0             | 72.2-100.0 | 100.0           | 72.2-100.0 | 90.0                | 59.6-98.2  | 100.0             | 72.2-100.0 |
| Terra         | Codex         | persistent   | medium | 40             | 40             | 0           | 56,127.5                       | 90.0              | 59.6-98.2  | 100.0           | 72.2-100.0 | 70.0                | 39.7-89.2  | 90.0              | 59.6-98.2  |
| Luna          | Codex         | persistent   | medium | 40             | 36             | 4           | 77,247.5                       | 40.0              | 16.8-68.7  | 70.0            | 39.7-89.2  | 10.0                | 1.8-40.4   | 50.0              | 23.7-76.3  |
| Claude Opus 5 | Custom runner | persistent   | medium | 20             | 20             | 0           | 321,510.5                      | 100.0             | 56.6-100.0 | 100.0           | 56.6-100.0 | 100.0               | 56.6-100.0 | 100.0             | 56.6-100.0 |
| Qwen 3.8 27B  | Biomni        | Biomni       | xhigh  | 40             | 34             | 6           | 81,632.5                       | 100.0             | 72.2-100.0 | 100.0           | 72.2-100.0 | 90.0                | 59.6-98.2  | 100.0             | 72.2-100.0 |
| Qwen 3.8 27B  | Custom runner | persistent   | medium | 40             | 38             | 2           | 183,076                        | 100.0             | 72.2-100.0 | 80.0            | 49.0-94.3  | 80.0                | 49.0-94.3  | 100.0             | 72.2-100.0 |
| Gemma 4 31B   | Custom runner | persistent   | medium | 40             | 37             | 3           | 199,780                        | 100.0             | 72.2-100.0 | 100.0           | 72.2-100.0 | 50.0                | 23.7-76.3  | 100.0             | 72.2-100.0 |
| Astra         | Codex         | sequential   | medium | 40             | 39             | 1           | 80,126                         | 100.0             | 72.2-100.0 | 100.0           | 72.2-100.0 | 100.0               | 72.2-100.0 | 100.0             | 72.2-100.0 |
| Astra         | Codex         | deliberative | medium | 40             | 40             | 0           | 255,276                        | 100.0             | 72.2-100.0 | 100.0           | 72.2-100.0 | 100.0               | 72.2-100.0 | 100.0             | 72.2-100.0 |
| Sol           | Codex         | sequential   | medium | 40             | 40             | 0           | 176,867.5                      | 90.0              | 59.6-98.2  | 100.0           | 72.2-100.0 | 80.0                | 49.0-94.3  | 100.0             | 72.2-100.0 |
| Sol           | Codex         | deliberative | medium | 40             | 38             | 2           | 547,345.5                      | 80.0              | 49.0-94.3  | 100.0           | 72.2-100.0 | 40.0                | 16.8-68.7  | 100.0             | 72.2-100.0 |
| Terra         | Codex         | sequential   | medium | 40             | 39             | 1           | 58,512                         | 100.0             | 72.2-100.0 | 100.0           | 72.2-100.0 | 70.0                | 39.7-89.2  | 100.0             | 72.2-100.0 |
| Terra         | Codex         | deliberative | medium | 40             | 39             | 1           | 160,931                        | 90.0              | 59.6-98.2  | 90.0            | 59.6-98.2  | 20.0                | 5.7-51.0   | 100.0             | 72.2-100.0 |
| Luna          | Codex         | sequential   | medium | 40             | 39             | 1           | 79,146                         | 10.0              | 1.8-40.4   | 80.0            | 49.0-94.3  | 0.0                 | 0.0-27.8   | 60.0              | 31.3-83.2  |
| Luna          | Codex         | deliberative | medium | 40             | 38             | 2           | 237,172.5                      | 10.0              | 1.8-40.4   | 80.0            | 49.0-94.3  | 0.0                 | 0.0-27.8   | 70.0              | 39.7-89.2  |
| Claude Opus 5 | Custom runner | sequential   | medium | 20             | 19             | 1           | 309,733                        | 100.0             | 56.6-100.0 | 100.0           | 56.6-100.0 | 100.0               | 56.6-100.0 | 100.0             | 56.6-100.0 |
| Claude Opus 5 | Custom runner | deliberative | medium | 20             | 18             | 2           | 1,222,954                      | 100.0             | 56.6-100.0 | 100.0           | 56.6-100.0 | 100.0               | 56.6-100.0 | 100.0             | 56.6-100.0 |
| Qwen 3.8 27B  | Custom runner | sequential   | medium | 40             | 40             | 0           | 265,093.5                      | 100.0             | 72.2-100.0 | 90.0            | 59.6-98.2  | 80.0                | 49.0-94.3  | 100.0             | 72.2-100.0 |
| Qwen 3.8 27B  | Custom runner | deliberative | medium | 40             | 40             | 0           | 822,696.5                      | 100.0             | 72.2-100.0 | 80.0            | 49.0-94.3  | 70.0                | 39.7-89.2  | 90.0              | 59.6-98.2  |
| Gemma 4 31B   | Custom runner | sequential   | medium | 40             | 40             | 0           | 310,695.5                      | 90.0              | 59.6-98.2  | 100.0           | 72.2-100.0 | 70.0                | 39.7-89.2  | 100.0             | 72.2-100.0 |
| Gemma 4 31B   | Custom runner | deliberative | medium | 40             | 33             | 7           | 810,756                        | 100.0             | 72.2-100.0 | 100.0           | 72.2-100.0 | 80.0                | 49.0-94.3  | 100.0             | 72.2-100.0 |

Cohort: all ended runs with reports, including runs that ended with errors. Counts and token medians span all four conditions; condition means use available scored runs. NA = unavailable, not zero. 95% CIs: whole-run bootstrap; Wilson for focal recovery. NA in a CI column means insufficient eligible runs or evidence. Exact model identifiers are in the appendix. Full-precision values and per-condition counts: results.csv and condition\_metrics.csv.

# Weighted condition score C

Persistent + Biomni: first 8 rows | Other workflows below the divider | Means and 95% CIs

| LLM           | Harness       | Workflow     | Effort | Attempted runs | Completed runs | Failed runs | Median output tokens / success | Expected unmasked |           | Expected masked |           | Surprising unmasked |           | Surprising masked |           |
|---------------|---------------|--------------|--------|----------------|----------------|-------------|--------------------------------|-------------------|-----------|-----------------|-----------|---------------------|-----------|-------------------|-----------|
|               |               |              |        |                |                |             |                                | Mean              | 95% CI    | Mean            | 95% CI    | Mean                | 95% CI    | Mean              | 95% CI    |
| Astra         | Codex         | persistent   | medium | 40             | 40             | 0           | 85,653                         | 86.3              | 85.6-87.2 | 83.9            | 82.6-84.9 | 82.8                | 81.3-84.2 | 84.6              | 83.4-85.9 |
| Sol           | Codex         | persistent   | medium | 40             | 40             | 0           | 172,152                        | 82.1              | 79.3-84.5 | 83.5            | 82.1-84.7 | 75.9                | 66.0-81.8 | 82.4              | 80.4-84.1 |
| Terra         | Codex         | persistent   | medium | 40             | 40             | 0           | 56,127.5                       | 74.2              | 62.9-81.7 | 83.0            | 78.9-85.5 | 73.0                | 64.1-81.0 | 80.0              | 72.6-85.2 |
| Luna          | Codex         | persistent   | medium | 40             | 36             | 4           | 77,247.5                       | 35.8              | 21.3-51.1 | 59.8            | 45.1-73.1 | 39.2                | 29.5-49.9 | 55.9              | 41.0-69.9 |
| Claude Opus 5 | Custom runner | persistent   | medium | 20             | 20             | 0           | 321,510.5                      | 82.4              | 78.8-85.6 | 79.5            | 77.7-82.4 | 78.9                | 78.2-79.5 | 78.5              | 75.2-81.2 |
| Qwen 3.8 27B  | Biomni        | Biomni       | xhigh  | 40             | 34             | 6           | 81,632.5                       | 77.3              | 73.6-79.4 | 78.7            | 77.7-79.4 | 69.7                | 55.6-78.3 | 79.1              | 78.2-79.7 |
| Qwen 3.8 27B  | Custom runner | persistent   | medium | 40             | 38             | 2           | 183,076                        | 76.5              | NA        | 68.8            | 56.3-78.1 | 70.6                | 58.7-79.0 | 73.4              | 68.5-77.7 |
| Gemma 4 31B   | Custom runner | persistent   | medium | 40             | 37             | 3           | 199,780                        | 68.6              | 66.1-71.4 | 77.6            | 76.0-78.7 | 59.6                | 48.0-71.2 | 78.5              | 78.0-78.9 |
| Astra         | Codex         | sequential   | medium | 40             | 39             | 1           | 80,126                         | 82.3              | 79.8-84.8 | 86.4            | 84.4-88.4 | 85.7                | 84.5-87.1 | 83.6              | 81.6-85.6 |
| Astra         | Codex         | deliberative | medium | 40             | 40             | 0           | 255,276                        | 86.1              | 85.8-86.5 | 84.6            | 83.8-85.4 | 85.5                | 84.8-85.9 | 84.6              | 83.9-85.4 |
| Sol           | Codex         | sequential   | medium | 40             | 40             | 0           | 176,867.5                      | 76.1              | 64.4-83.2 | 82.1            | 80.9-83.3 | 74.0                | 60.2-83.8 | 80.9              | 78.6-83.2 |
| Sol           | Codex         | deliberative | medium | 40             | 38             | 2           | 547,345.5                      | 70.2              | 55.1-81.6 | 82.4            | 79.6-84.5 | 53.8                | 38.2-69.7 | 82.5              | 80.4-84.2 |
| Terra         | Codex         | sequential   | medium | 40             | 39             | 1           | 58,512                         | 79.0              | 74.3-83.4 | 83.4            | 80.2-85.4 | 70.3                | 59.7-79.1 | 83.7              | 82.2-85.0 |
| Terra         | Codex         | deliberative | medium | 40             | 39             | 1           | 160,931                        | 73.8              | 62.2-80.8 | 79.9            | 72.4-84.7 | 51.7                | 40.7-63.4 | 82.1              | 79.6-84.4 |
| Luna          | Codex         | sequential   | medium | 40             | 39             | 1           | 79,146                         | 24.3              | 17.1-33.6 | 68.2            | 60.1-75.2 | 24.8                | 19.9-30.4 | 55.7              | 40.3-70.9 |
| Luna          | Codex         | deliberative | medium | 40             | 38             | 2           | 237,172.5                      | 23.2              | 14.8-34.3 | 63.3            | 50.9-74.3 | 26.9                | 21.5-32.4 | 63.8              | 51.3-75.0 |
| Claude Opus 5 | Custom runner | sequential   | medium | 20             | 19             | 1           | 309,733                        | 79.9              | 79.0-81.3 | 78.4            | 77.6-79.7 | 80.3                | 78.2-83.9 | 79.3              | 76.0-83.0 |
| Claude Opus 5 | Custom runner | deliberative | medium | 20             | 18             | 2           | 1,222,954                      | 80.3              | 78.7-83.4 | 79.0            | 77.2-80.3 | 79.8                | 78.8-81.3 | 81.1              | 77.9-83.8 |
| Qwen 3.8 27B  | Custom runner | sequential   | medium | 40             | 40             | 0           | 265,093.5                      | 72.1              | NA        | 71.3            | 61.0-77.4 | 66.8                | 54.6-76.1 | 75.8              | 72.4-78.2 |
| Qwen 3.8 27B  | Custom runner | deliberative | medium | 40             | 40             | 0           | 822,696.5                      | 74.3              | NA        | 64.4            | 50.4-75.2 | 66.9                | NA        | 68.0              | 57.5-76.2 |
| Gemma 4 31B   | Custom runner | sequential   | medium | 40             | 40             | 0           | 310,695.5                      | 67.5              | 57.1-74.4 | 78.0            | 77.4-78.6 | 65.0                | 53.2-75.6 | 78.5              | 77.8-79.1 |
| Gemma 4 31B   | Custom runner | deliberative | medium | 40             | 33             | 7           | 810,756                        | 75.7              | 72.4-78.6 | 77.5            | 76.3-78.4 | 69.8                | 59.7-77.6 | 79.0              | 78.6-79.3 |

Cohort: all ended runs with reports, including runs that ended with errors. Counts and token medians span all four conditions; condition means use available scored runs. NA = unavailable, not zero. 95% CIs: whole-run bootstrap; Wilson for focal recovery. NA in a CI column means insufficient eligible runs or evidence. Exact model identifiers are in the appendix. Full-precision values and per-condition counts: results.csv and condition\_metrics.csv.

# Summary and separate focal interaction

Persistent + Biomni: first 8 rows | Other workflows below the divider | 95% confidence intervals

| LLM           | Harness       | Workflow     | Effort | Attempted runs | Completed runs | Failed runs | Median output tokens / success | Summary S (0-100) | 95% CI       | Interaction I (pp) | 95% CI        |
|---------------|---------------|--------------|--------|----------------|----------------|-------------|--------------------------------|-------------------|--------------|--------------------|---------------|
| Astra         | Codex         | persistent   | medium | 40             | 40             | 0           | 85,653                         | 84.4              | 83.7 to 84.9 | +0.0               | -39.2 to 39.2 |
| Sol           | Codex         | persistent   | medium | 40             | 40             | 0           | 172,152                        | 80.9              | 77.9 to 82.8 | +10.0              | -30.1 to 51.2 |
| Terra         | Codex         | persistent   | medium | 40             | 40             | 0           | 56,128                         | 77.5              | 73.2 to 80.9 | +10.0              | -37.1 to 52.7 |
| Luna          | Codex         | persistent   | medium | 40             | 36             | 4           | 77,248                         | 46.6              | 38.9 to 53.1 | +10.0              | -40.3 to 60.1 |
| Claude Opus 5 | Custom runner | persistent   | medium | 20             | 20             | 0           | 321,510                        | 79.8              | 78.5 to 81.1 | +0.0               | -61.4 to 61.4 |
| Qwen 3.8 27B  | Biomni        | Biomni       | xhigh  | 40             | 34             | 6           | 81,632                         | 76.1              | 71.7 to 78.7 | +10.0              | -30.1 to 51.2 |
| Qwen 3.8 27B  | Custom runner | persistent   | medium | 40             | 38             | 2           | 183,076                        | 72.3              | NA           | +40.0              | -4.2 to 83.8  |
| Gemma 4 31B   | Custom runner | persistent   | medium | 40             | 37             | 3           | 199,780                        | 70.6              | 66.9 to 74.0 | +50.0              | 2.7 to 88.3   |
| Astra         | Codex         | sequential   | medium | 40             | 39             | 1           | 80,126                         | 84.5              | 83.5 to 85.5 | +0.0               | -39.2 to 39.2 |
| Astra         | Codex         | deliberative | medium | 40             | 40             | 0           | 255,276                        | 85.2              | 84.9 to 85.5 | +0.0               | -39.2 to 39.2 |
| Sol           | Codex         | sequential   | medium | 40             | 40             | 0           | 176,868                        | 78.2              | 73.2 to 81.9 | +10.0              | -33.6 to 52.4 |
| Sol           | Codex         | deliberative | medium | 40             | 38             | 2           | 547,346                        | 71.2              | 64.4 to 77.0 | +40.0              | -10.6 to 78.9 |
| Terra         | Codex         | sequential   | medium | 40             | 39             | 1           | 58,512                         | 78.9              | 75.5 to 81.7 | +30.0              | -13.7 to 71.1 |
| Terra         | Codex         | deliberative | medium | 40             | 39             | 1           | 160,931                        | 70.7              | 65.6 to 75.3 | +80.0              | 27.8 to 114.6 |
| Luna          | Codex         | sequential   | medium | 40             | 39             | 1           | 79,146                         | 38.9              | 33.9 to 43.8 | -10.0              | -53.2 to 39.2 |
| Luna          | Codex         | deliberative | medium | 40             | 38             | 2           | 237,172                        | 39.8              | 34.4 to 45.2 | +0.0               | -44.3 to 47.5 |
| Claude Opus 5 | Custom runner | sequential   | medium | 20             | 19             | 1           | 309,733                        | 79.5              | 78.3 to 80.8 | +0.0               | -61.4 to 61.4 |
| Claude Opus 5 | Custom runner | deliberative | medium | 20             | 18             | 2           | 1,222,954                      | 80.1              | 79.0 to 81.1 | +0.0               | -61.4 to 61.4 |
| Qwen 3.8 27B  | Custom runner | sequential   | medium | 40             | 40             | 0           | 265,094                        | 71.4              | NA           | +30.0              | -12.6 to 73.4 |
| Qwen 3.8 27B  | Custom runner | deliberative | medium | 40             | 40             | 0           | 822,696                        | 68.3              | NA           | +40.0              | -7.6 to 84.1  |
| Gemma 4 31B   | Custom runner | sequential   | medium | 40             | 40             | 0           | 310,696                        | 72.0              | 67.4 to 75.5 | +20.0              | -25.4 to 61.9 |
| Gemma 4 31B   | Custom runner | deliberative | medium | 40             | 33             | 7           | 810,756                        | 75.4              | 72.5 to 77.6 | +20.0              | -21.8 to 61.6 |

**S**: equal-weight geometric mean of the four weighted condition scores; higher is better. **I**: (focal EU - SU) - (focal EM - SM), in percentage points; positive means a larger surprise penalty with recognizable labels. I is a signed contrast, not a performance score. See page 8 for formulas and limitations.

95% CIs: joint whole-run bootstrap for S; Wilson-based MOVER for I. NA = unavailable, not zero. Missing CIs indicate insufficient eligible runs/evidence. Counts and tokens span all four conditions. Full precision and CI availability reasons: cross\_condition\_scores.csv.

# Estimated run costs

Standard API-equivalent charges and modeled electricity on existing GPUs | USD

**Totals include known usage from all attempted runs, including failures.** API prices include recorded cache discounts; GPU estimates cover active electricity only. Input tokens include cached input. M = million tokens.

| Model / harness               | Cost basis      | Attempted / completed | Input, M | Output, M | Total cost |
|-------------------------------|-----------------|-----------------------|----------|-----------|------------|
| Astra / Codex                 | Standard API    | 120 / 119             | 678.96   | 16.79     | \$7,124.15 |
| Sol / Codex                   | Standard API    | 120 / 118             | 809.86   | 36.33     | \$3,773.27 |
| Terra / Codex                 | Standard API    | 120 / 118             | 505.53   | 11.18     | \$1,078.86 |
| Luna / Codex                  | Standard API    | 120 / 113             | 591.31   | 15.99     | \$137.10   |
| Claude Opus 5 / Custom runner | Standard API    | 60 / 57               | 697.23   | 36.84     | \$4,407.25 |
| Qwen 3.8 27B / Biomni         | GPU electricity | 40 / 34               | 402.35   | 3.77      | \$10.69    |
| Qwen 3.8 27B / Custom runner  | GPU electricity | 120 / 118             | 574.83   | 54.27     | \$23.42    |
| Gemma 4 31B / Custom runner   | GPU electricity | 120 / 110             | 674.57   | 53.92     | \$25.85    |

**Mean recorded cost per completed run**, averaged across all four dataset conditions. Failed runs are excluded from these means. Dash = workflow not used.

| Model / harness               | Persistent | Sequential | Deliberative | Biomni |
|-------------------------------|------------|------------|--------------|--------|
| Astra / Codex                 | \$57.23    | \$30.93    | \$89.99      | -      |
| Sol / Codex                   | \$23.56    | \$17.88    | \$52.82      | -      |
| Terra / Codex                 | \$10.19    | \$4.53     | \$12.23      | -      |
| Luna / Codex                  | \$1.09     | \$0.63     | \$1.71       | -      |
| Claude Opus 5 / Custom runner | \$46.33    | \$37.62    | \$137.00     | -      |
| Qwen 3.8 27B / Biomni         | -          | -          | -            | \$0.28 |
| Qwen 3.8 27B / Custom runner  | \$0.17     | \$0.10     | \$0.32       | -      |
| Gemma 4 31B / Custom runner   | \$0.18     | \$0.11     | \$0.34       | -      |

Electricity: \$0.15/kWh; 1 GPU at 600 W; 1000 input and 150 output tokens/s. These electricity costs are not full serving costs. Calculation and limitations follow.

# How costs are estimated

Reproducible token accounting | explicit rate and hardware assumptions

**API pricing snapshot: 2026-09-12.** Standard global rates in USD per million tokens, without fast, batch, flex, or regional adjustments. Prices are frozen in presentation\_costs.json; update that file when rates change.

| Model         | Input   | Cached input | Cache write | Output  |
|---------------|---------|--------------|-------------|---------|
| Astra         | \$10.00 | \$1.00       | \$12.50     | \$50.00 |
| Sol           | \$4.00  | \$0.40       | \$5.00      | \$20.00 |
| Terra         | \$2.00  | \$0.20       | \$2.50      | \$12.00 |
| Luna          | \$0.20  | \$0.02       | \$0.25      | \$1.20  |
| Claude Opus 5 | \$5.00  | \$0.50       | \$6.25      | \$25.00 |

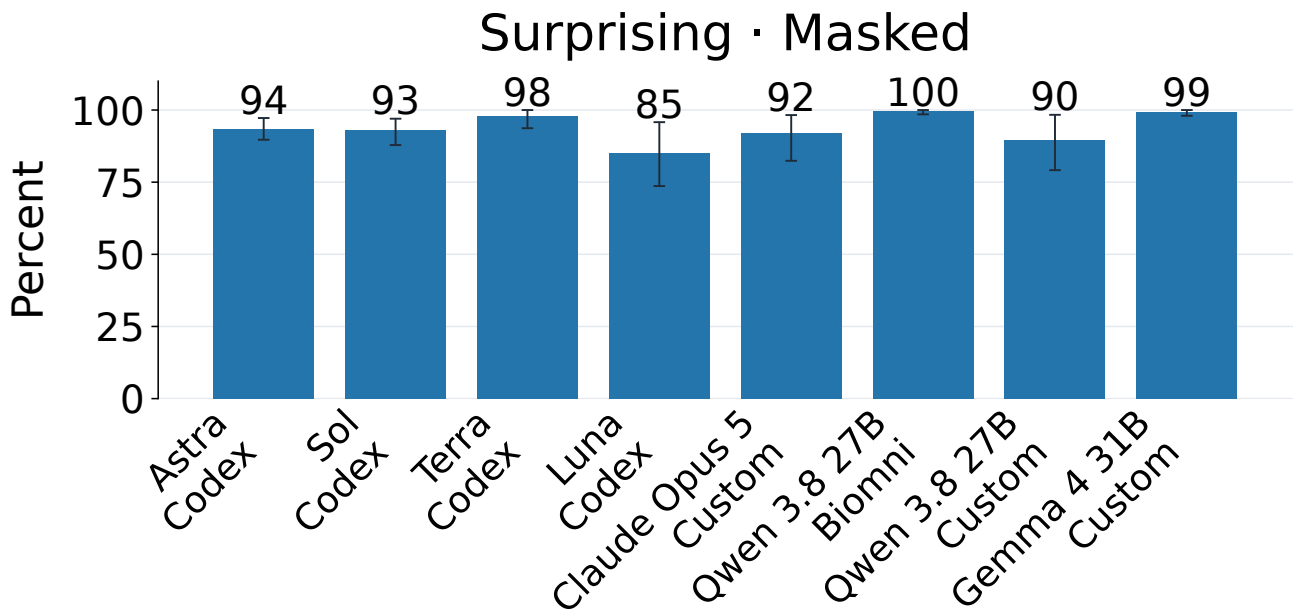
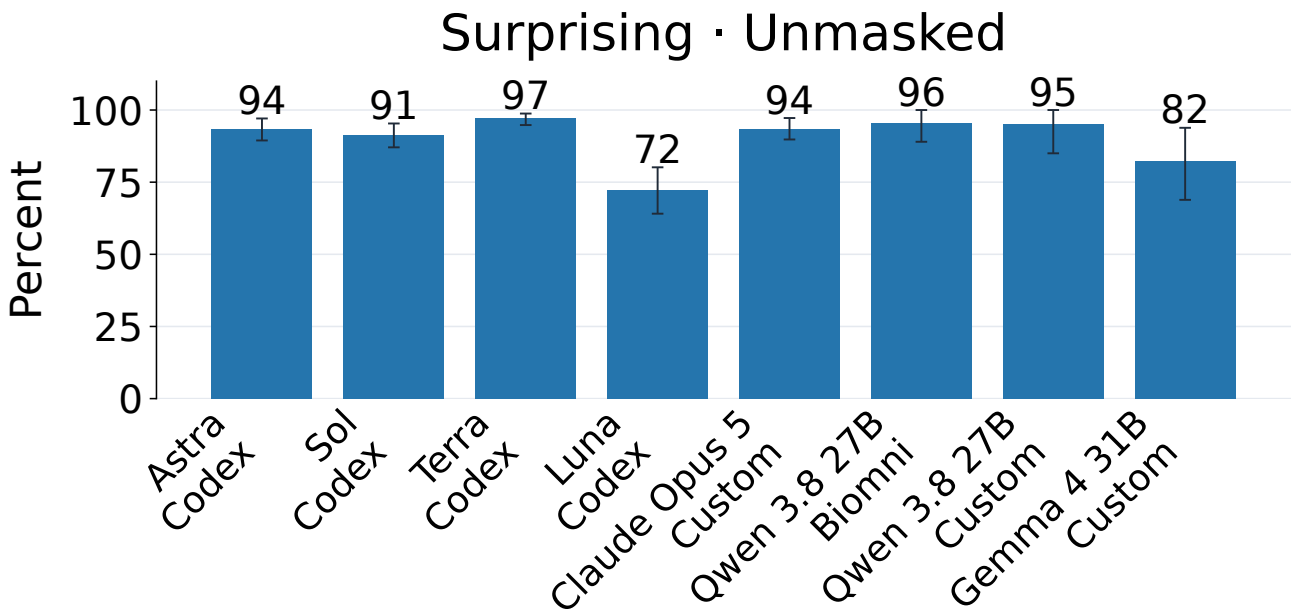
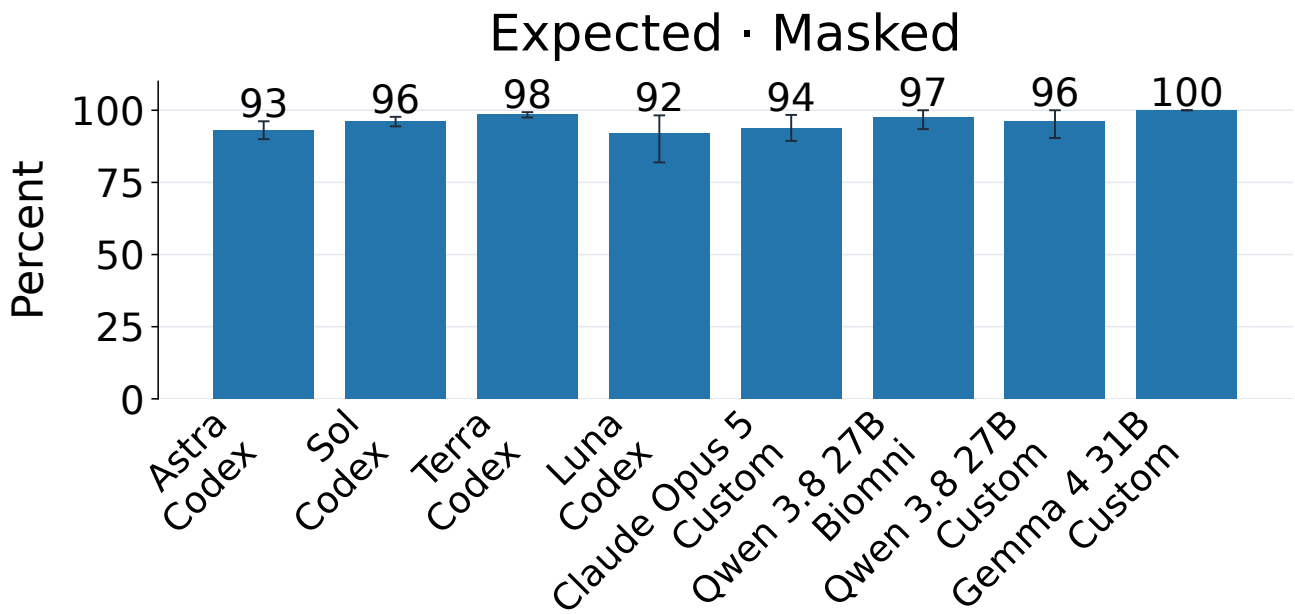
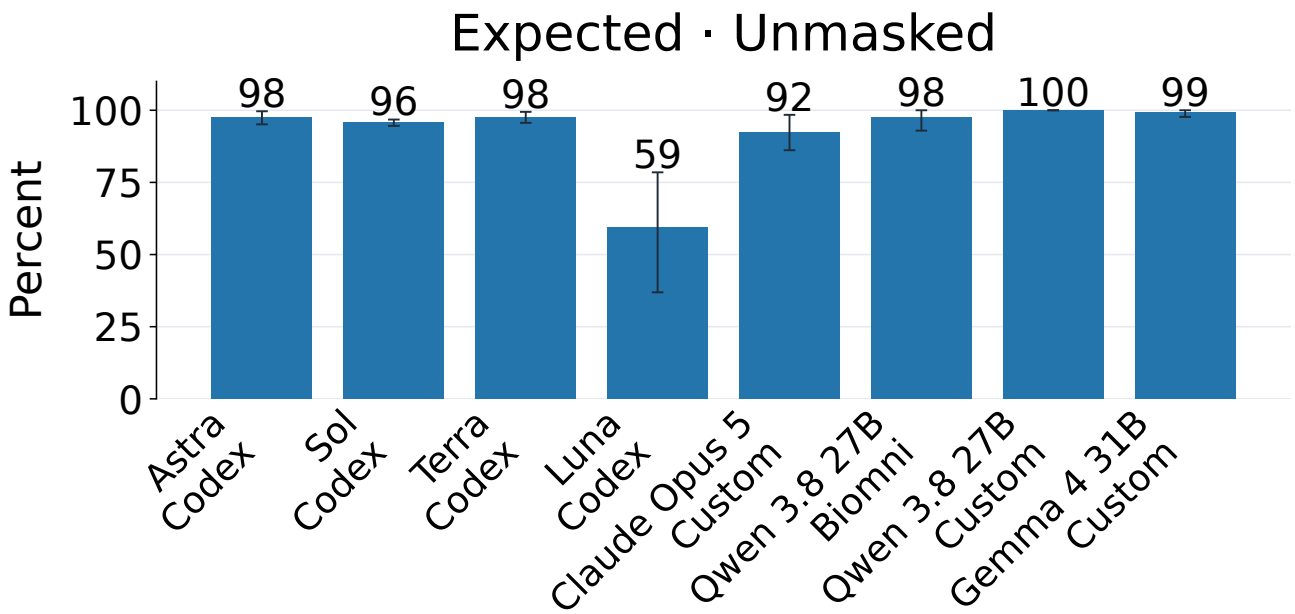
**API formula:** [(input - cached - cache-write tokens) x input rate + cached tokens x cached rate + cache-write tokens x write rate + output tokens x output rate] / 1,000,000. Output already includes reasoning tokens. OpenAI requests above 272,000 input tokens use 2x input/cache rates and 1.5x output rates. Claude's adapter does not request prompt caching; its recorded input is priced as uncached.

**Electricity formula:** hours = (input tokens / 1000 + output tokens / 150) / 3,600; kWh = hours x 0.6 kW x 1 GPU; cost = kWh x \$0.15. Prefill and generation times are added. The same assumed throughput and power apply to Gemma, Qwen custom runner, and Qwen Biomni. This is modeled active energy, not measured wall-clock energy. It excludes idle time, CPU/RAM, cooling, hardware depreciation and other infrastructure; assumes no batching or prompt-cache speedup. Cached-input discounts apply only to API pricing.

**Coverage and reproduction:** OpenAI token buckets come from provider audit files; Claude known usage is reconstructed from call journals. Local-model electricity uses the run inventory's input/output totals; Biomni uses its exported usage. Unknown token usage is not imputed. Missing calls and unlogged retries can make estimates lower bounds, including for completed runs. Costs use the selected run inventory, independent of the scientific-score cohort. Unchanged terminal API usage is cached by the complete run row; use --refresh-costs after changing audit files in place. Configurable assumptions: --cost-assumptions PATH. Outputs: cost\_estimates.json, run\_costs.csv, cost\_totals.csv, cost\_by\_workflow.csv and cost\_by\_condition.csv.

**Rate sources:** [OpenAI API pricing](#); [Anthropic API pricing](#). These are API-equivalent estimates, not invoice totals. An external GCP billing check was approximately \$4,400 for completed Opus runs; this is not used to calibrate the token-based calculation.

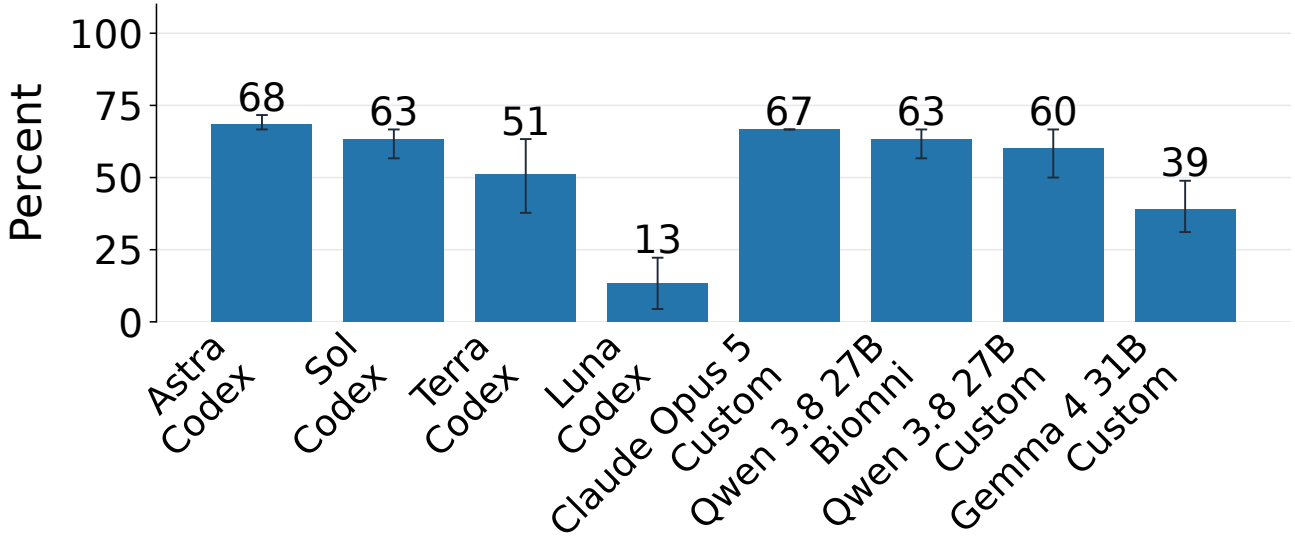
# Precision (%) · Persistent + Biomni



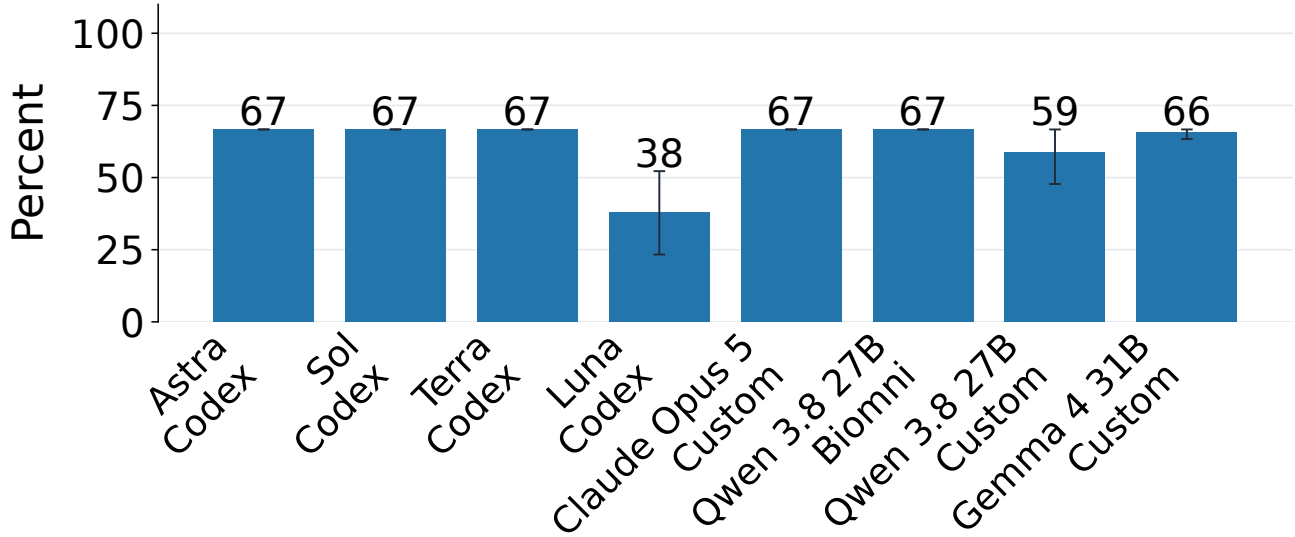
Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium reasoning | Biomni: xhigh reasoning  
95% CIs, fixed dataset: bootstrap; Wilson for focal recovery. NA / missing CIs: insufficient evidence.

# Recall (%) · Persistent + Biomni

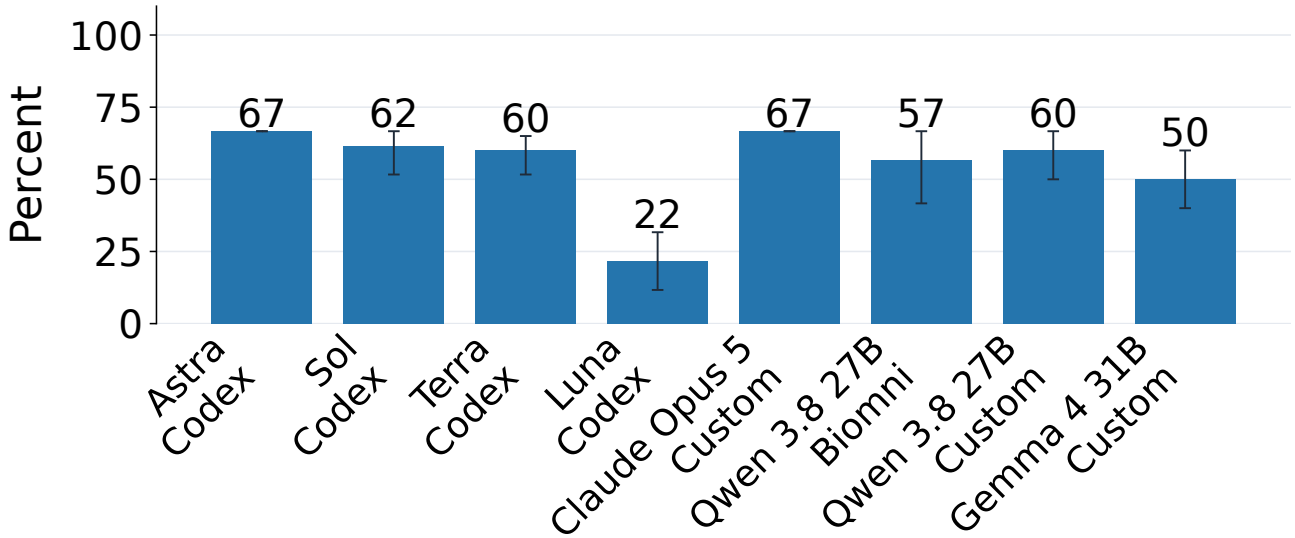
Expected · Unmasked



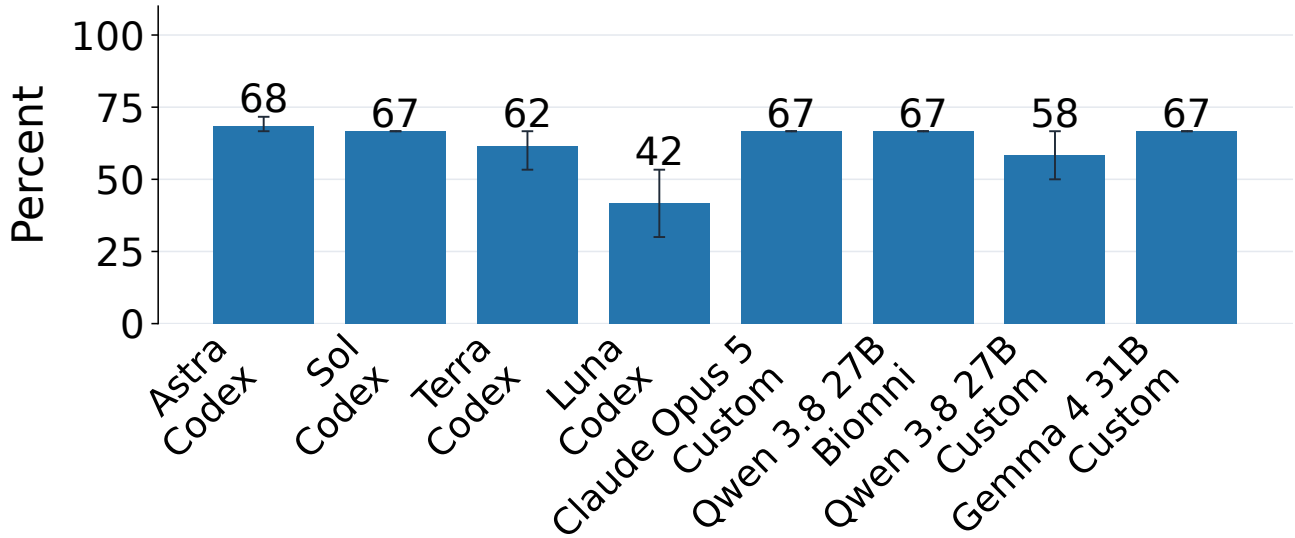
Expected · Masked



Surprising · Unmasked



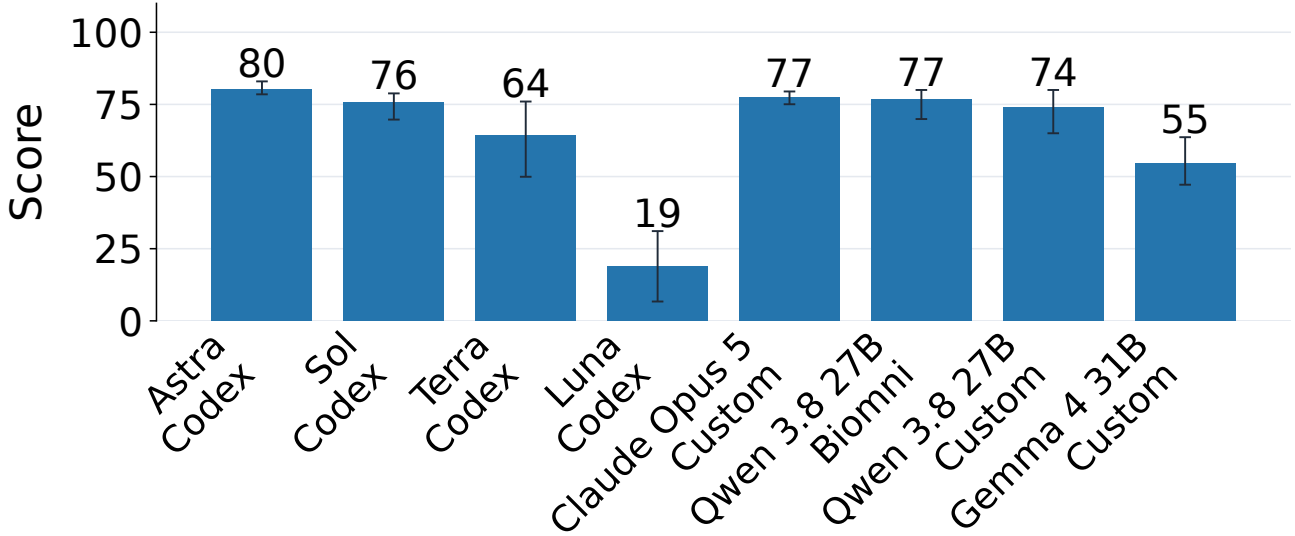
Surprising · Masked



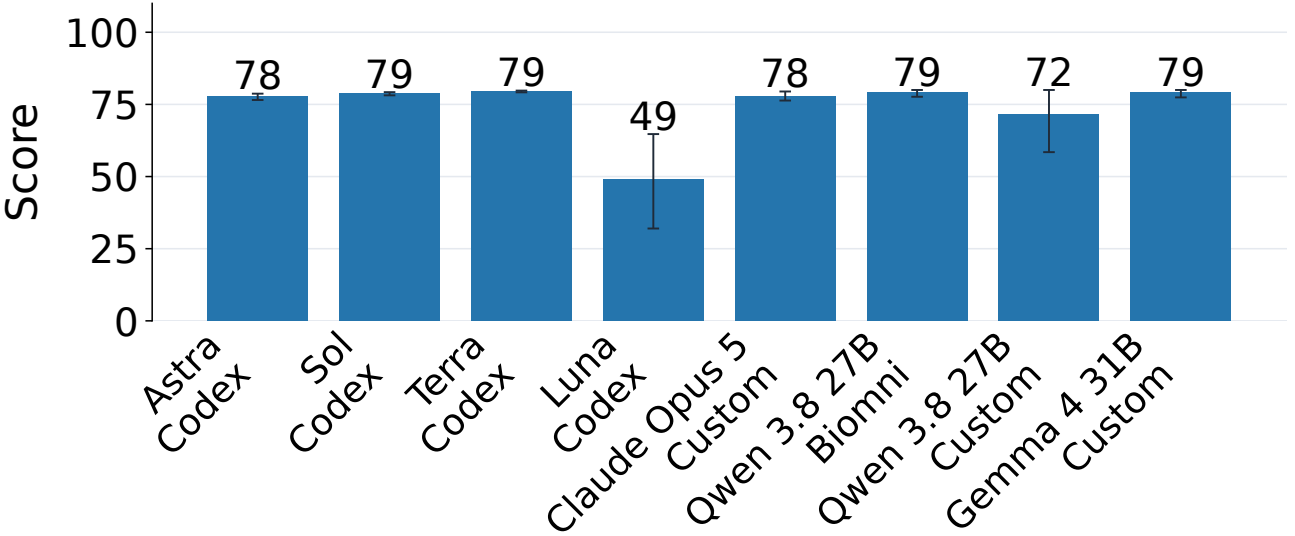
Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium reasoning | Biomni: xhigh reasoning  
95% CIs, fixed dataset: bootstrap; Wilson for focal recovery. NA / missing CIs: insufficient evidence.

# Scientific F1\* · Persistent + Biomni

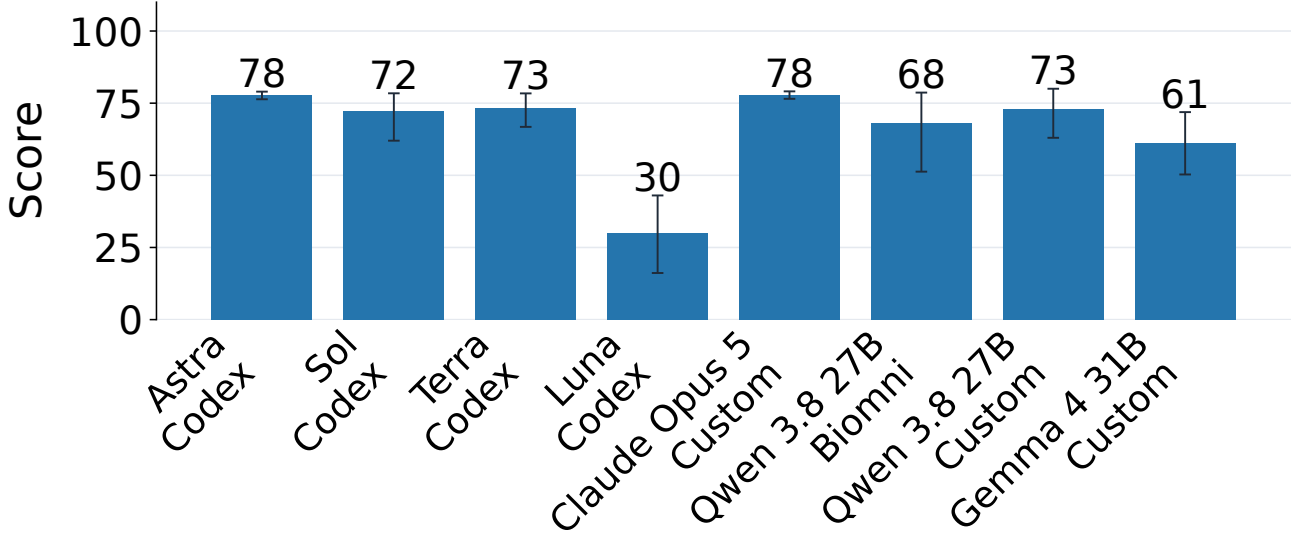
Expected · Unmasked



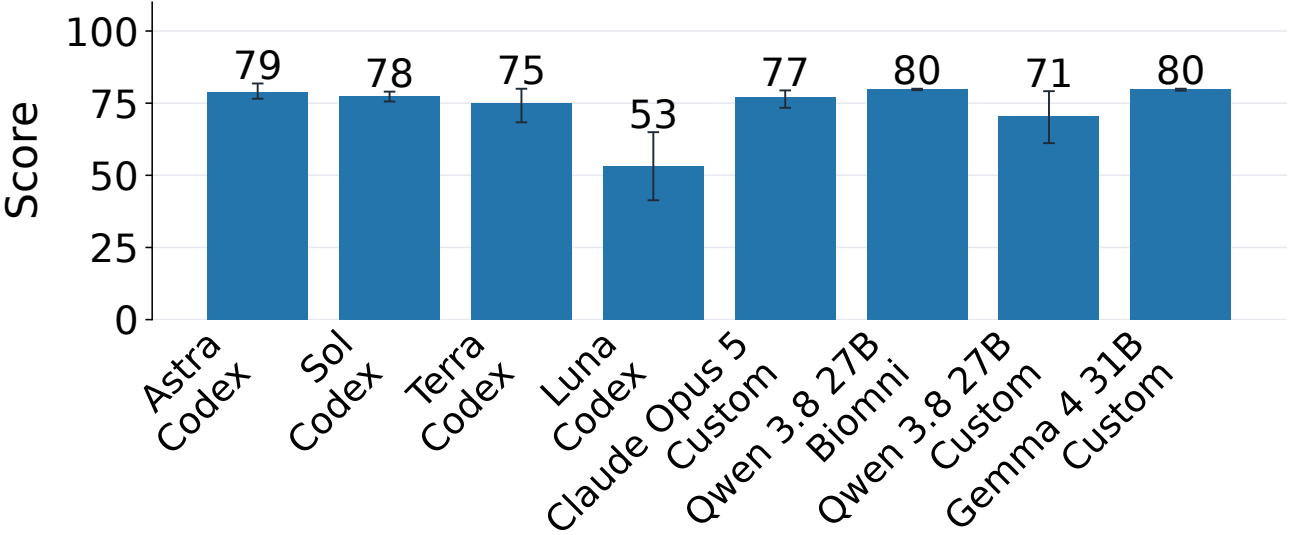
Expected · Masked



Surprising · Unmasked



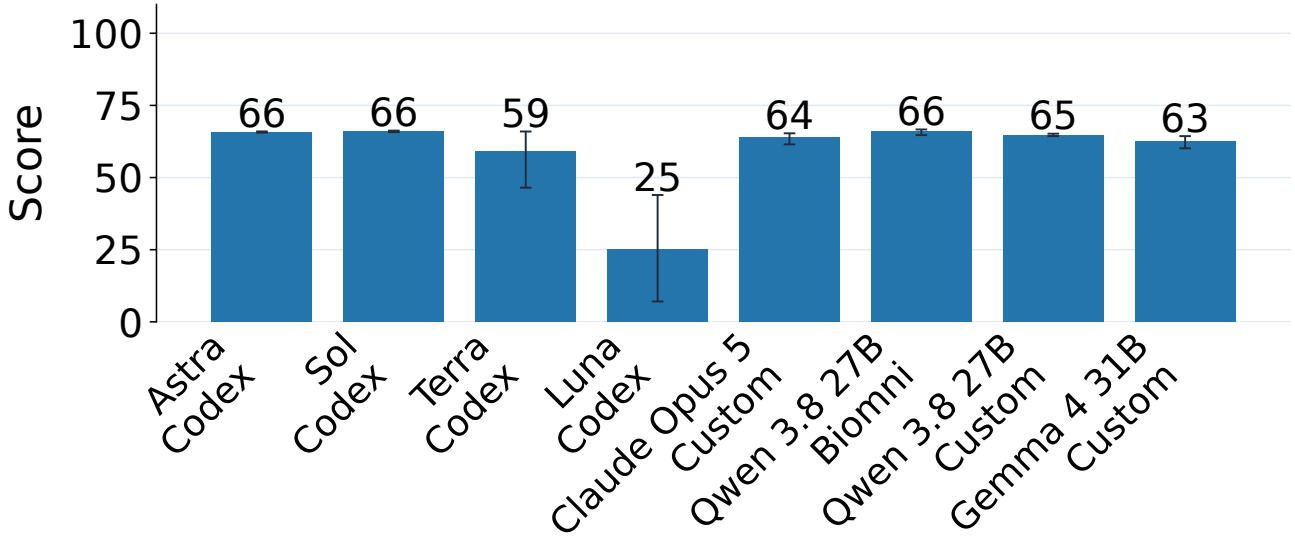
Surprising · Masked



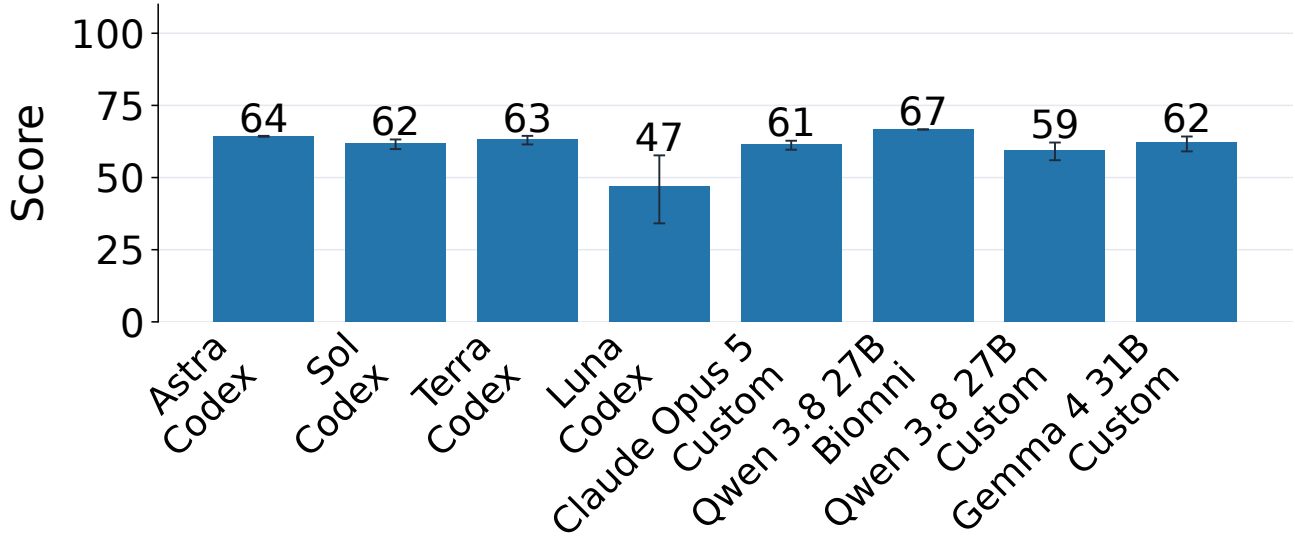
Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium reasoning | Biomni: xhigh reasoning  
95% CIs, fixed dataset: bootstrap; Wilson for focal recovery. NA / missing CIs: insufficient evidence.

# Exploration E · Persistent + Biomni

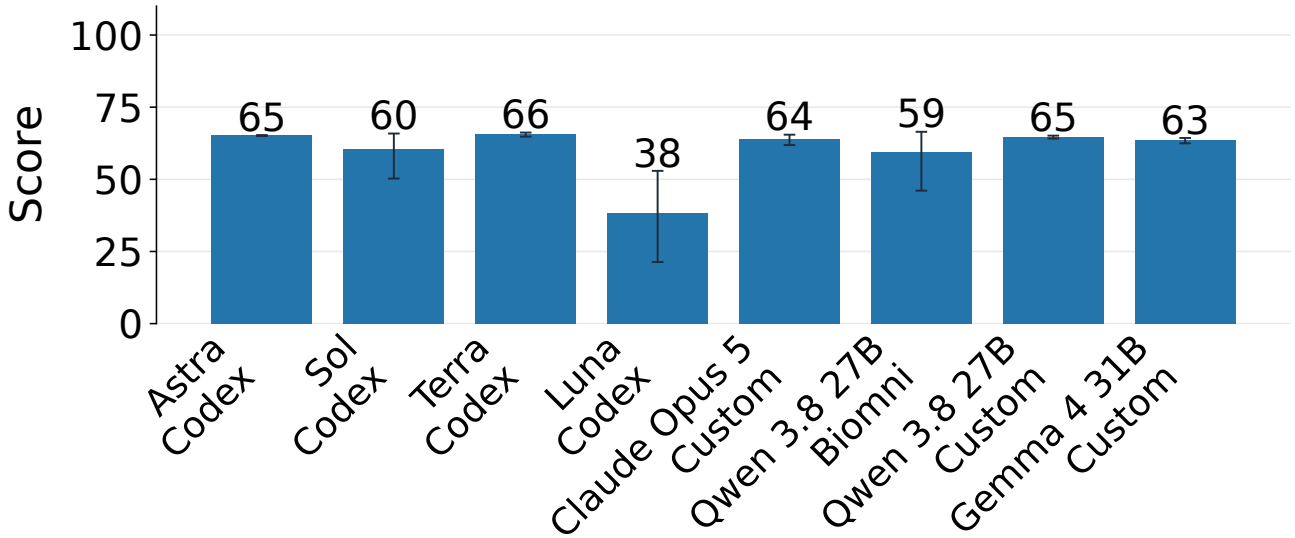
Expected · Unmasked



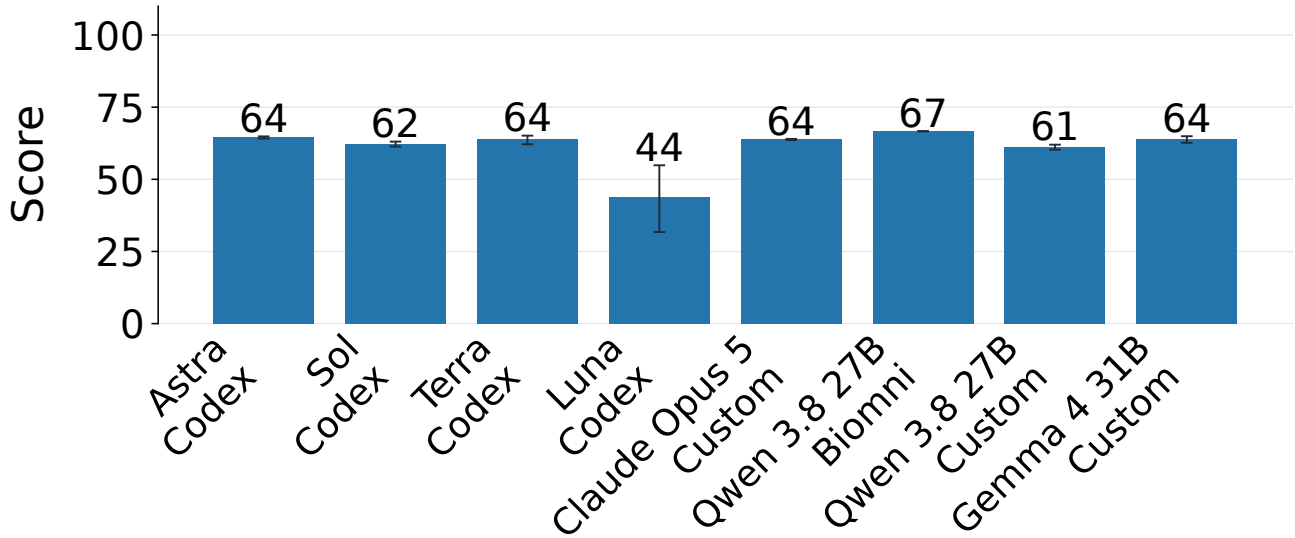
Expected · Masked



Surprising · Unmasked

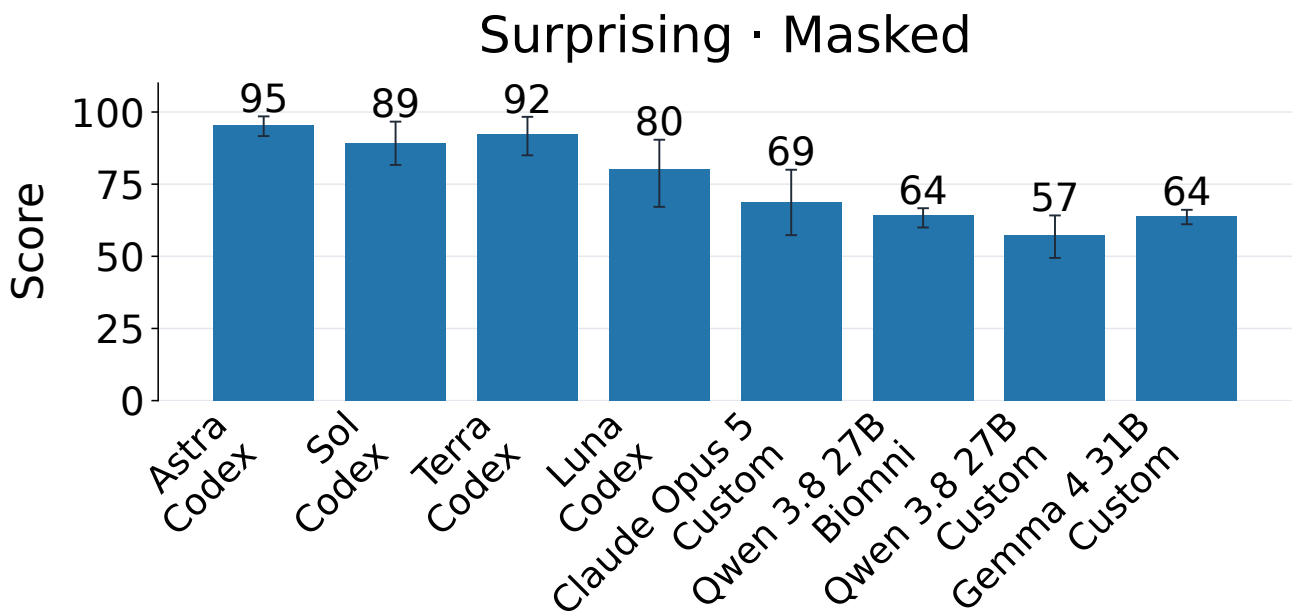
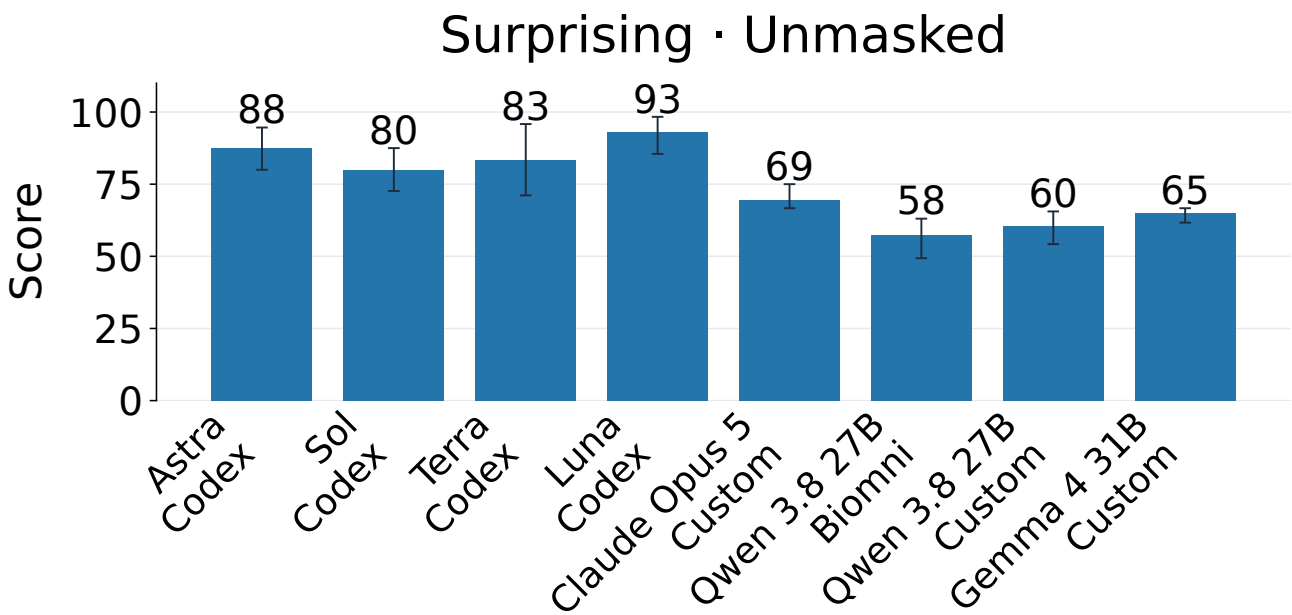
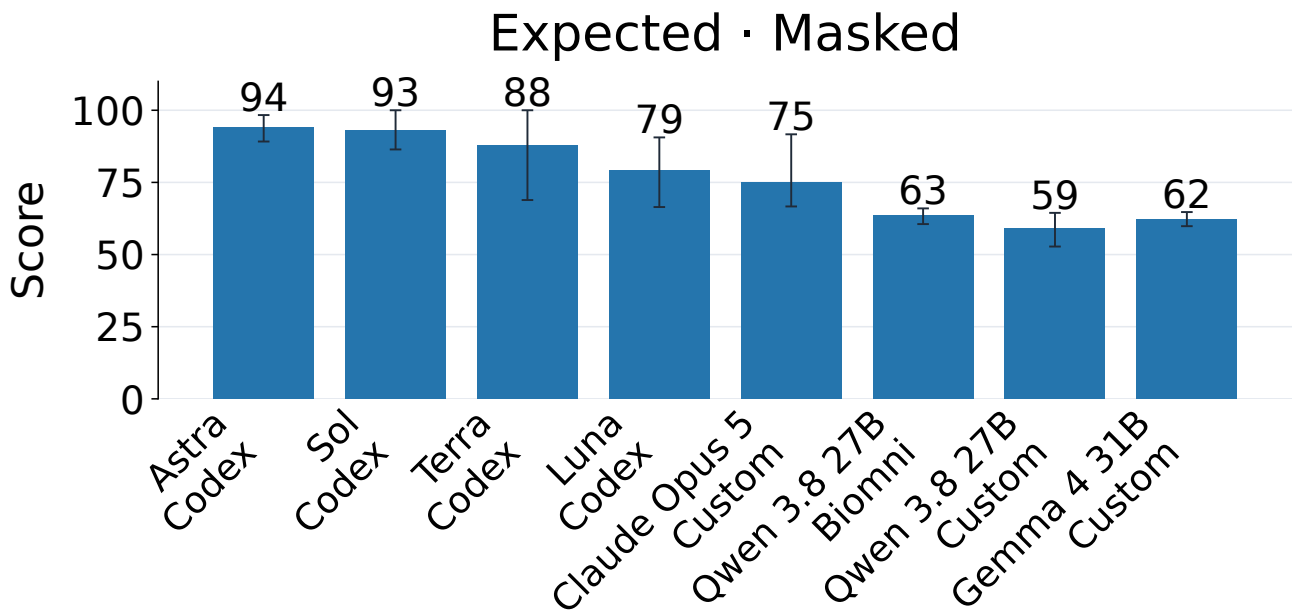
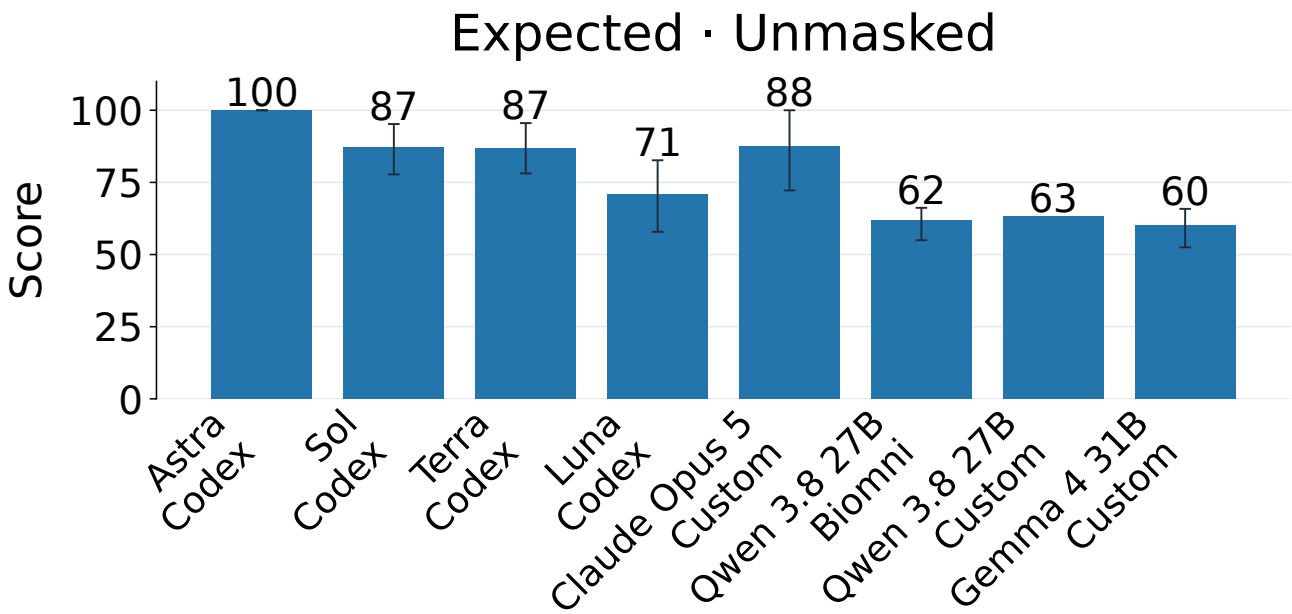


Surprising · Masked



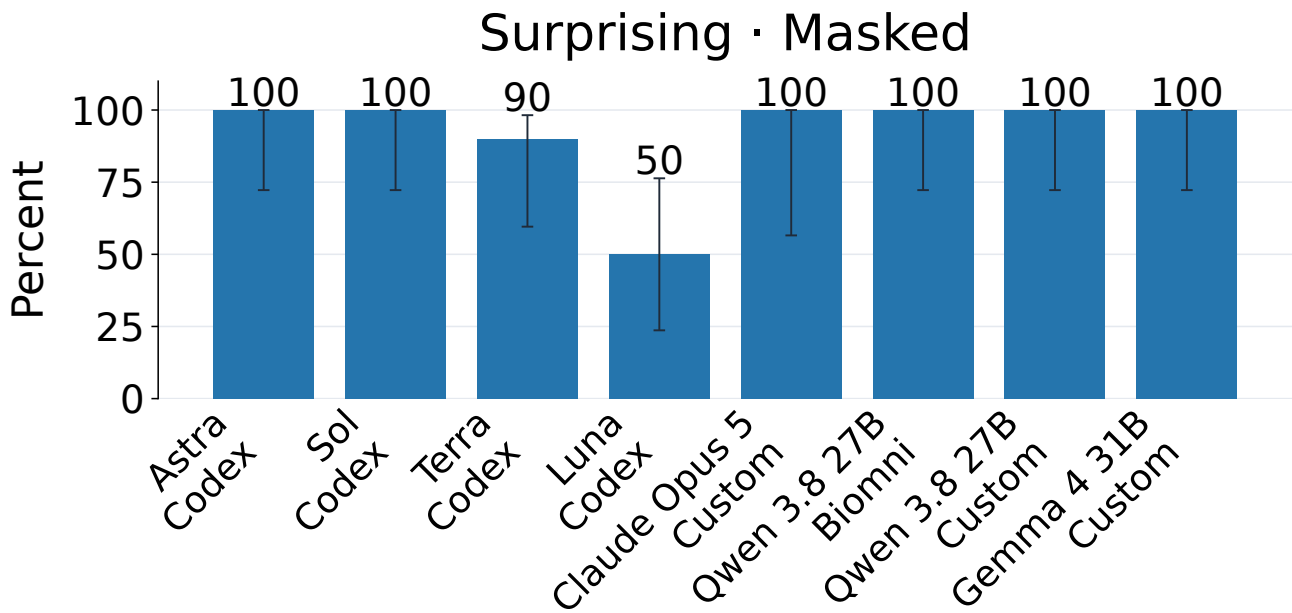
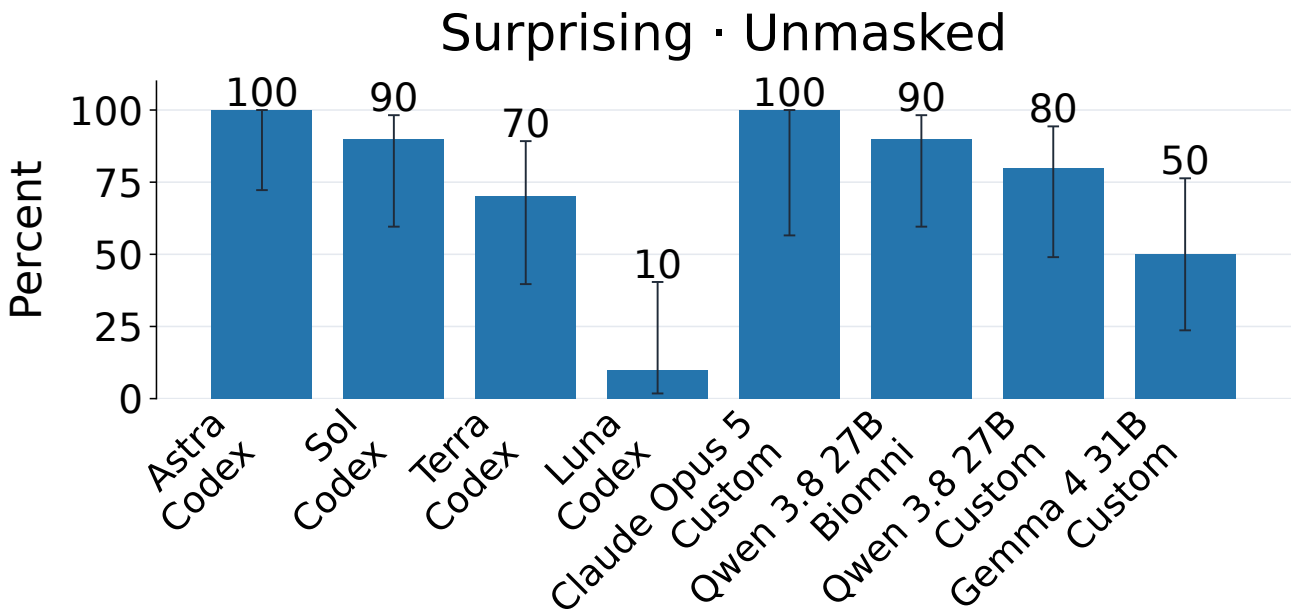
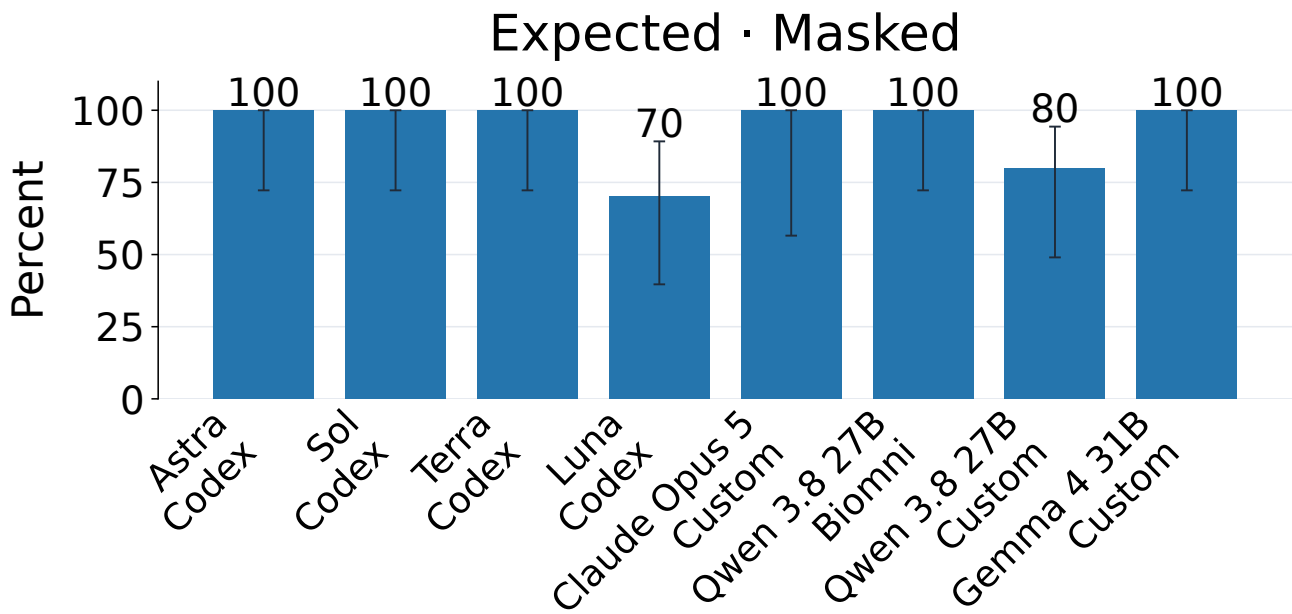
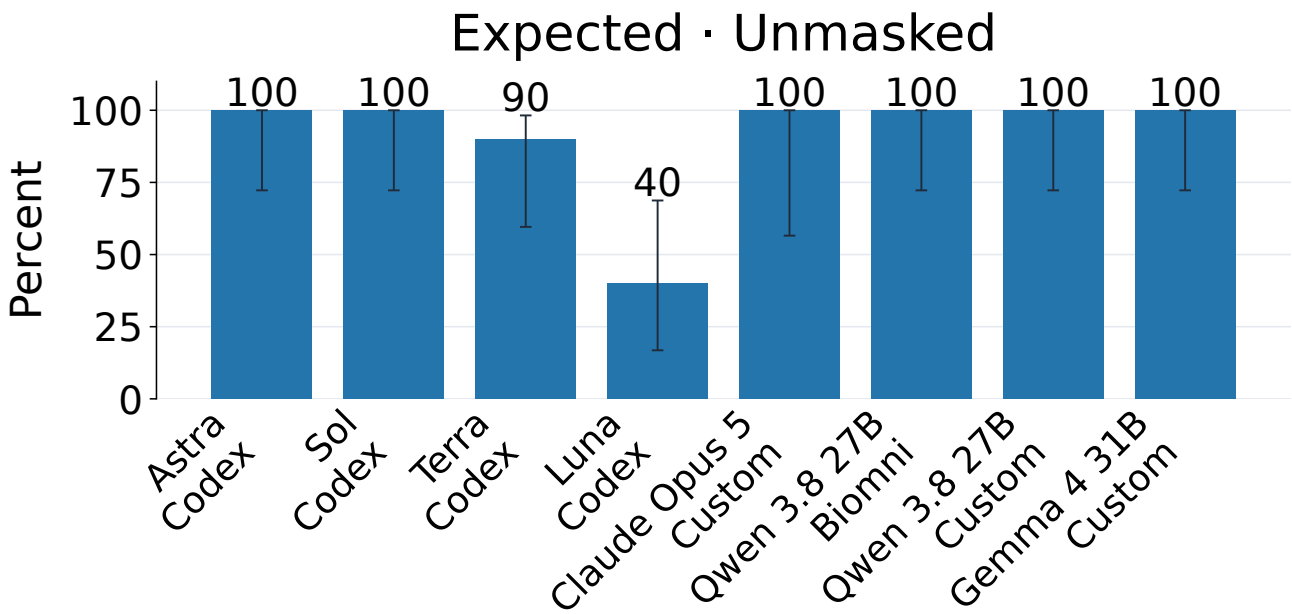
Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium reasoning | Biomni: xhigh reasoning  
95% CIs, fixed dataset: bootstrap; Wilson for focal recovery. NA / missing CIs: insufficient evidence.

# Evidence responsiveness B · Persistent + Biomni



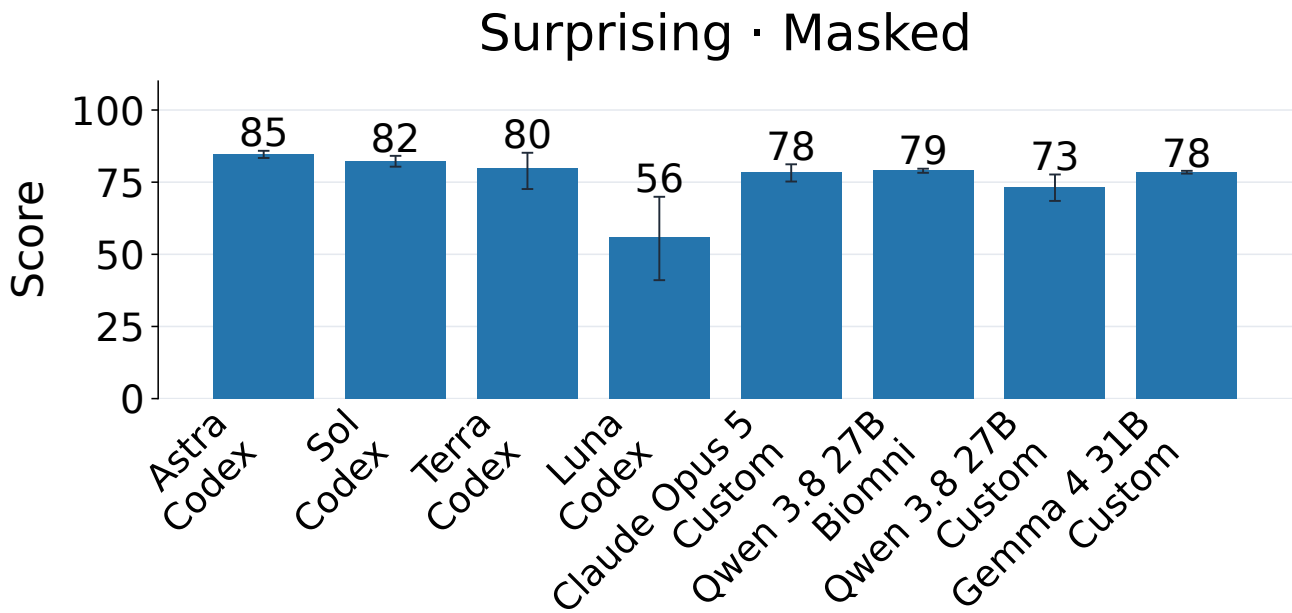
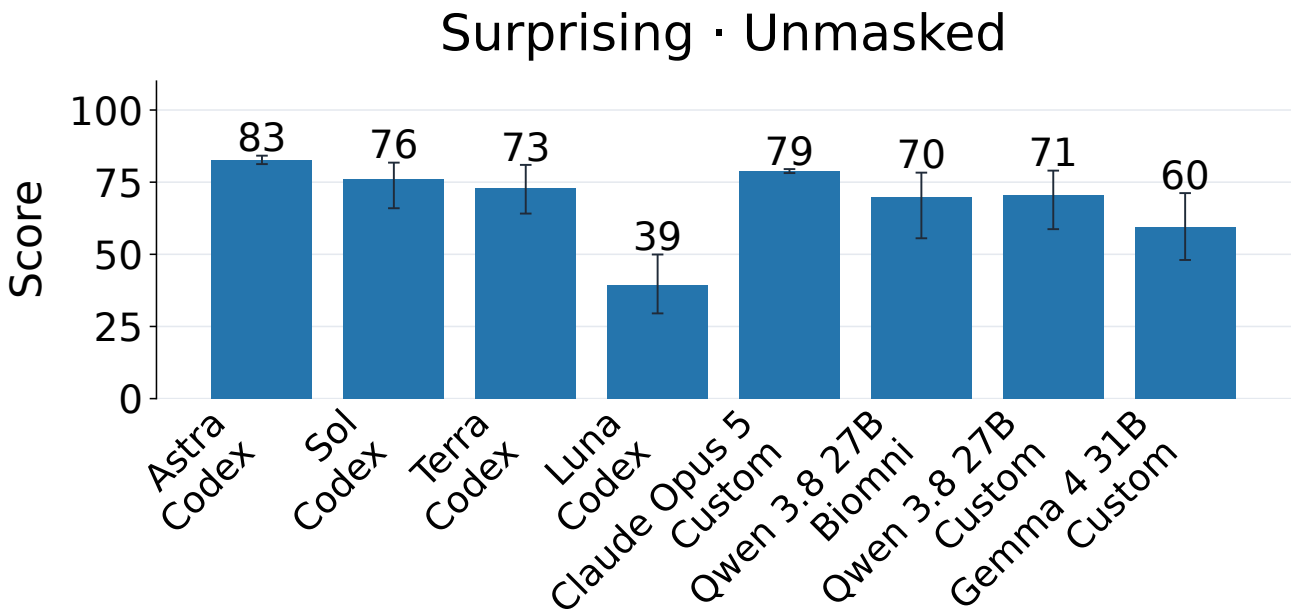
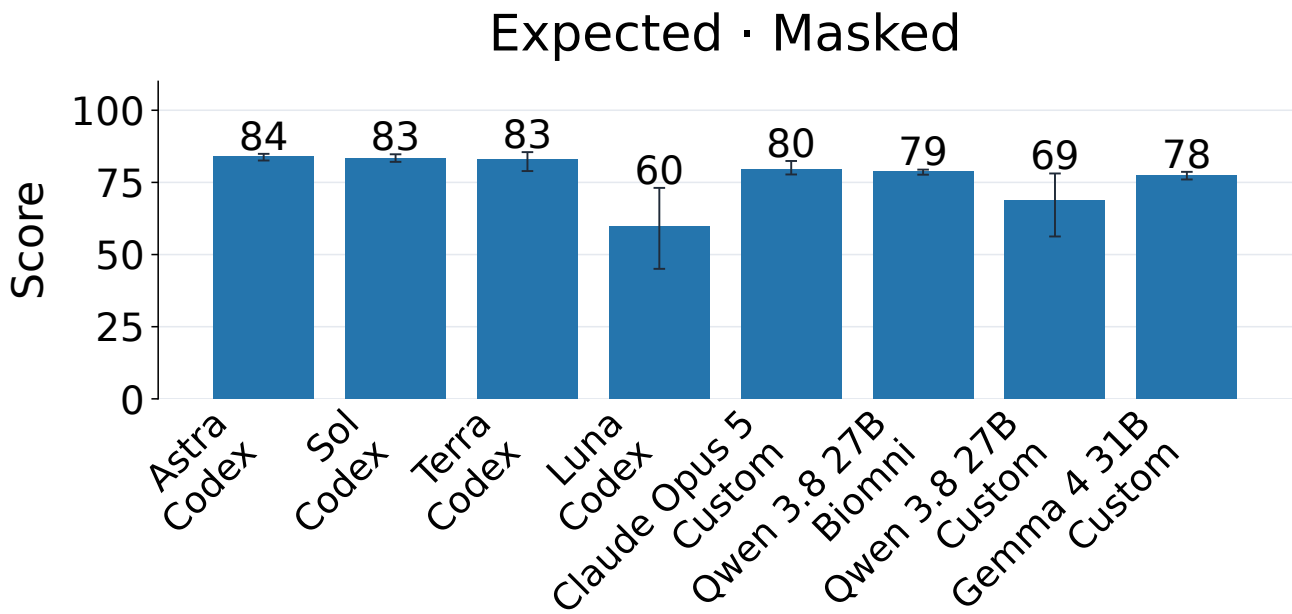
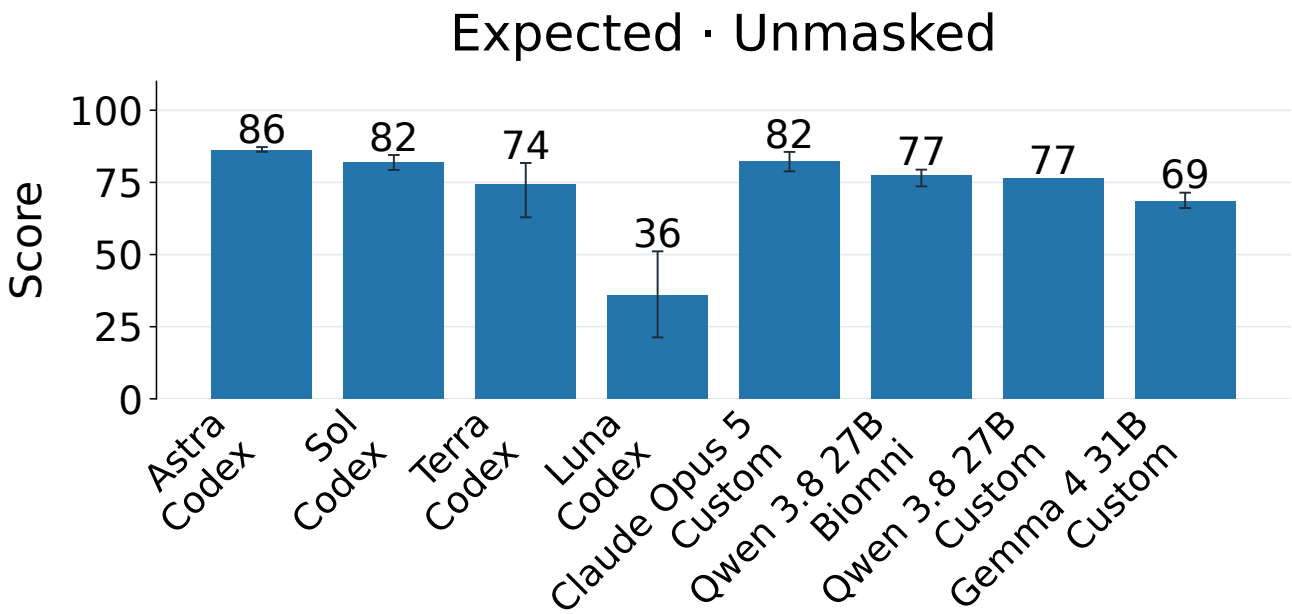
Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium reasoning | Biomni: xhigh reasoning  
95% CIs, fixed dataset: bootstrap; Wilson for focal recovery. NA / missing CIs: insufficient evidence.

# Focal recovery (%) · Persistent + Biomni



Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium reasoning | Biomni: xhigh reasoning  
95% CIs, fixed dataset: bootstrap; Wilson for focal recovery. NA / missing CIs: insufficient evidence.

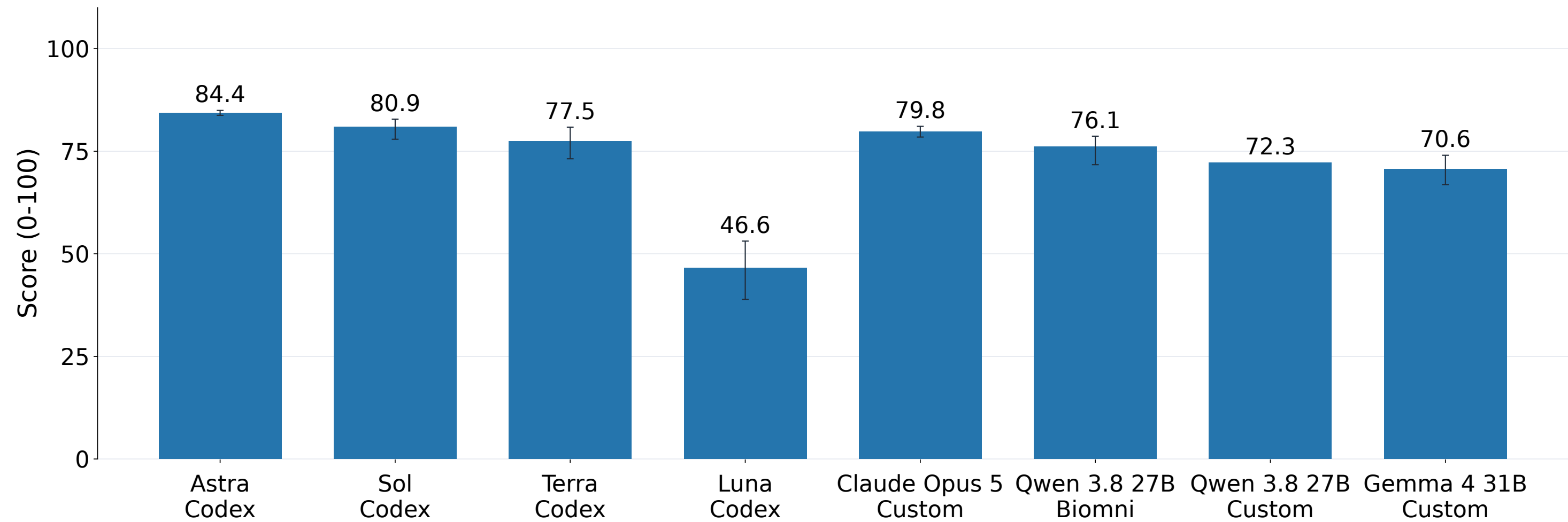
# Weighted condition score C · Persistent + Biomni



Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium reasoning | Biomni: xhigh reasoning  
95% CIs, fixed dataset: bootstrap; Wilson for focal recovery. NA / missing CIs: insufficient evidence.

# Grounded discovery and adaptation S - Persistent + Biomni

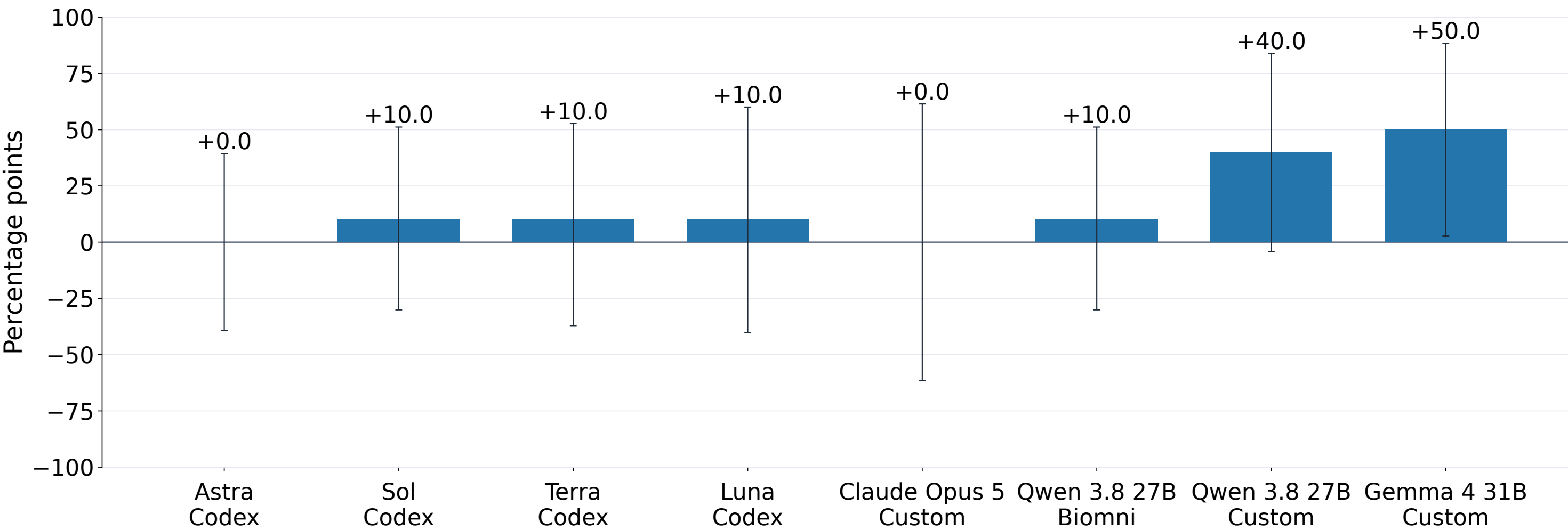
Equal-weight geometric mean across four conditions; higher is better (0-100).



Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium | Biomni: xhigh | Fixed dataset pair  
95% CIs: run bootstrap (S); Wilson-based MOVER (I). NA / missing CIs: insufficient data.

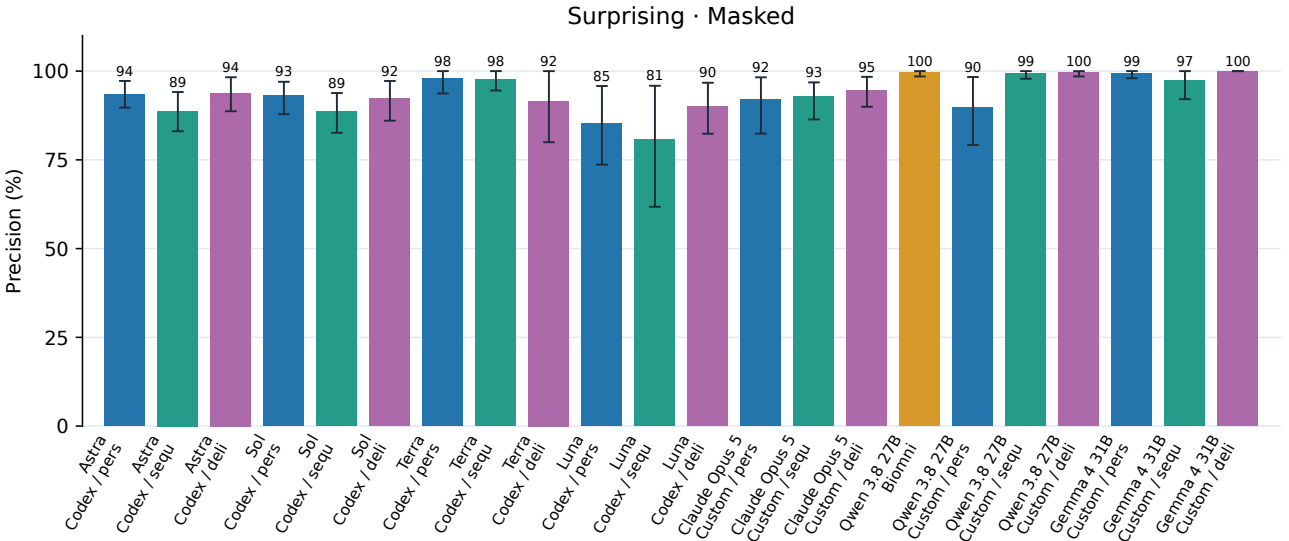
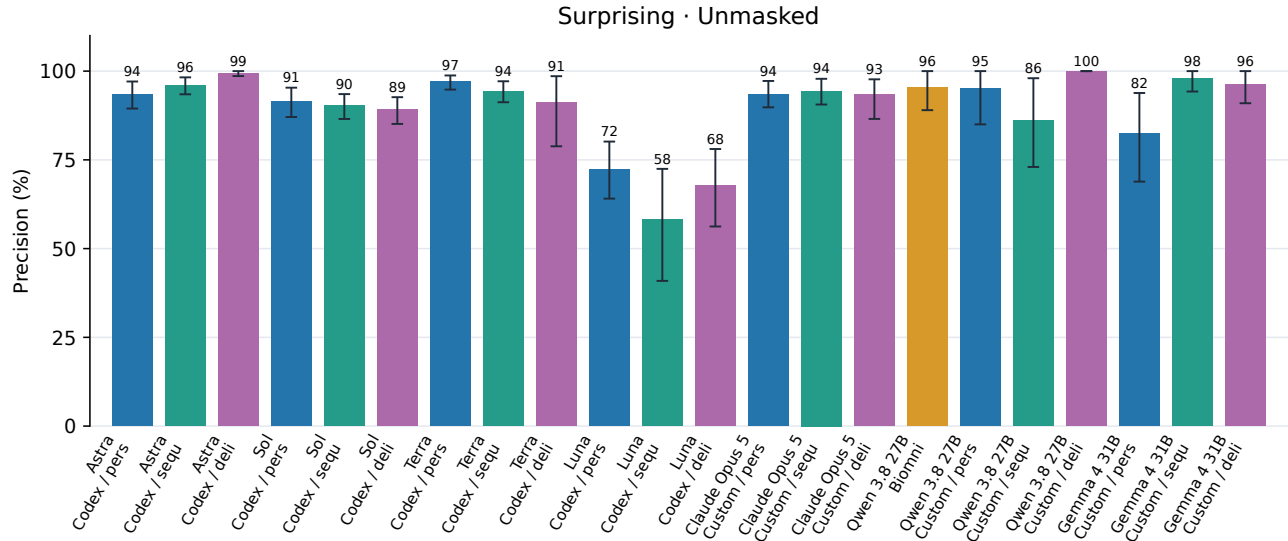
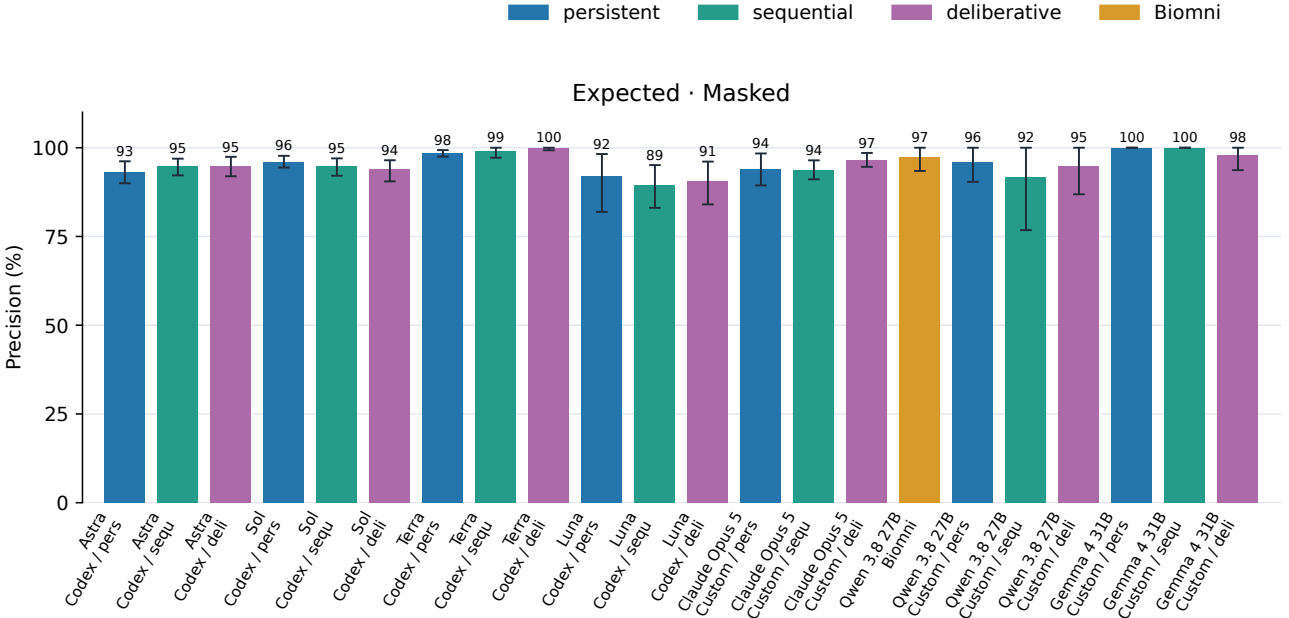
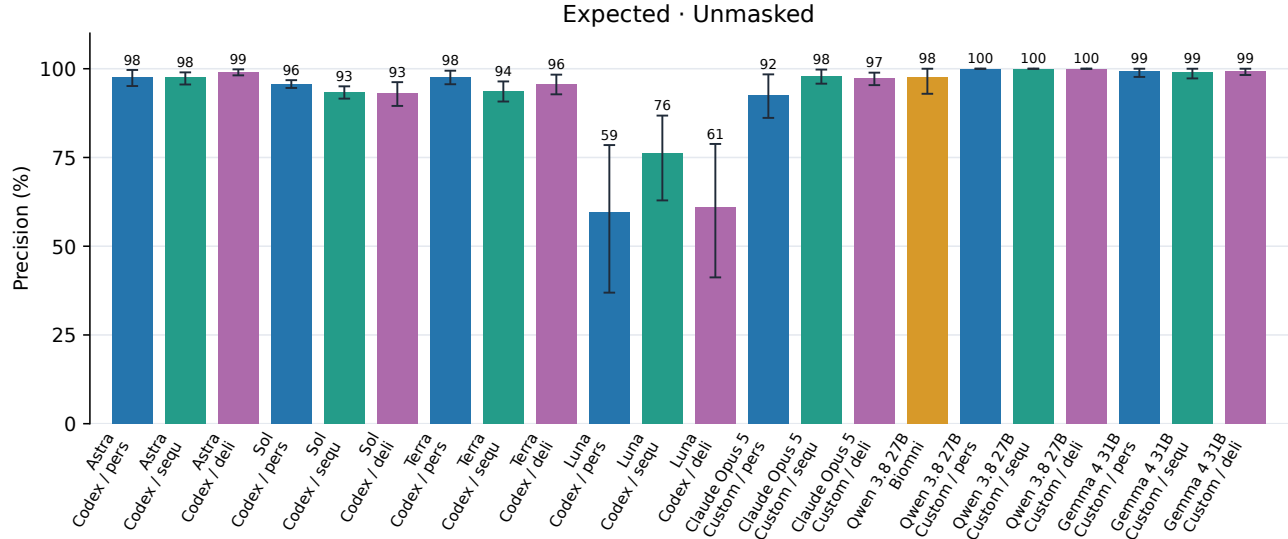
# Label-by-surprise interaction I (pp) - Persistent + Biomni

$I = (\text{Focal EU} - \text{Focal SU}) - (\text{Focal EM} - \text{Focal SM})$   
Positive = larger surprise penalty with recognizable labels.



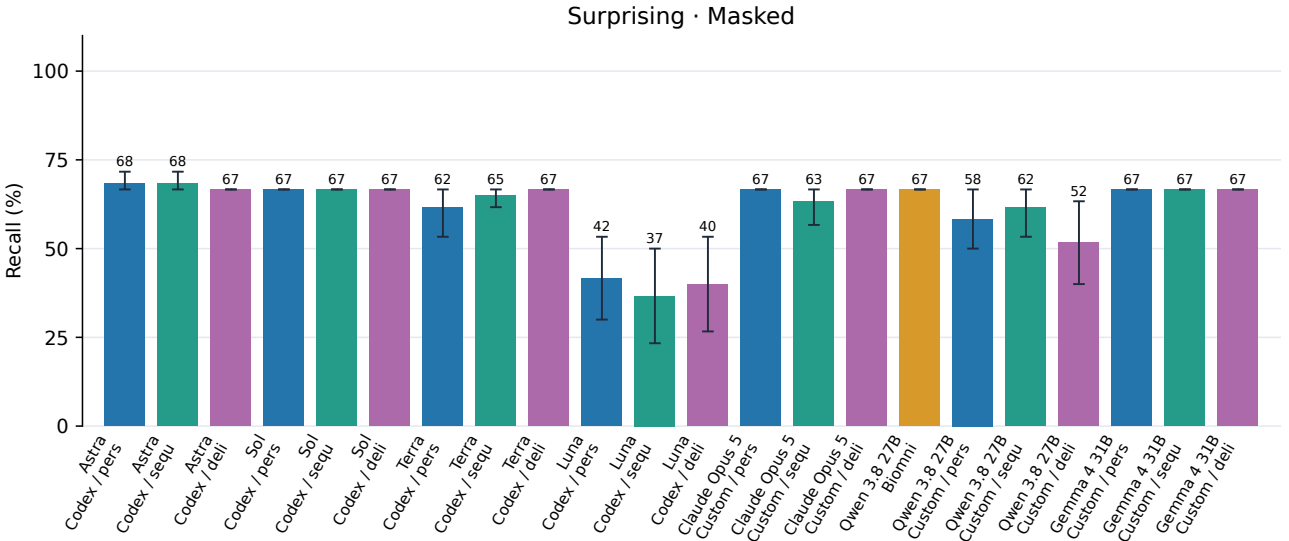
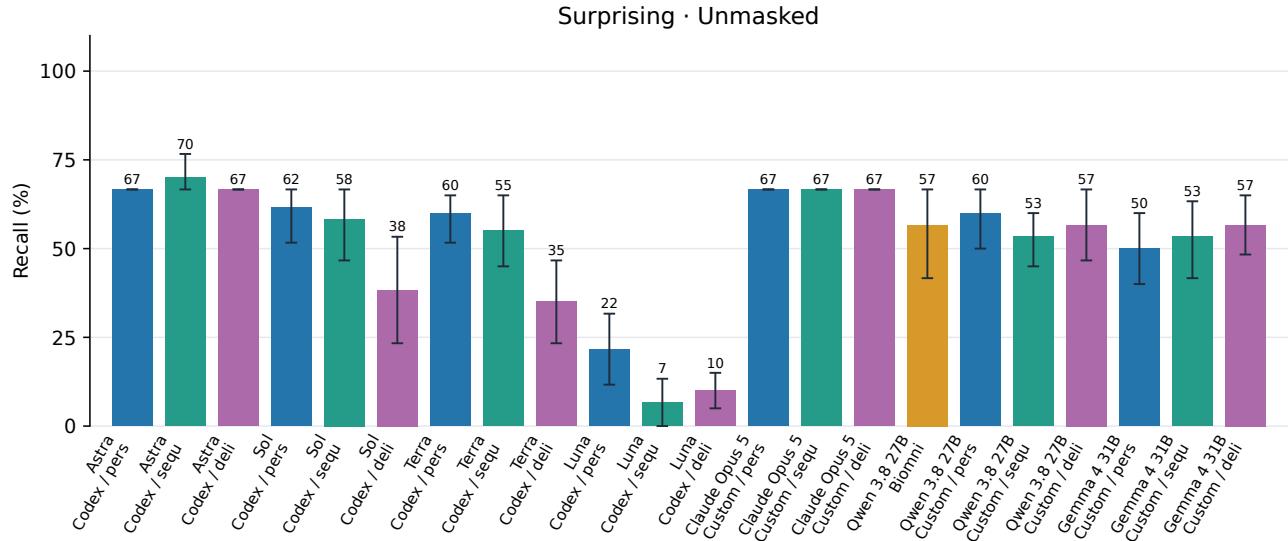
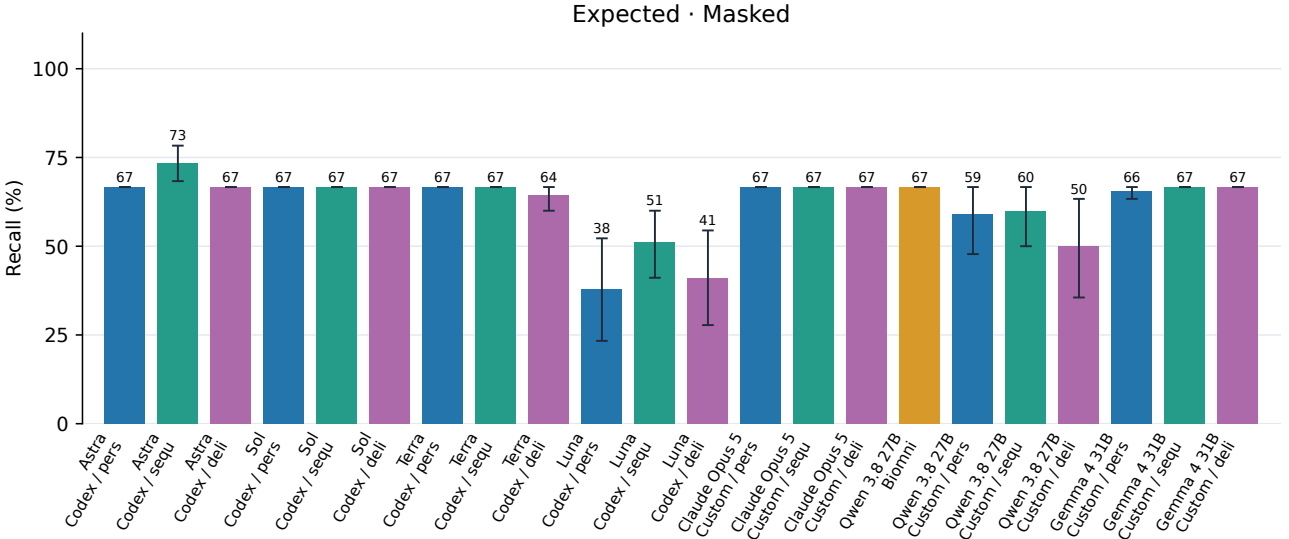
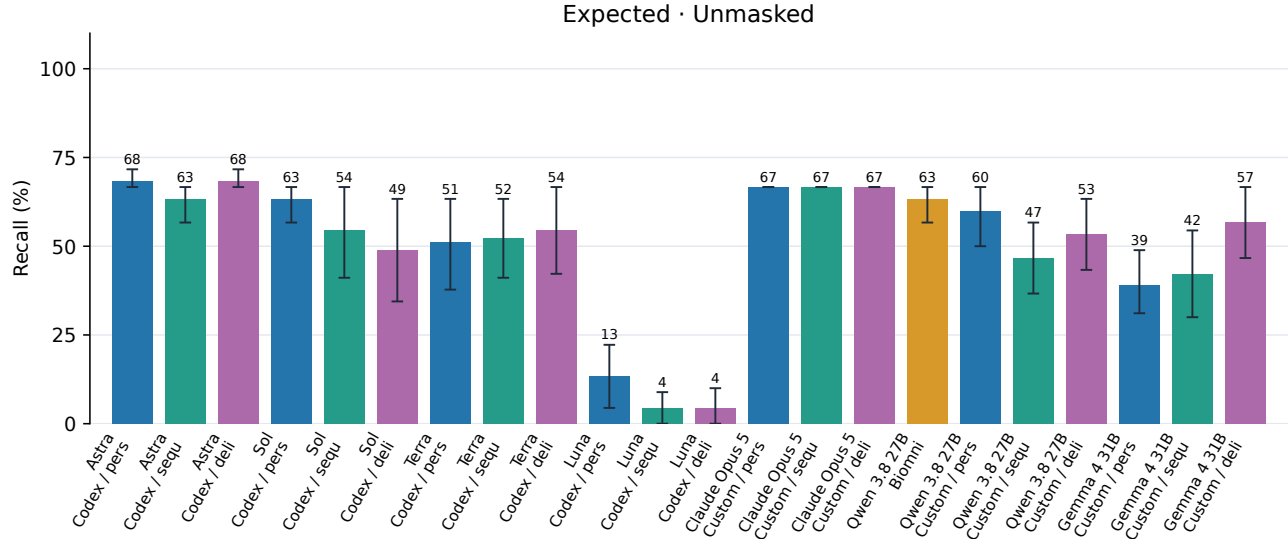
Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors  
Persistent: medium | Biomni: xhigh | Fixed dataset pair  
95% CIs: run bootstrap (S); Wilson-based MOVER (I). NA / missing CIs: insufficient data.

Precision (%)



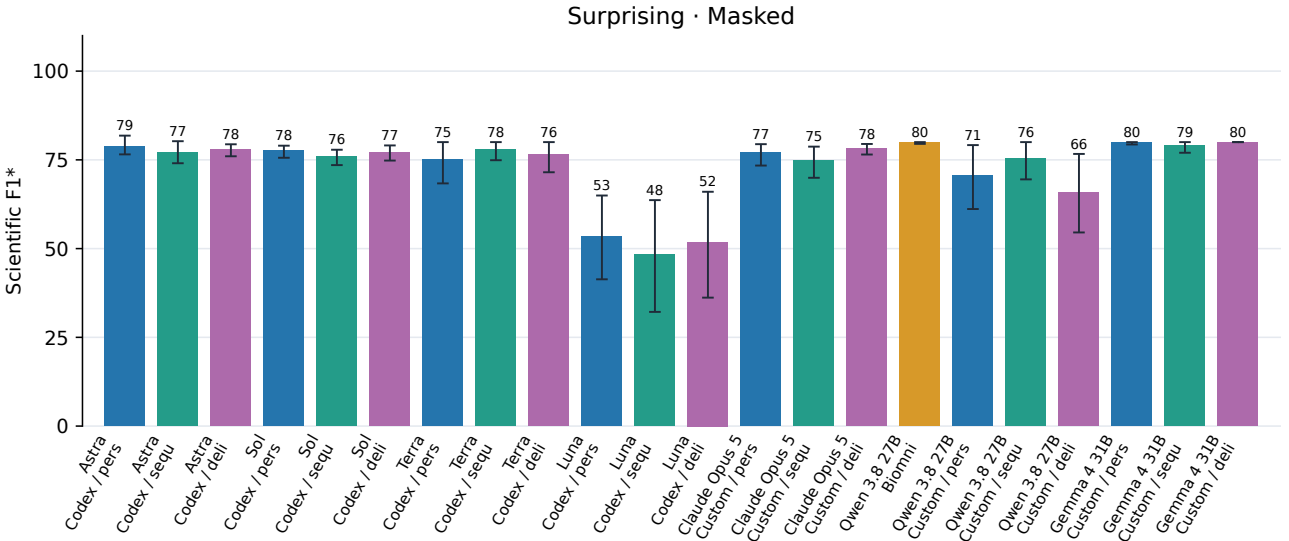
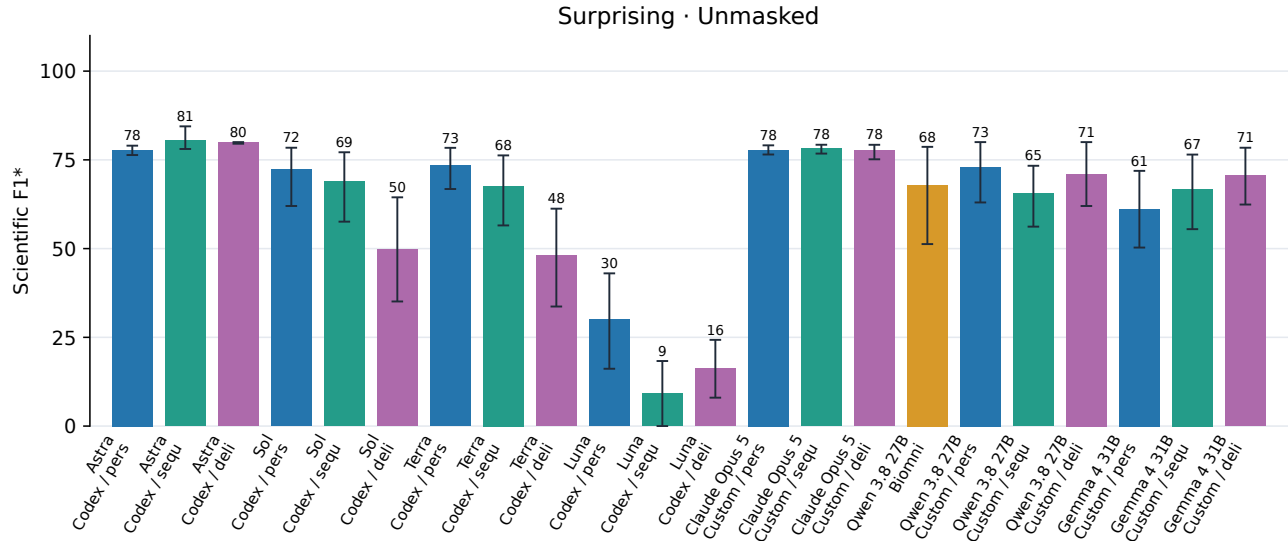
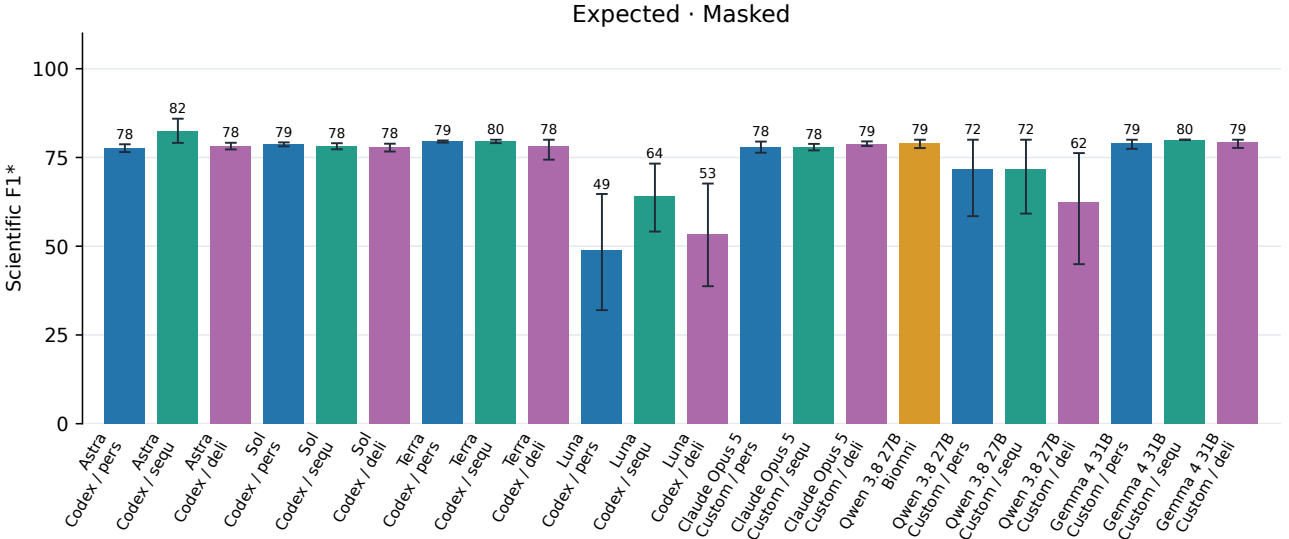
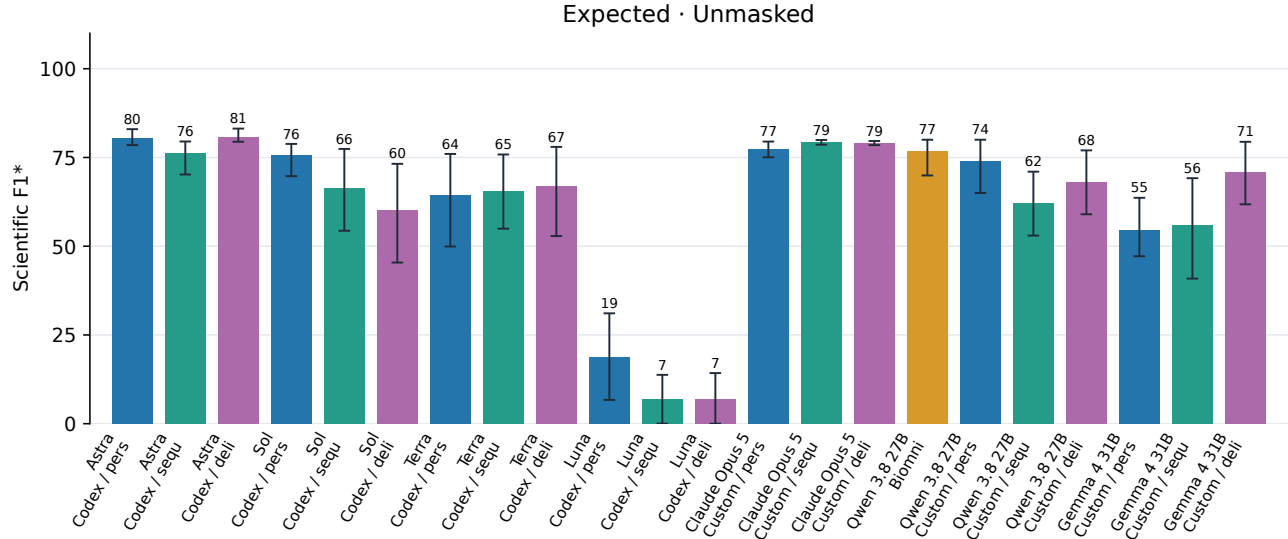
Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · 95% run-level CIs on a fixed dataset pair · NA = unavailable Bootstrap CIs; Wilson for focal recovery. Missing CIs: insufficient runs/evidence. Native Biomni uses xhigh reasoning; other runs use medium.

Recall (%)



Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · 95% run-level CIs on a fixed dataset pair · NA = unavailable Bootstrap CIs; Wilson for focal recovery. Missing CIs: insufficient runs/evidence. Native Biomni uses xhigh reasoning; other runs use medium.

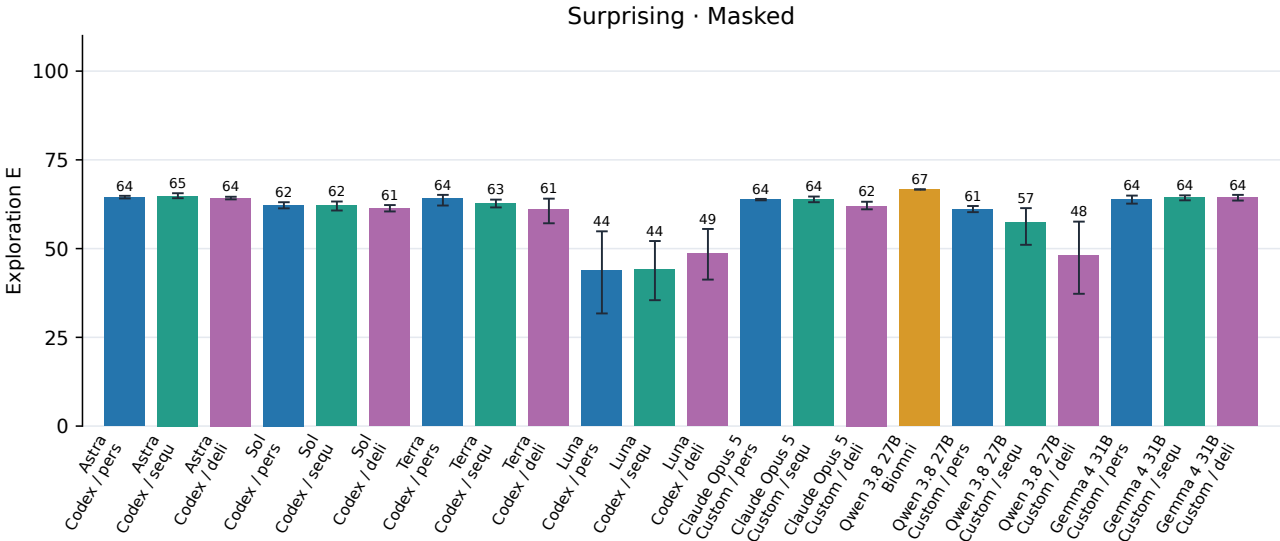
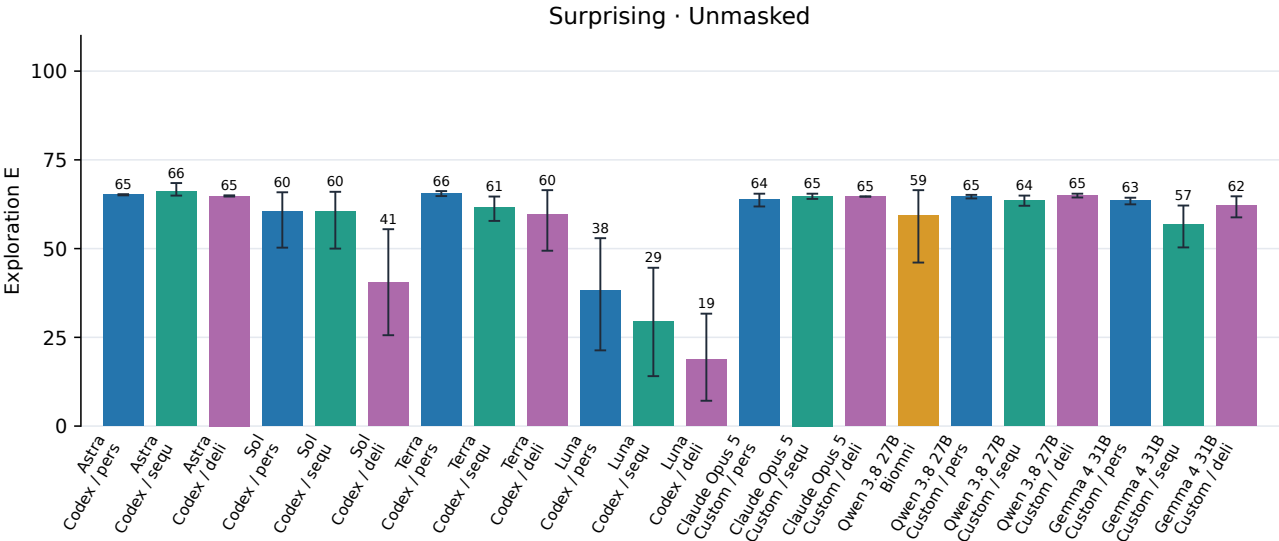
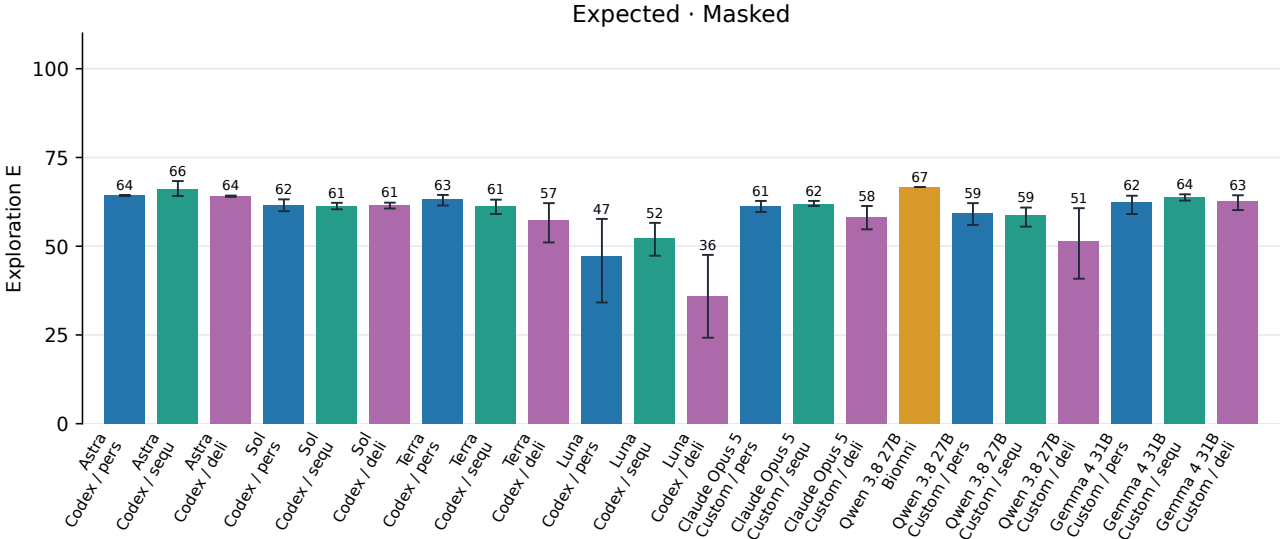
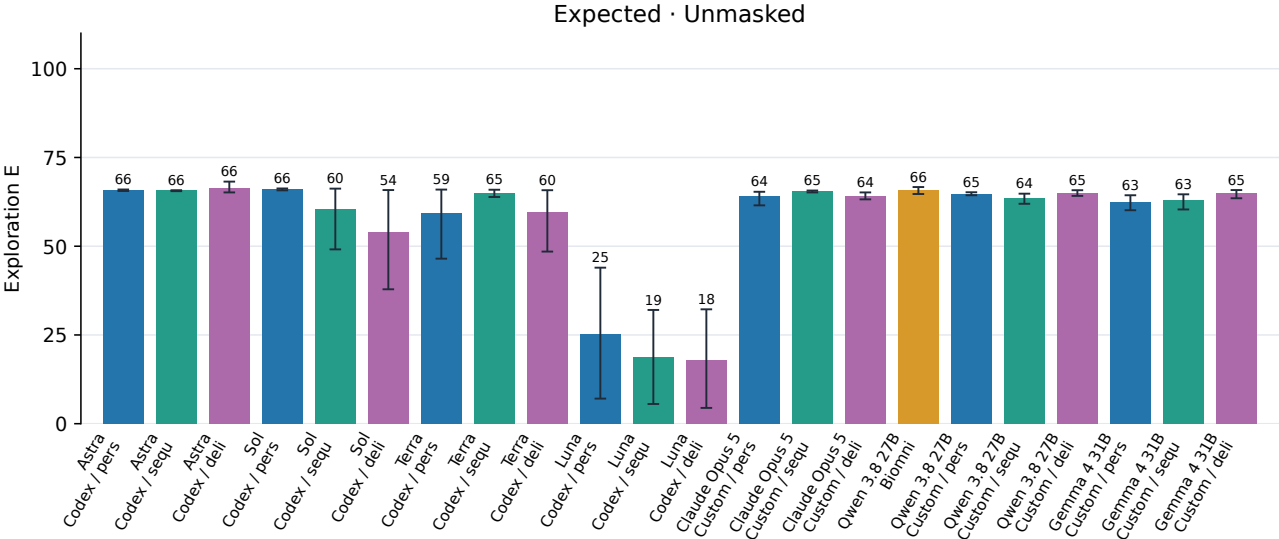
Scientific F1\*



Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · 95% run-level CIs on a fixed dataset pair · NA = unavailable Bootstrap CIs; Wilson for focal recovery. Missing CIs: insufficient runs/evidence. Native Biomni uses xhigh reasoning; other runs use medium.

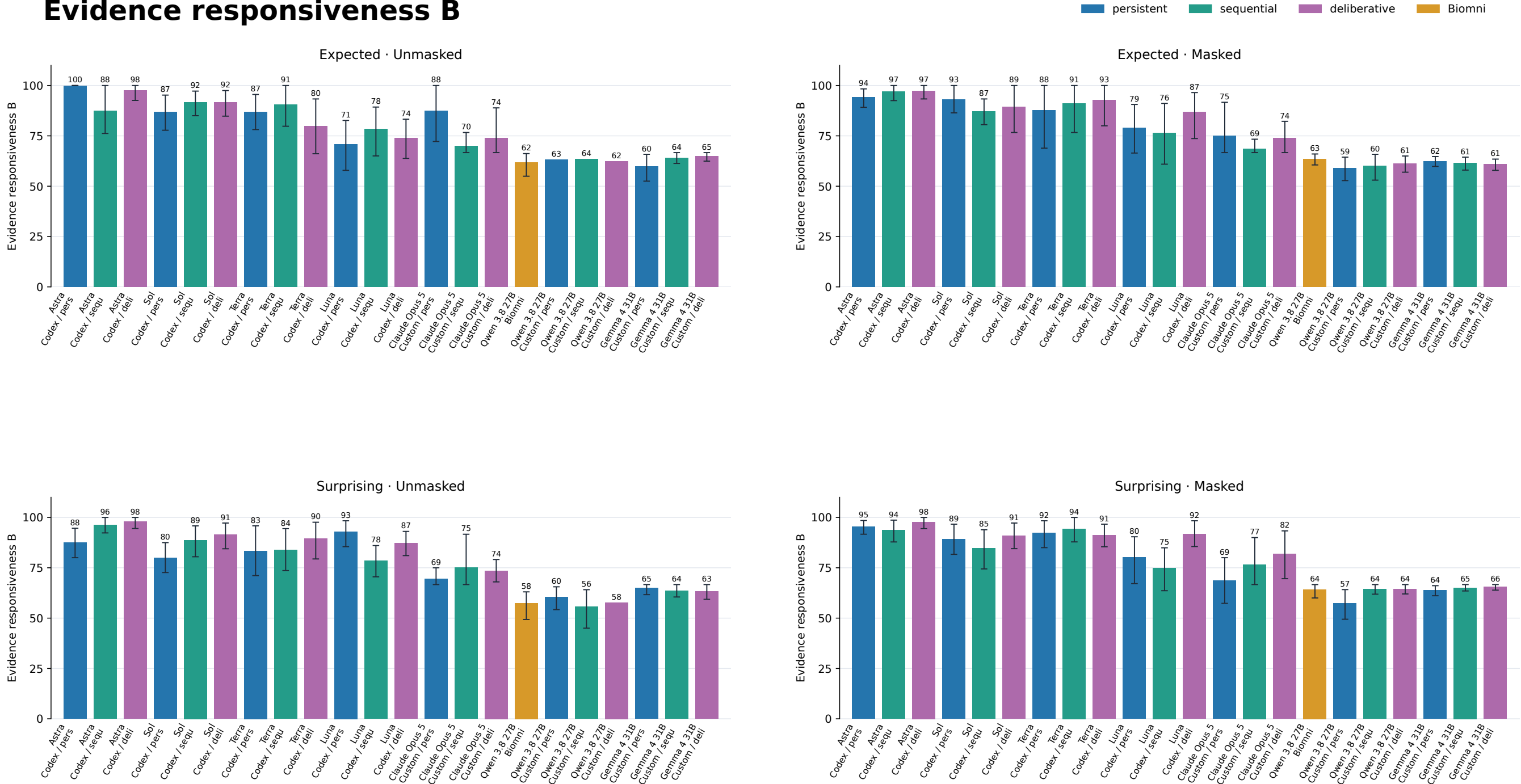
# Exploration E

persistent sequential deliberative Biomni



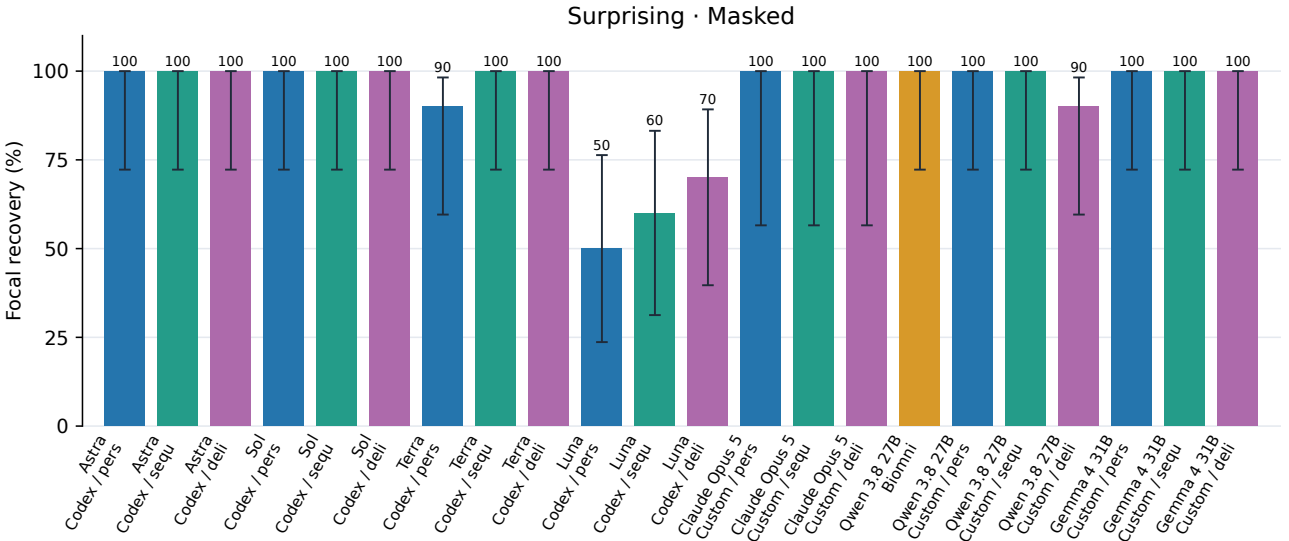
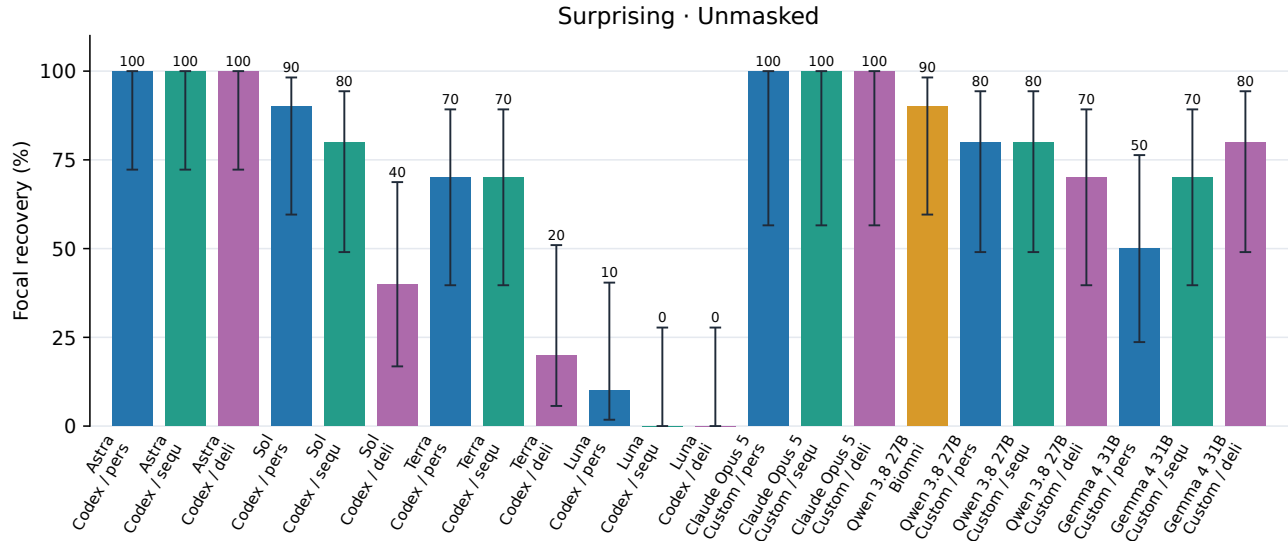
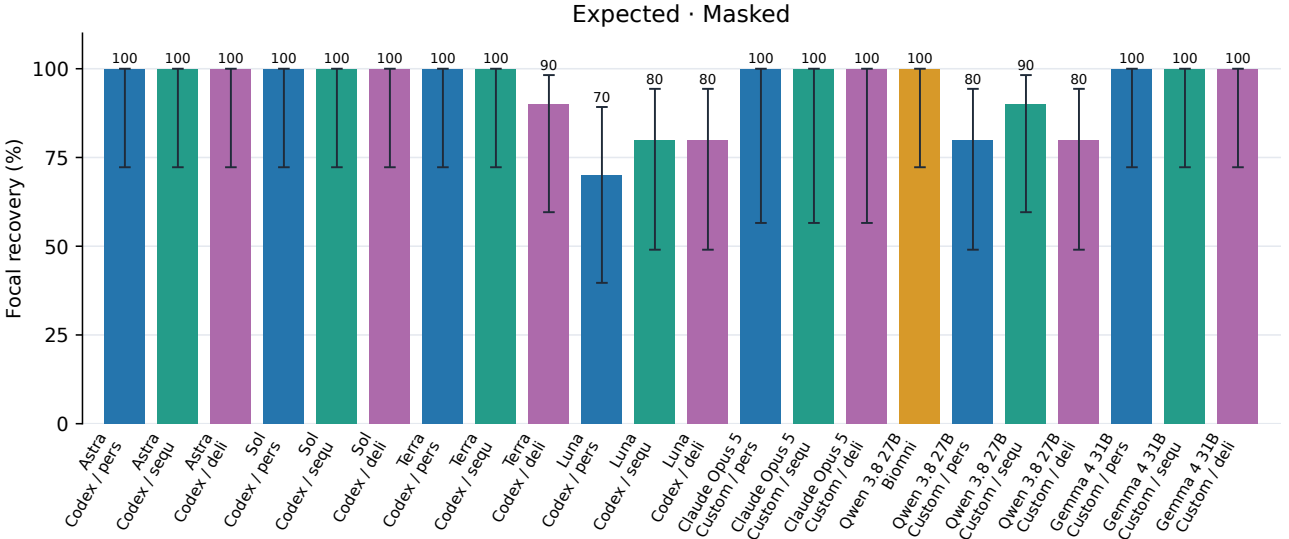
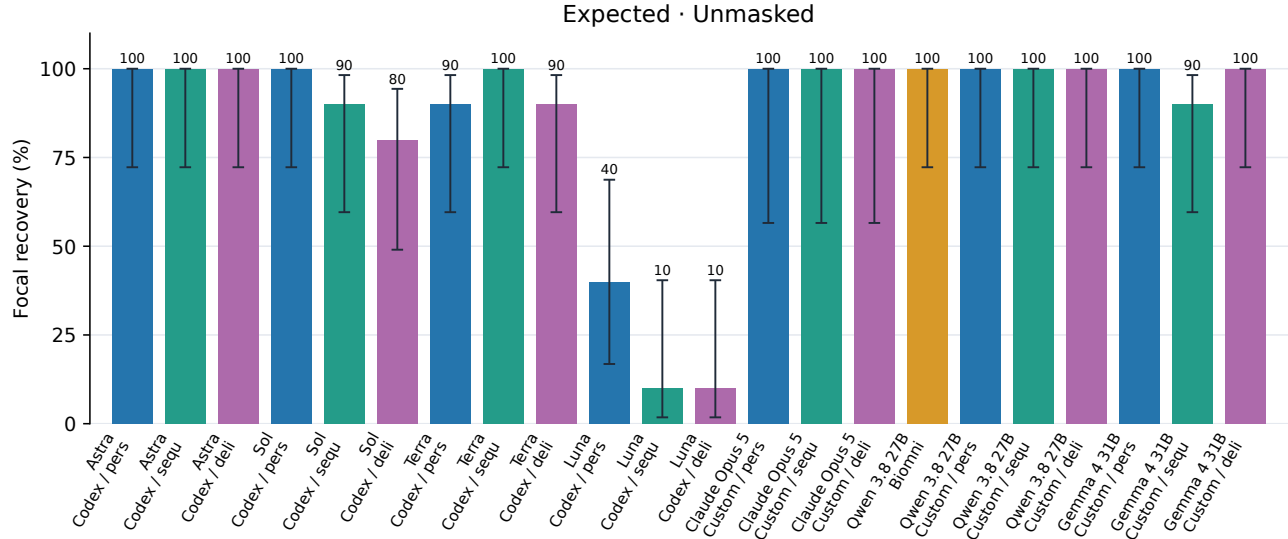
Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · 95% run-level CIs on a fixed dataset pair · NA = unavailable Bootstrap CIs; Wilson for focal recovery. Missing CIs: insufficient runs/evidence. Native Biomni uses xhigh reasoning; other runs use medium.

# Evidence responsiveness B



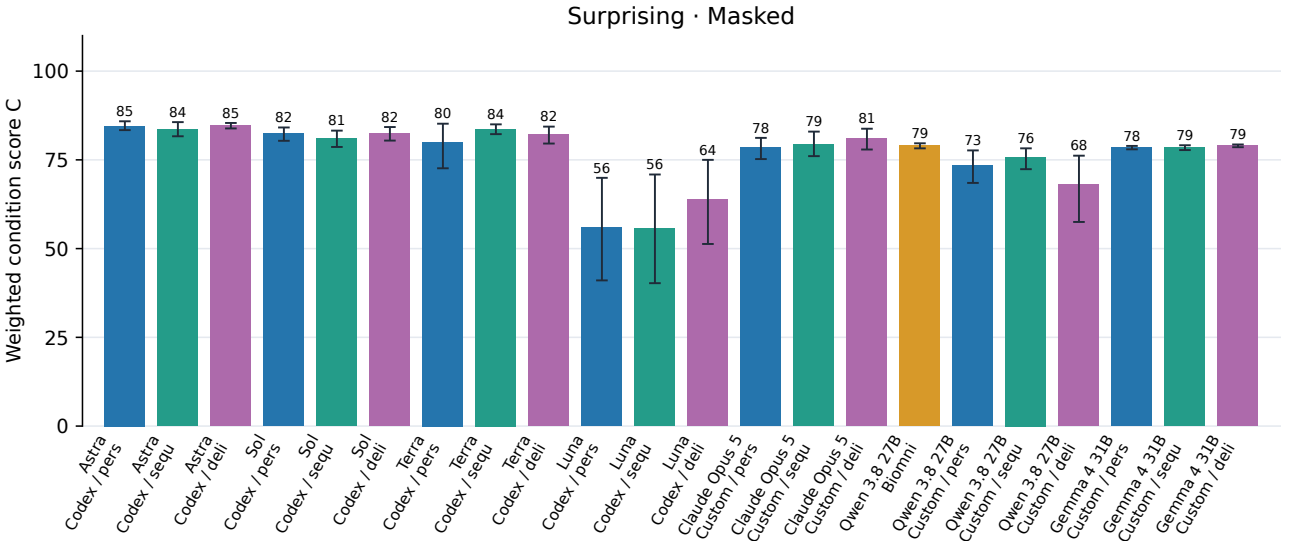
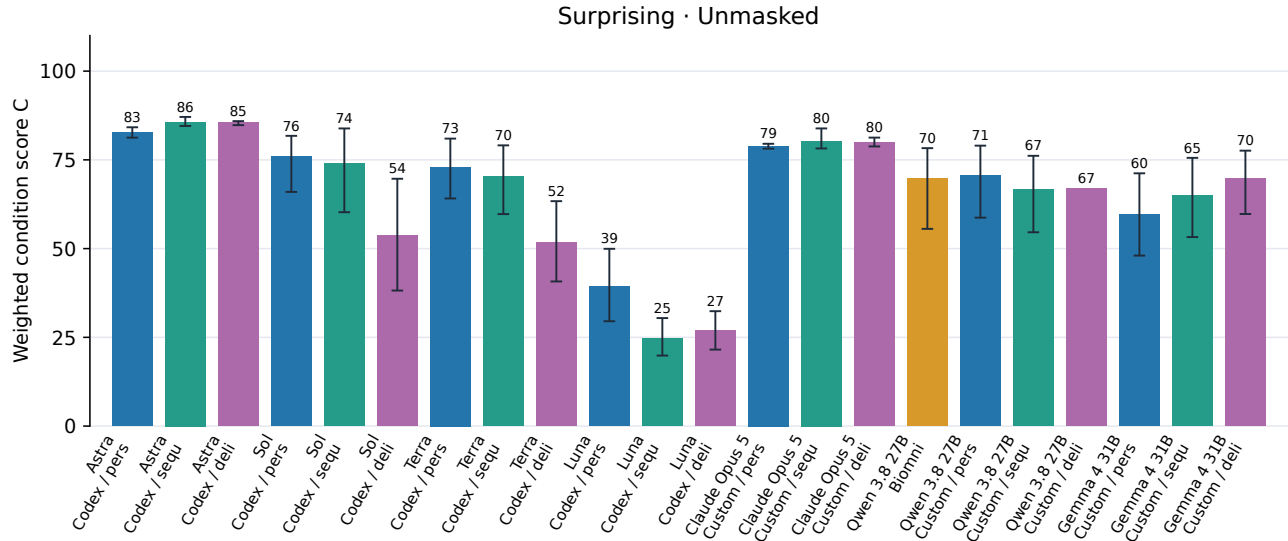
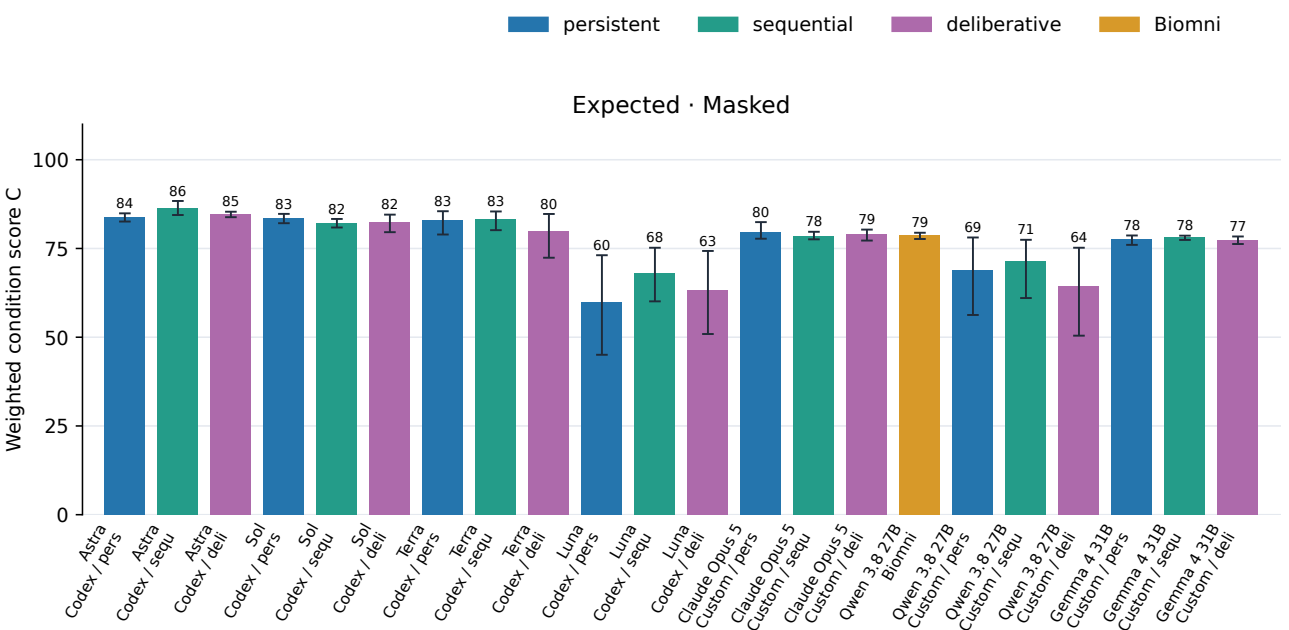
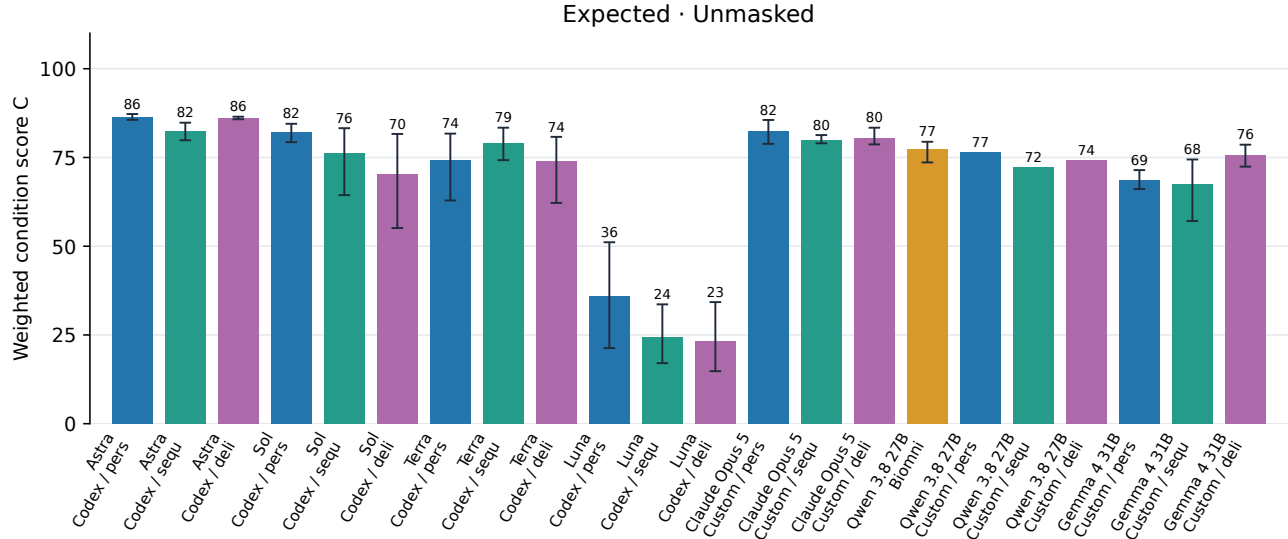
Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · 95% run-level CIs on a fixed dataset pair · NA = unavailable Bootstrap CIs; Wilson for focal recovery. Missing CIs: insufficient runs/evidence. Native Biomni uses xhigh reasoning; other runs use medium.

Focal recovery (%)



Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · 95% run-level CIs on a fixed dataset pair · NA = unavailable Bootstrap CIs; Wilson for focal recovery. Missing CIs: insufficient runs/evidence. Native Biomni uses xhigh reasoning; other runs use medium.

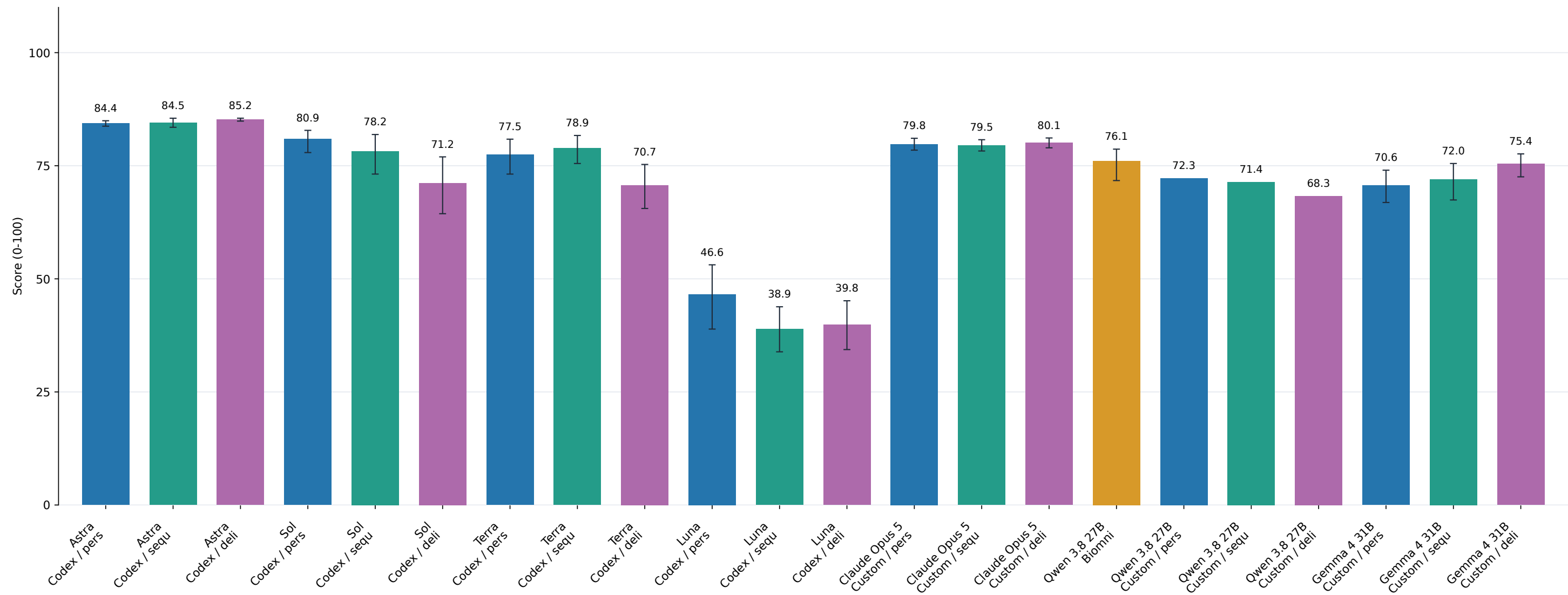
Weighted condition score C



Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · 95% run-level CIs on a fixed dataset pair · NA = unavailable Bootstrap CIs; Wilson for focal recovery. Missing CIs: insufficient runs/evidence. Native Biomni uses xhigh reasoning; other runs use medium.

# Grounded discovery and adaptation S

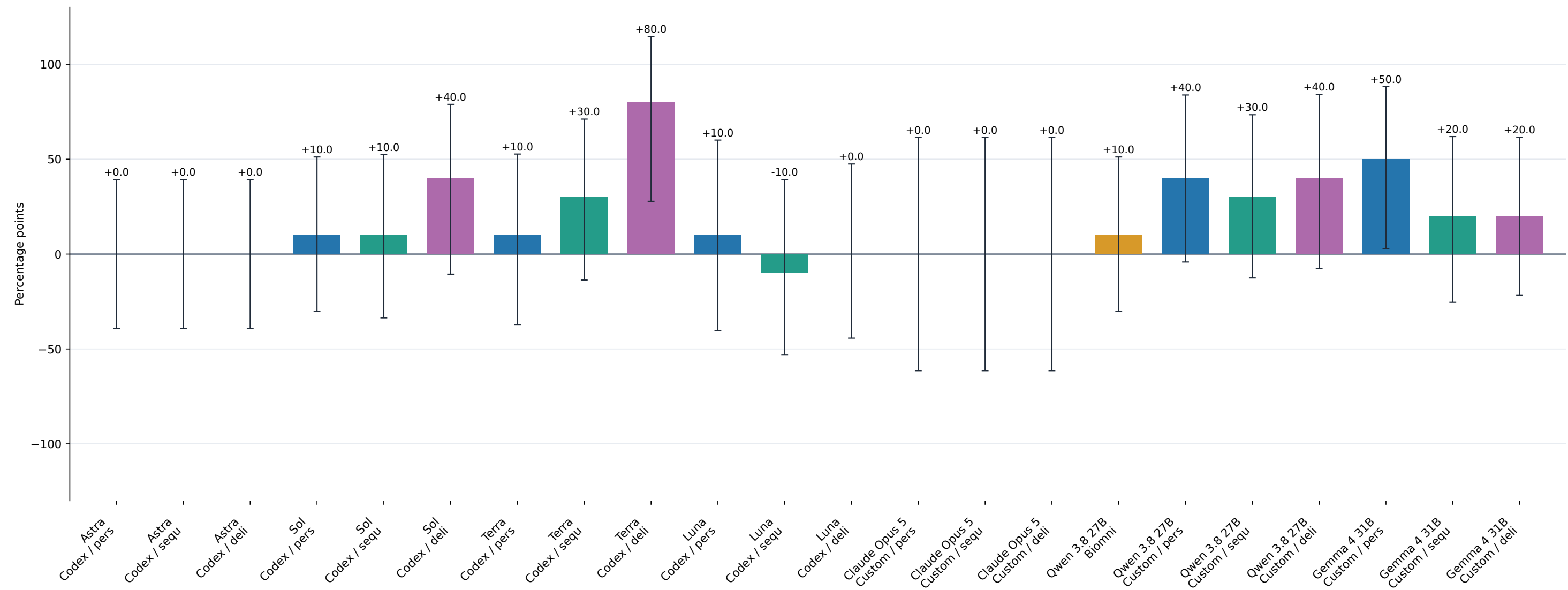
Equal-weight geometric mean across four conditions; higher is better (0-100).



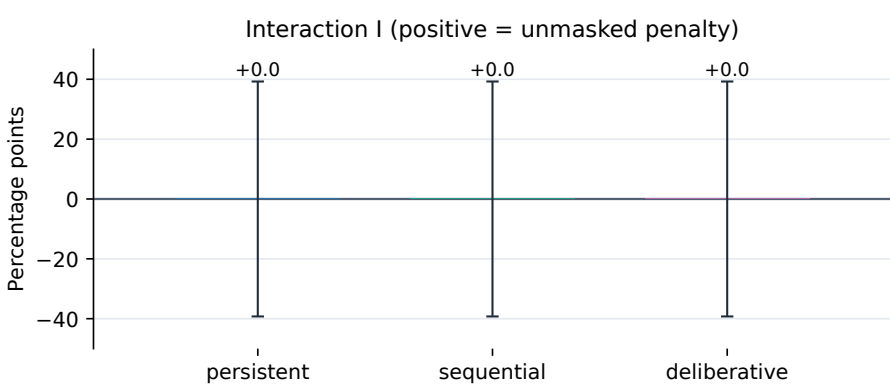
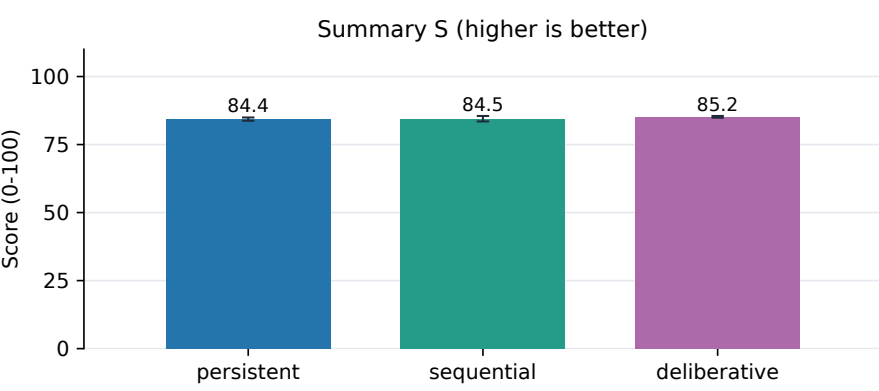
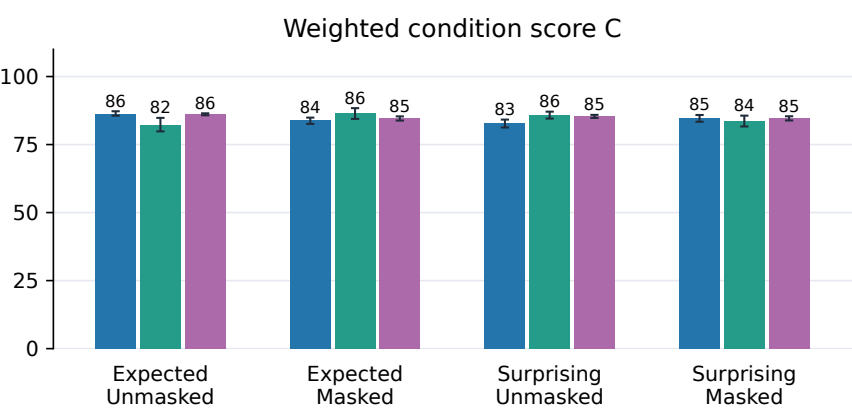
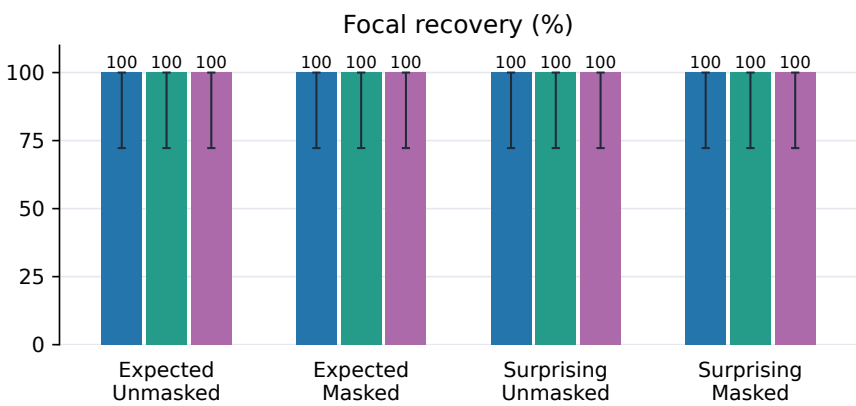
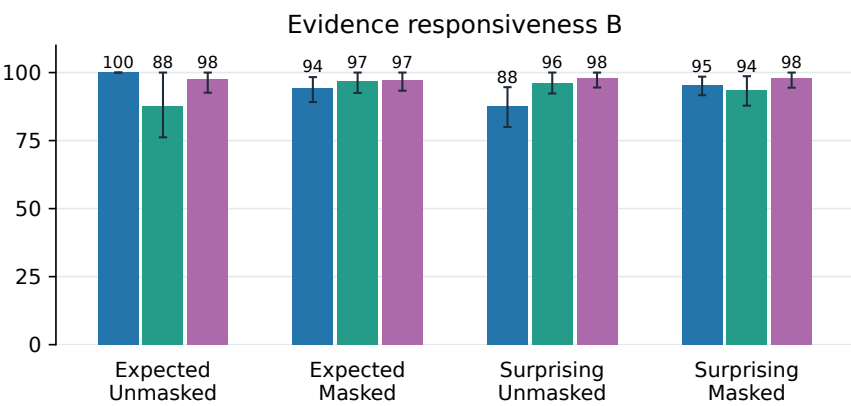
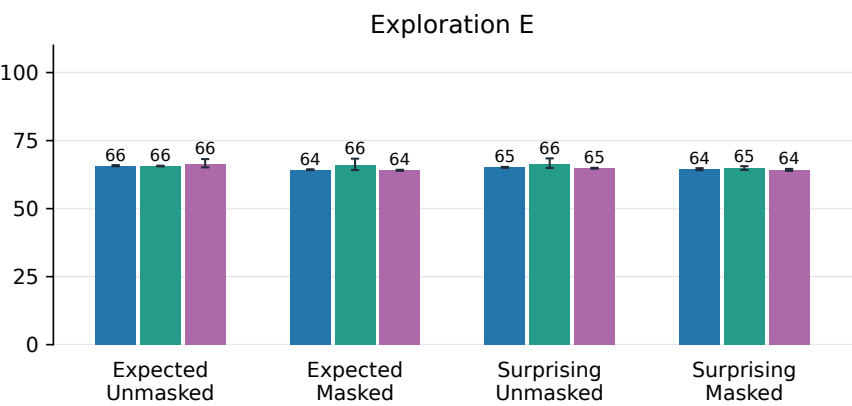
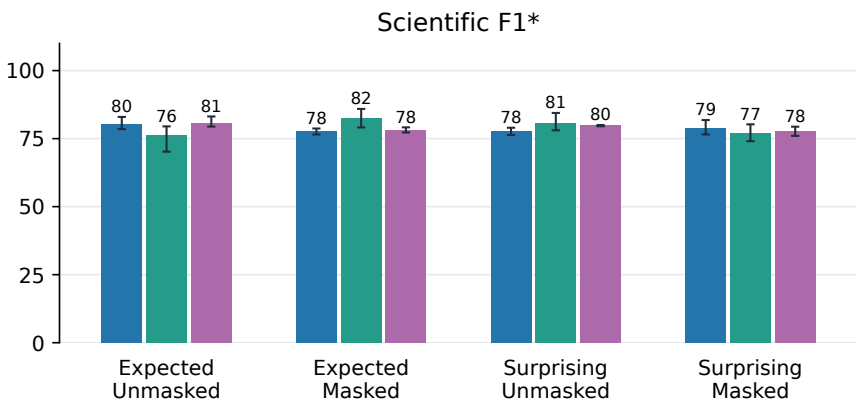
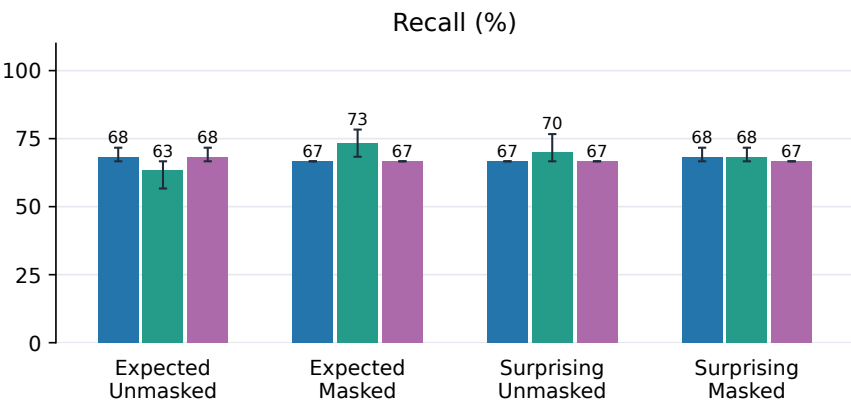
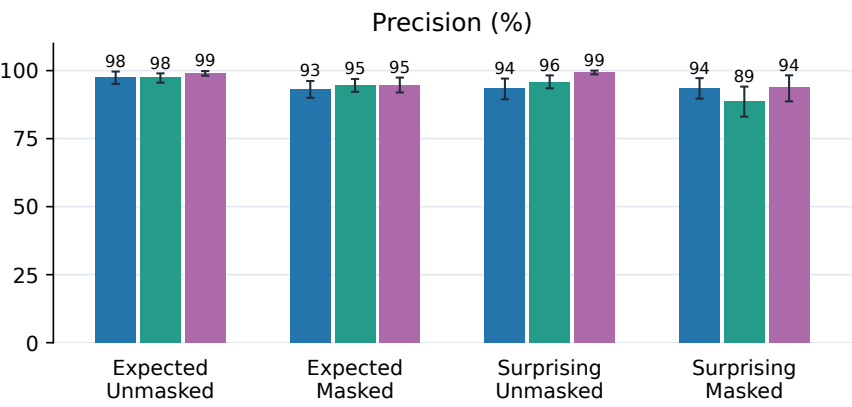
Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors | 95% CIs conditional on the fixed dataset pair.  
S: joint whole-run bootstrap; I: Wilson-based MOVER. NA = unavailable; absent intervals indicate insufficient runs/evidence.

# Label-by-surprise interaction I (pp)

(Focal EU - SU) - (Focal EM - SM); positive = larger surprise penalty with recognizable labels.



Snapshot 2026-09-12T20:27 UTC | Ended runs, including runs with errors | 95% CIs conditional on the fixed dataset pair.  
S: joint whole-run bootstrap; I: Wilson-based MOVER. NA = unavailable; absent intervals indicate insufficient runs/evidence.

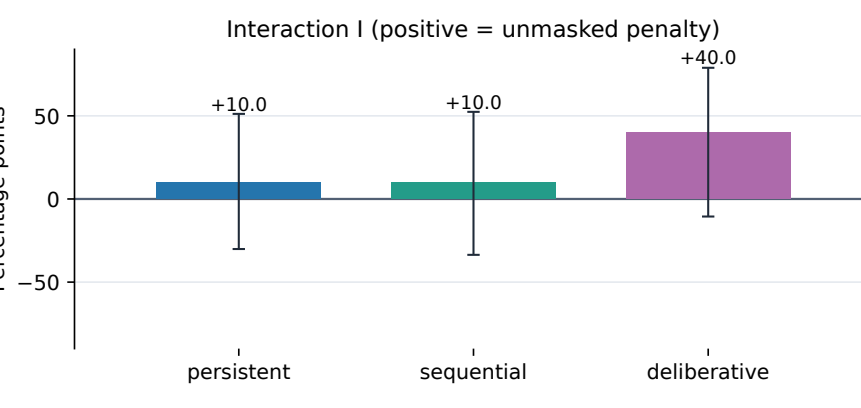
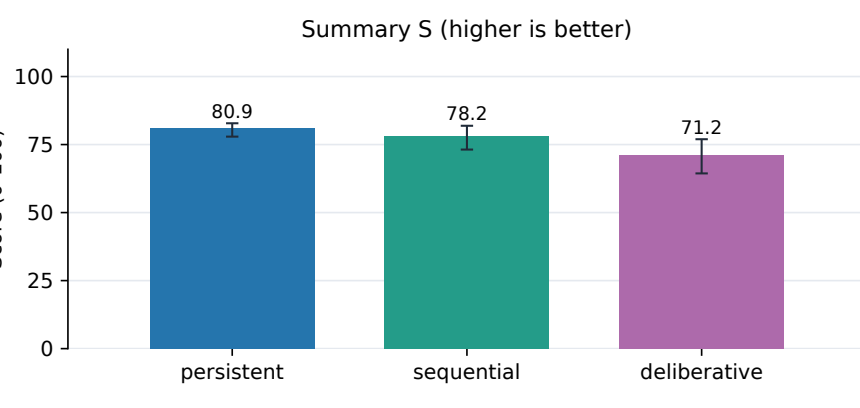
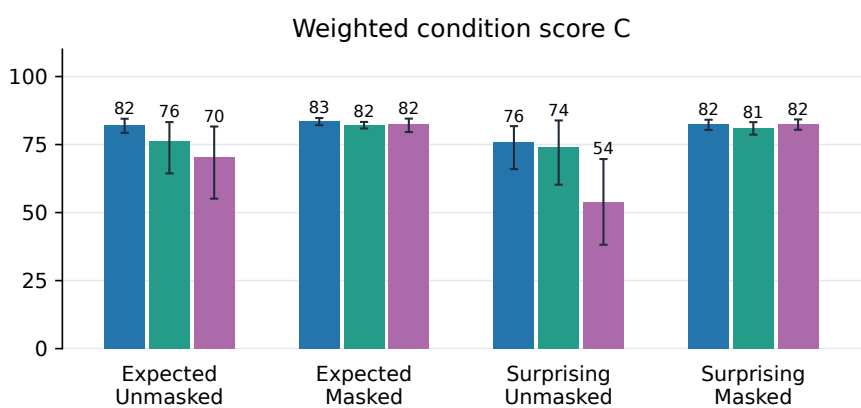
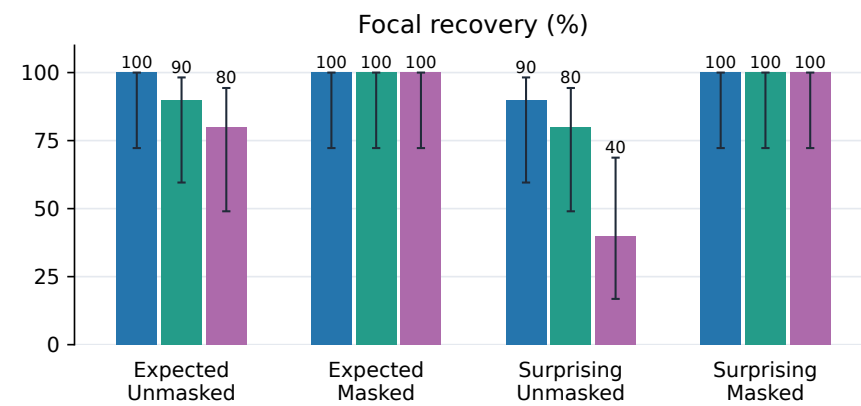
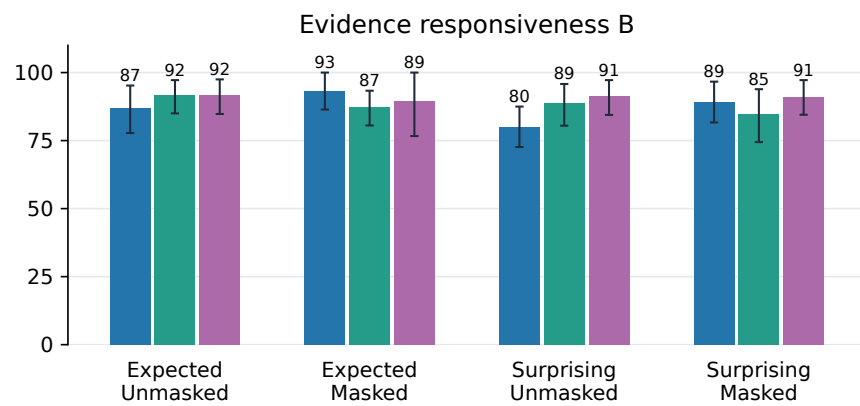
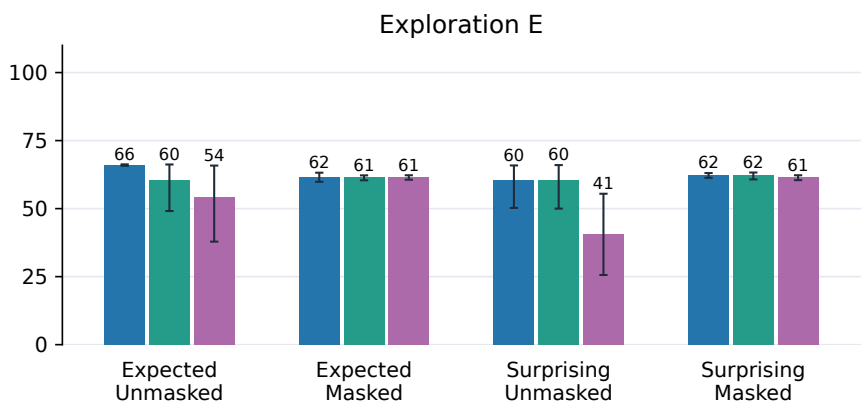
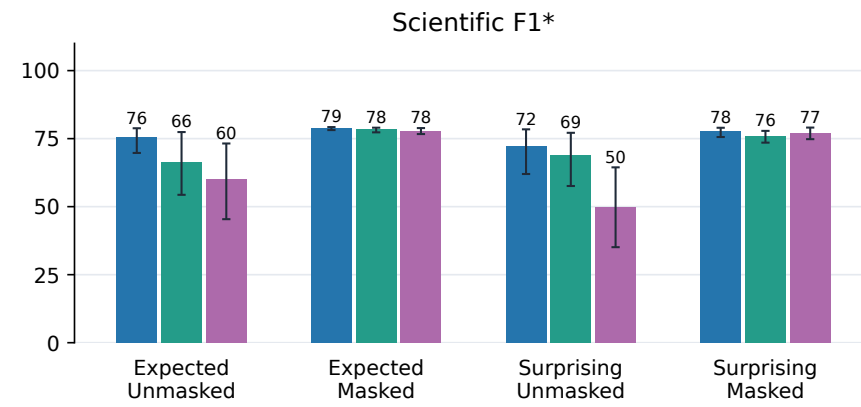
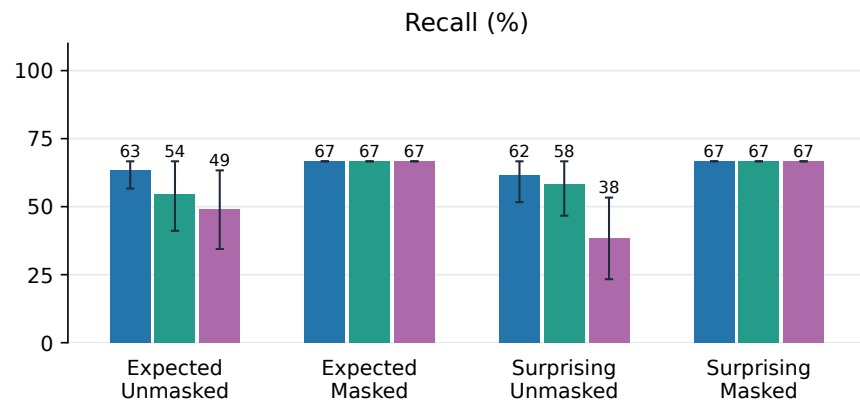
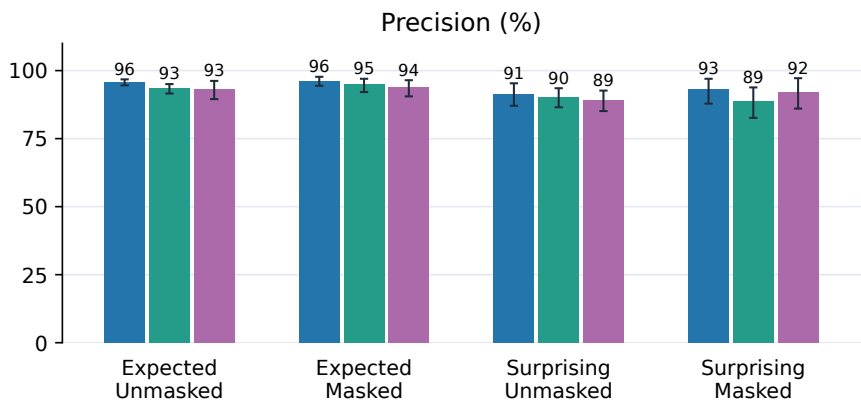


Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · S: equal condition weights; I: signed percentage points  
95% CIs: bootstrap; Wilson for focal recovery; Wilson-based MOVER for I. Fixed dataset pair. NA = unavailable; missing CIs indicate insufficient runs/evidence.

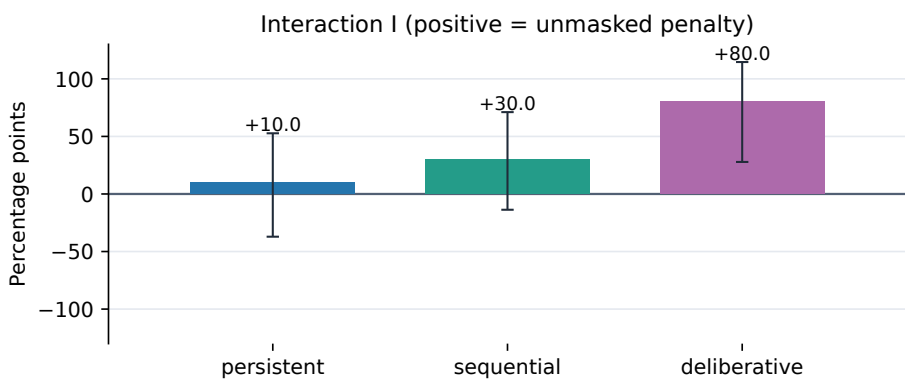
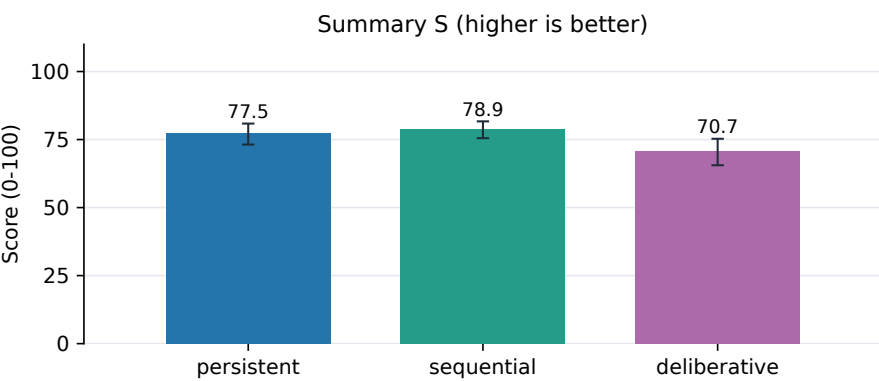
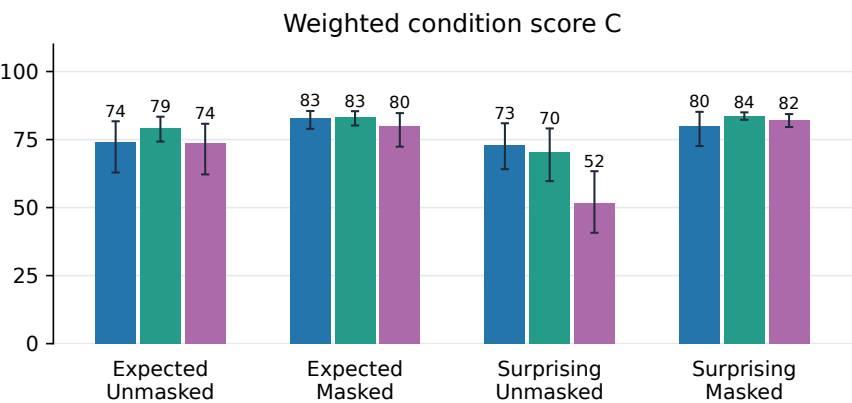
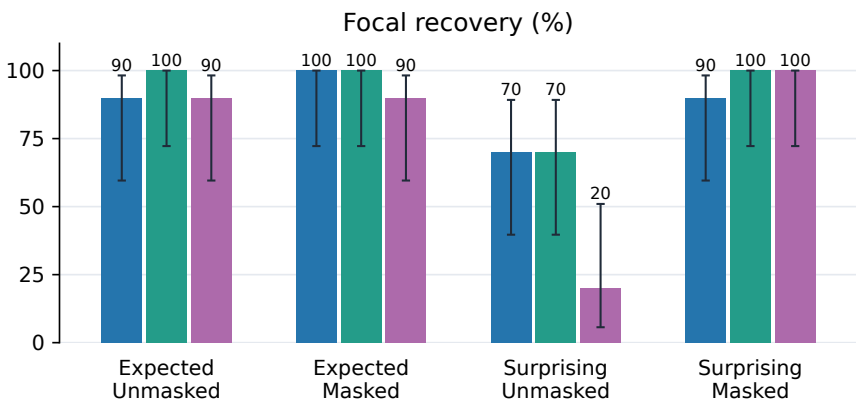
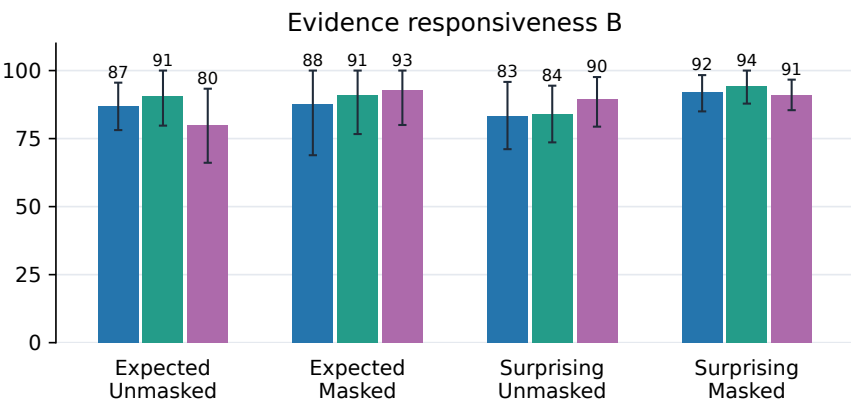
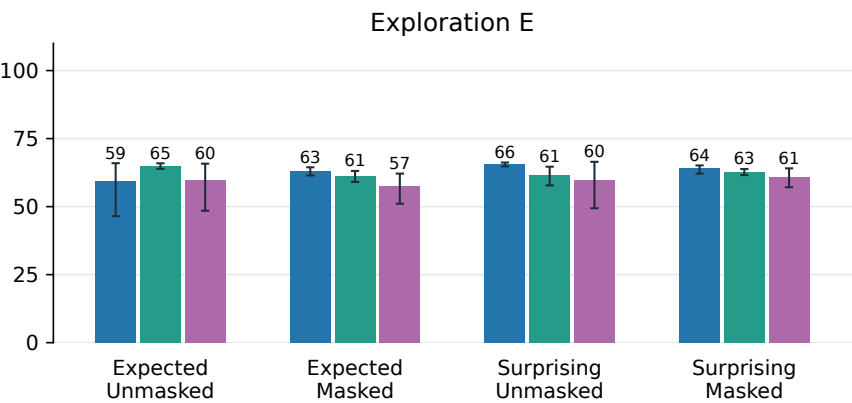
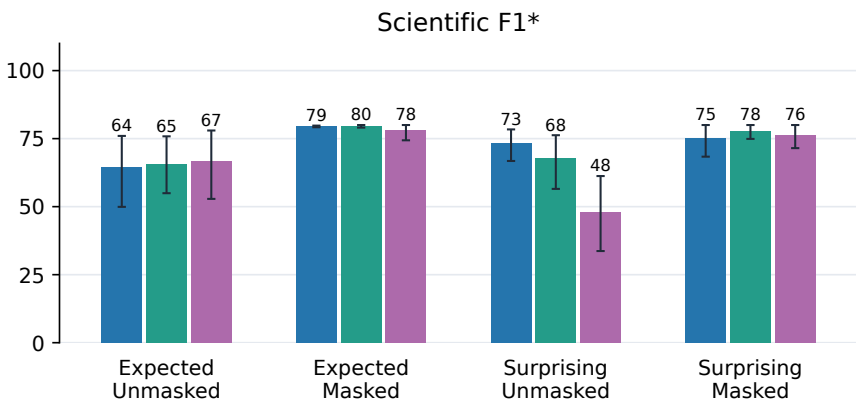
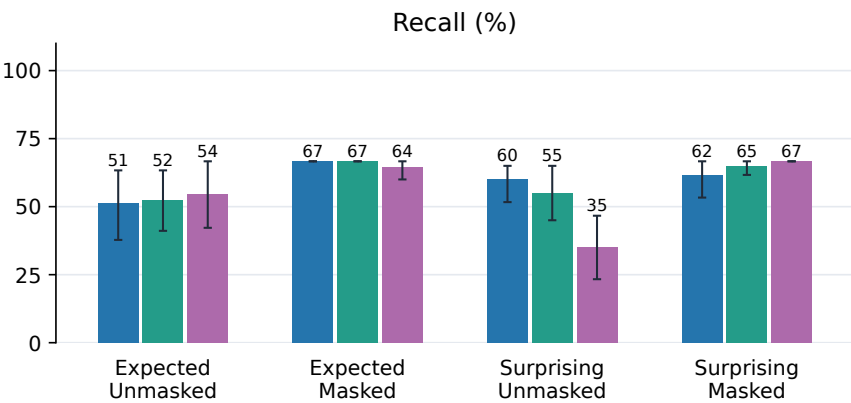
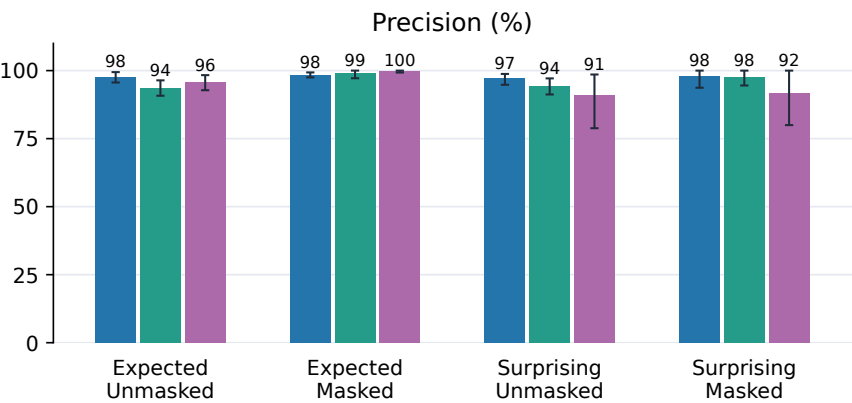
# Sol · Codex

Workflow and condition comparison · medium reasoning

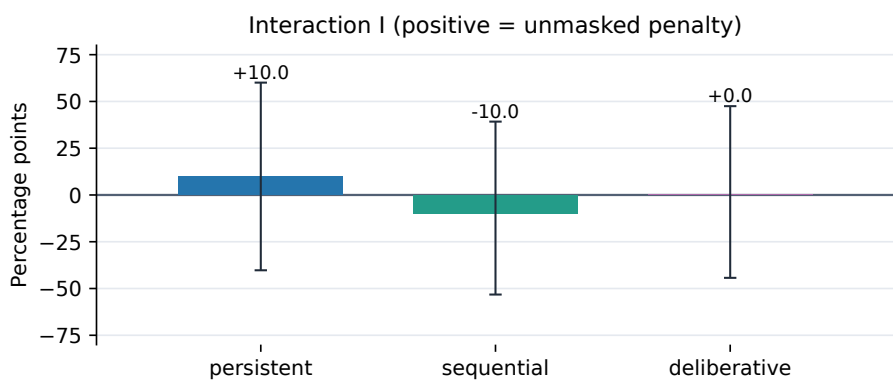
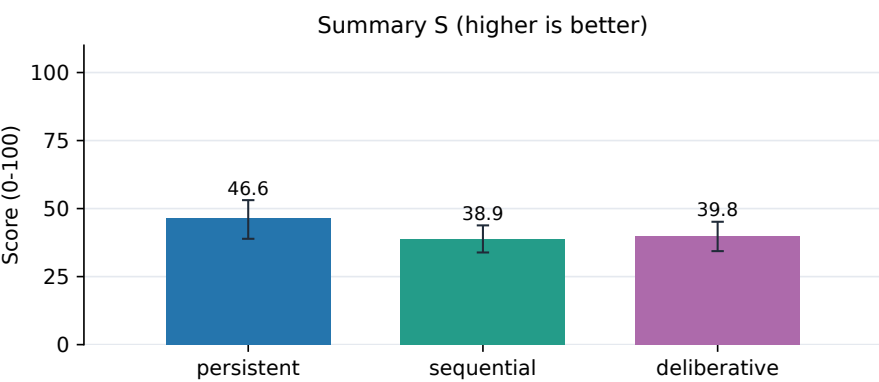
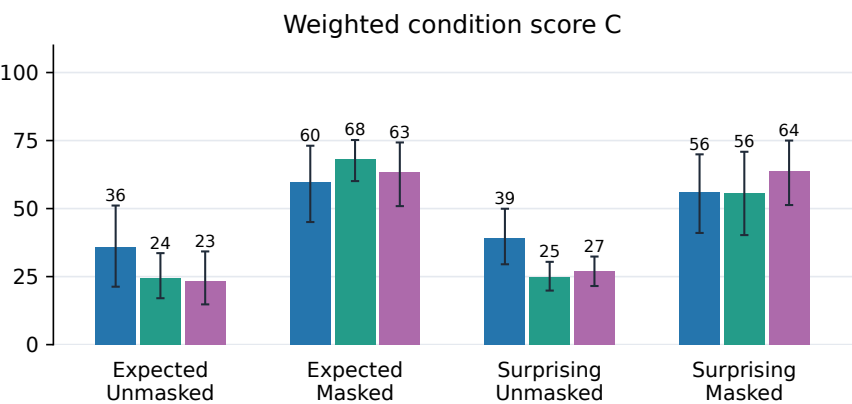
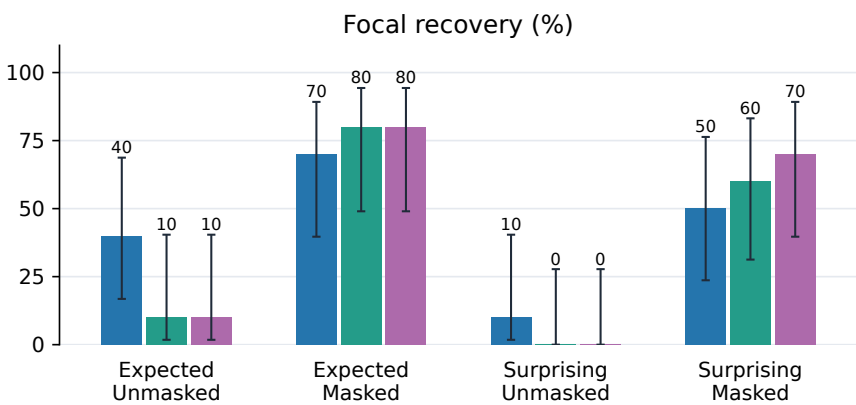
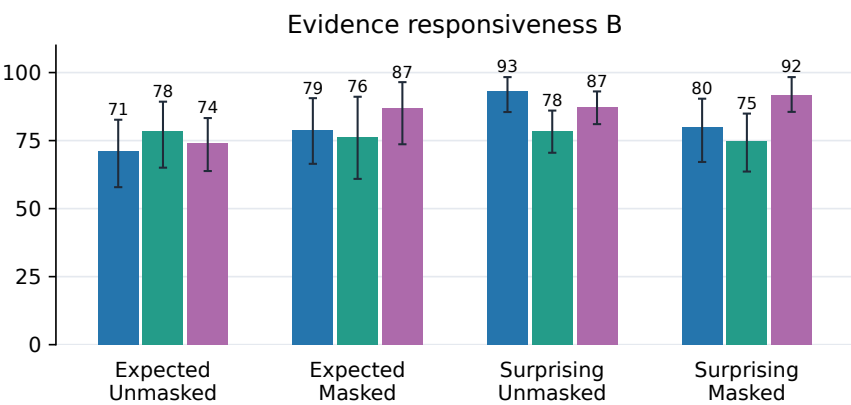
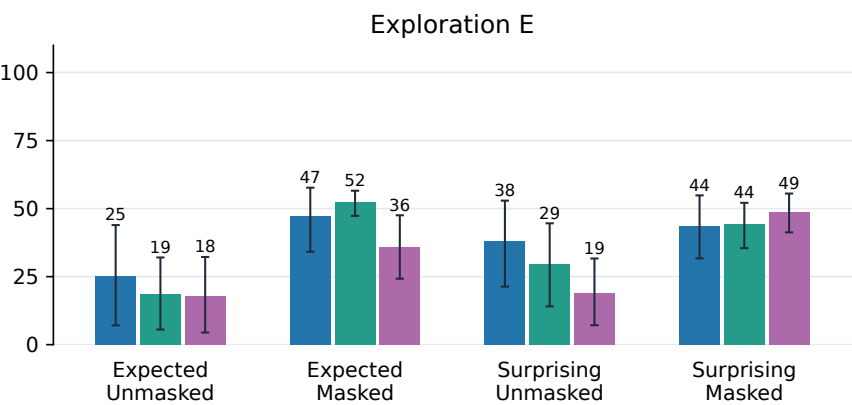
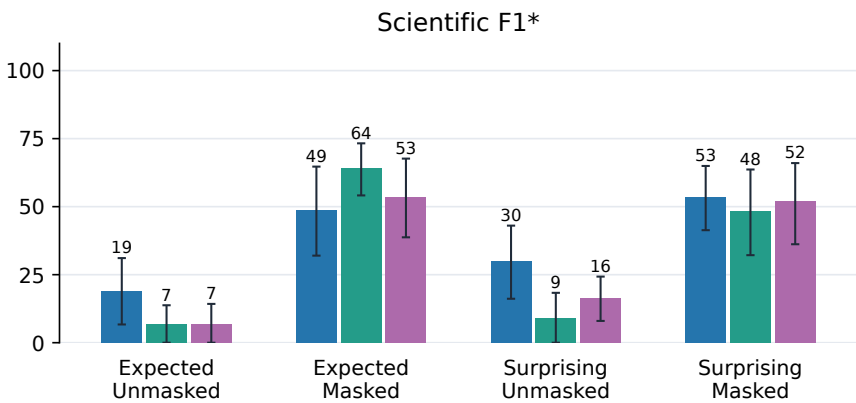
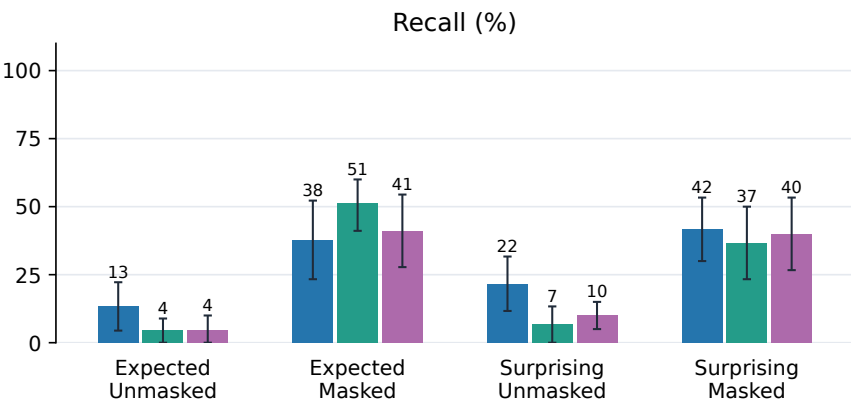
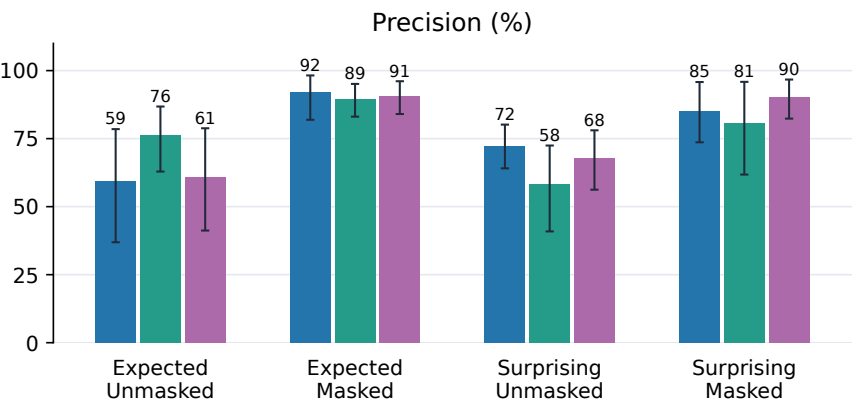
■ persistent ■ sequential ■ deliberative



Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · S: equal condition weights; I: signed percentage points  
95% CIs: bootstrap; Wilson for focal recovery; Wilson-based MOVER for I. Fixed dataset pair. NA = unavailable; missing CIs indicate insufficient runs/evidence.



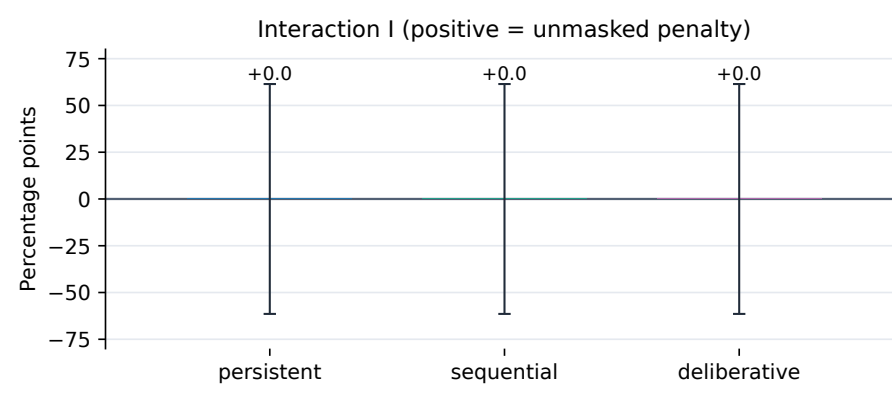
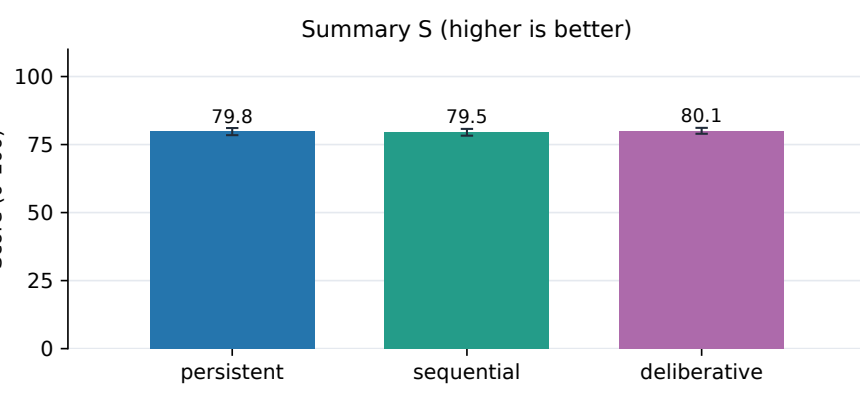
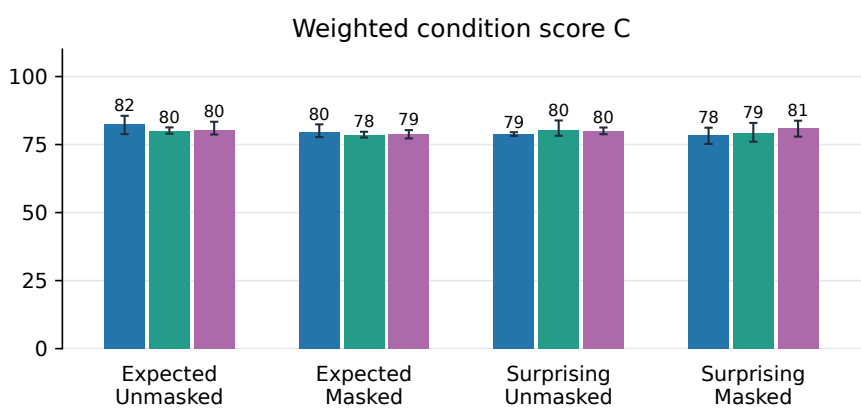
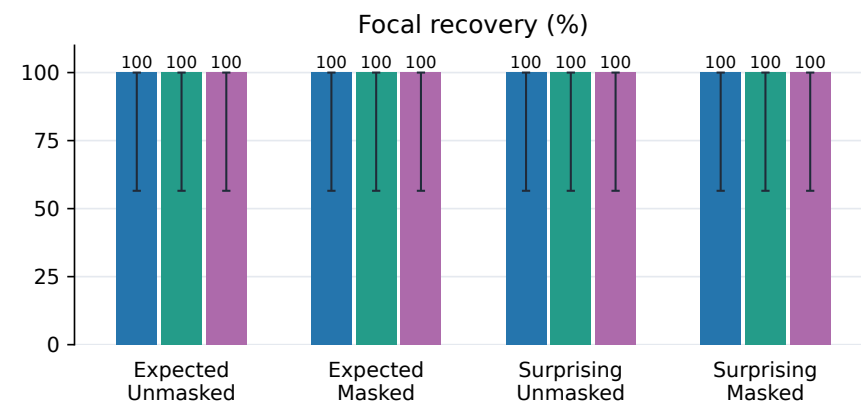
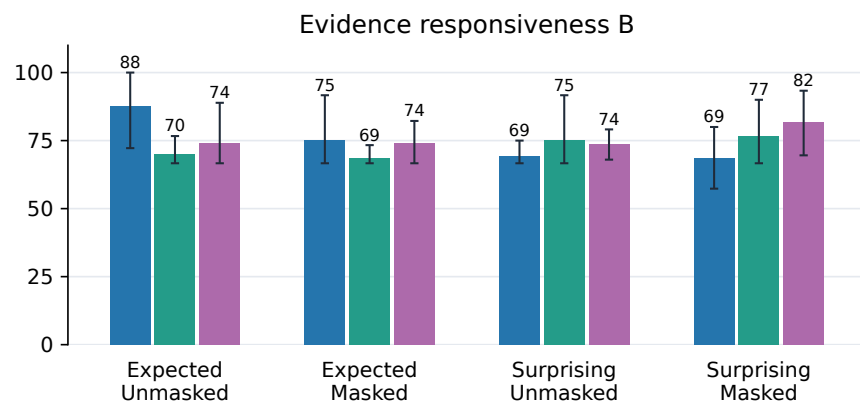
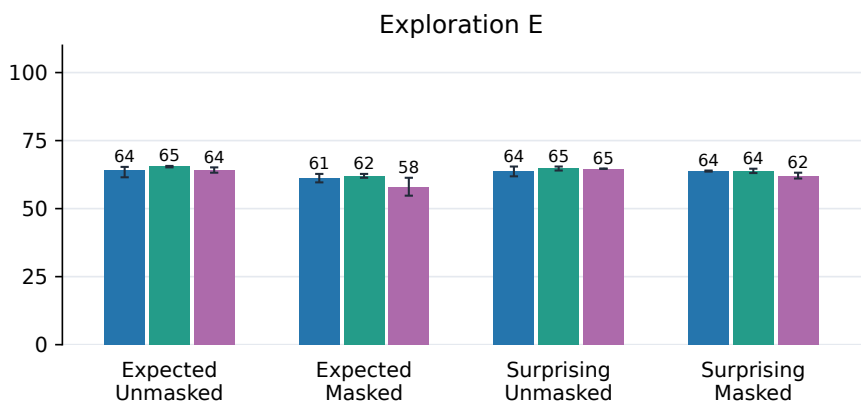
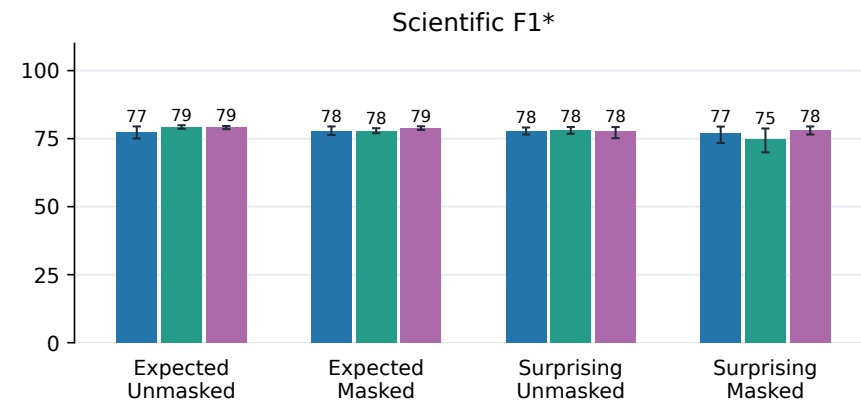
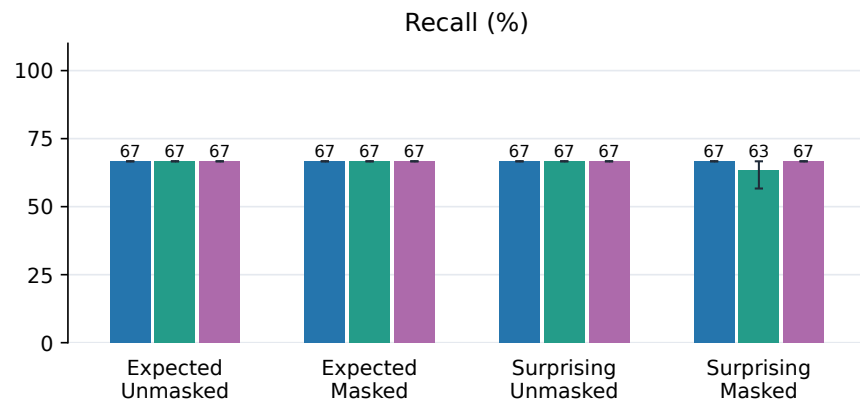
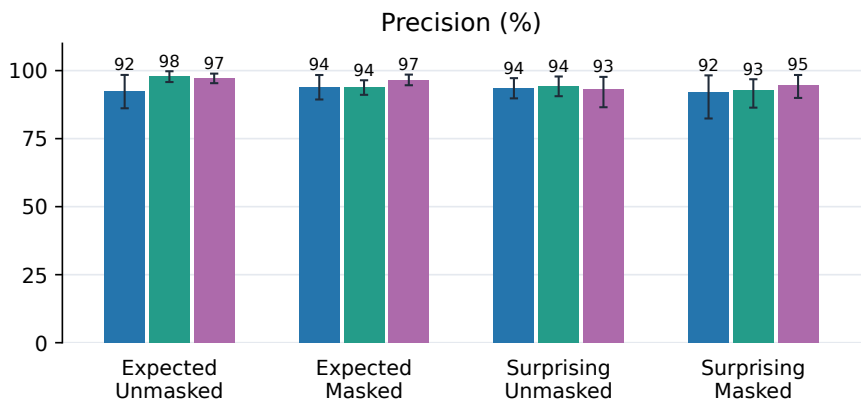
Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · S: equal condition weights; I: signed percentage points  
95% CIs: bootstrap; Wilson for focal recovery; Wilson-based MOVER for I. Fixed dataset pair. NA = unavailable; missing CIs indicate insufficient runs/evidence.



# Claude Opus 5 - Custom runner

Workflow and condition comparison · medium reasoning

persistent sequential deliberative

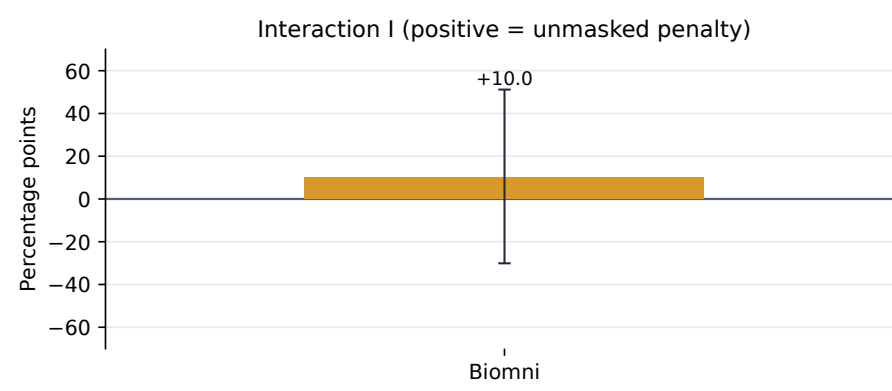
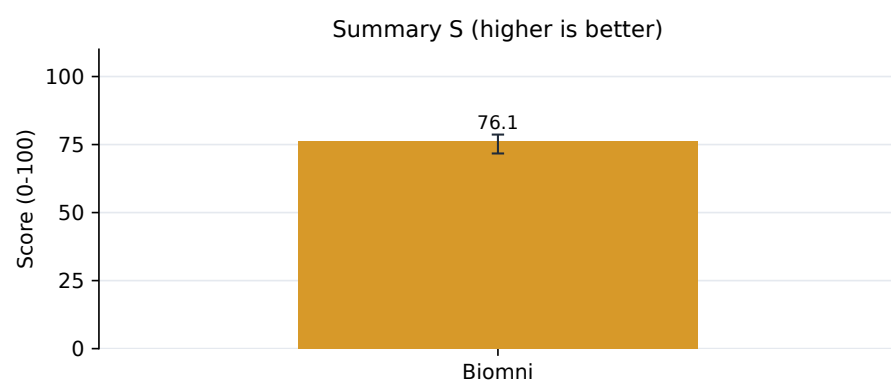
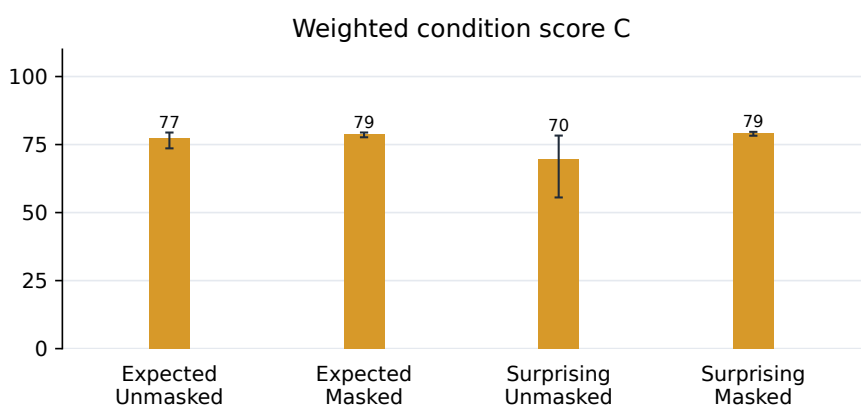
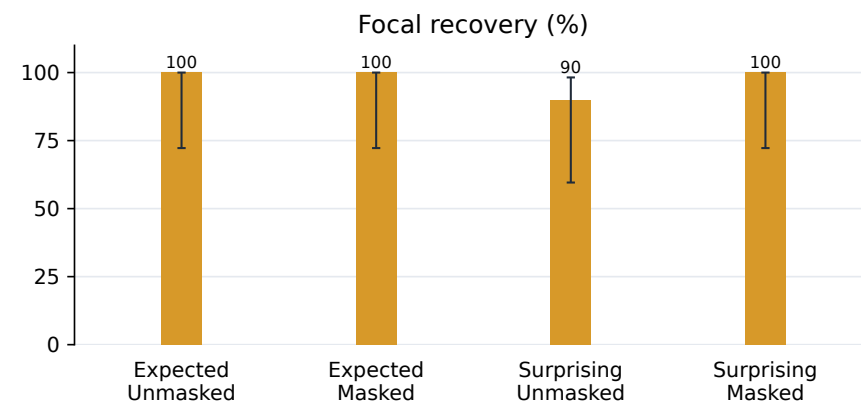
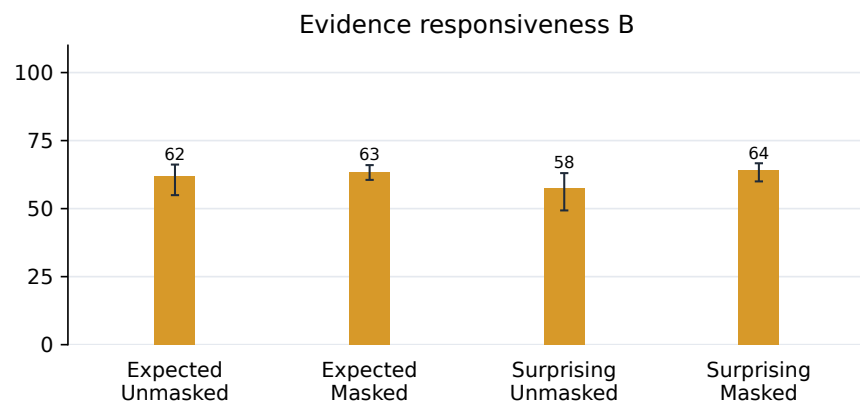
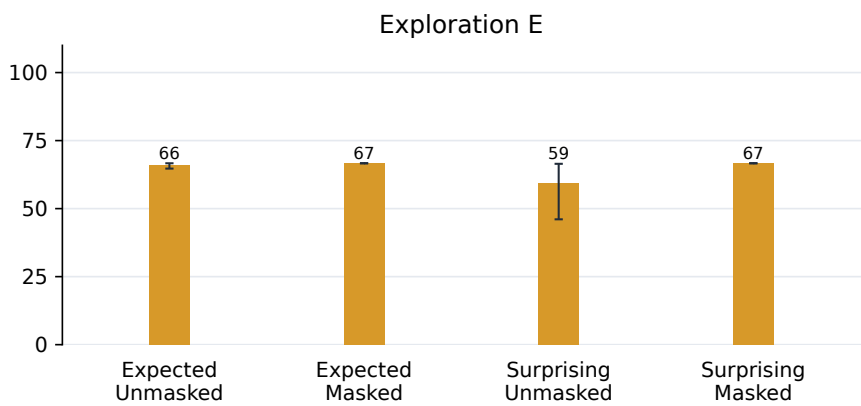
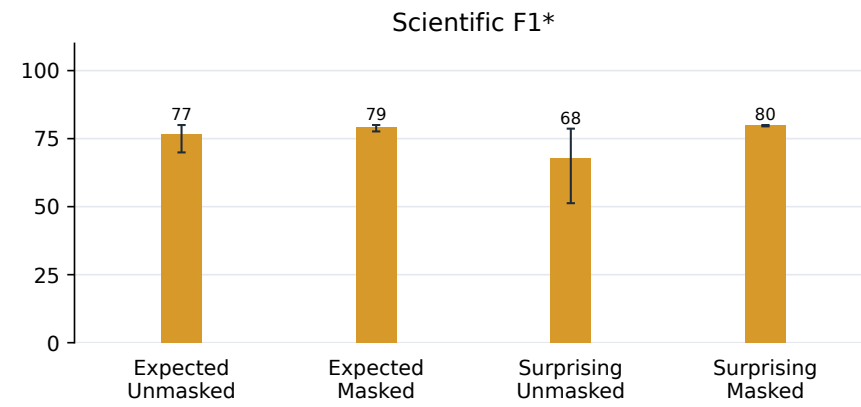
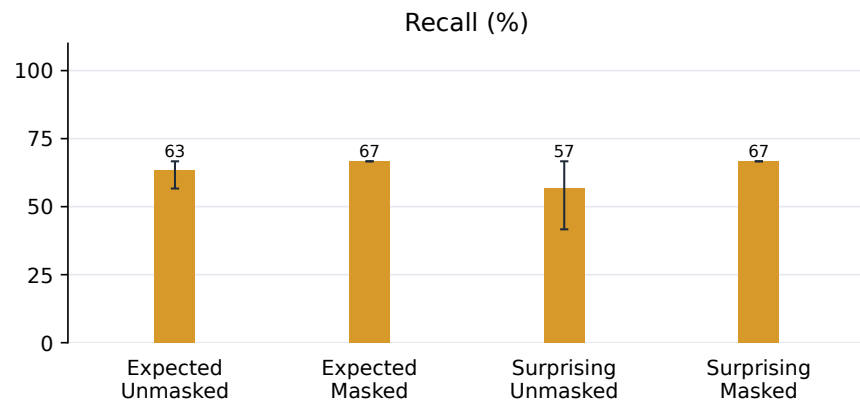
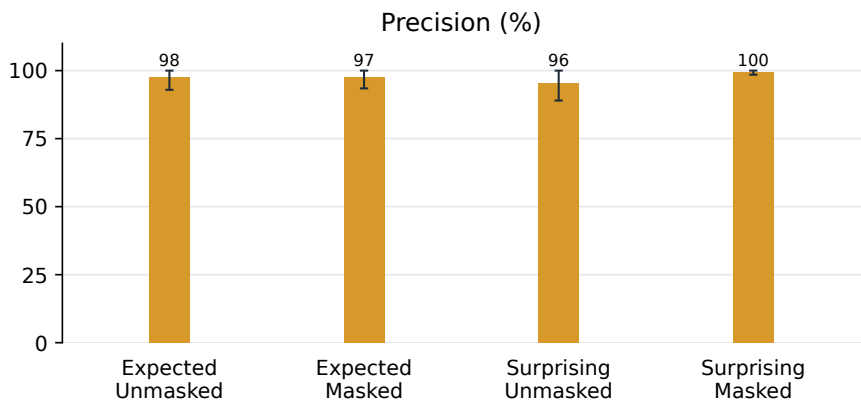


Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · S: equal condition weights; I: signed percentage points  
95% CIs: bootstrap; Wilson for focal recovery; Wilson-based MOVER for I. Fixed dataset pair. NA = unavailable; missing CIs indicate insufficient runs/evidence.

# Qwen 3.8 27B · Biomni

Workflow and condition comparison · xhigh reasoning

Biomni

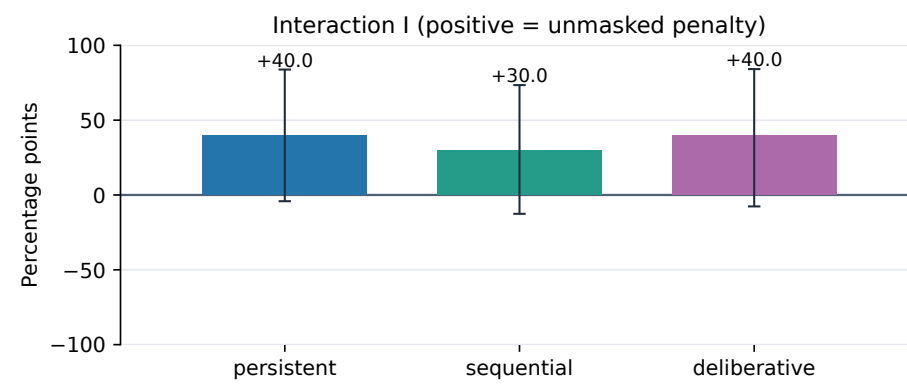
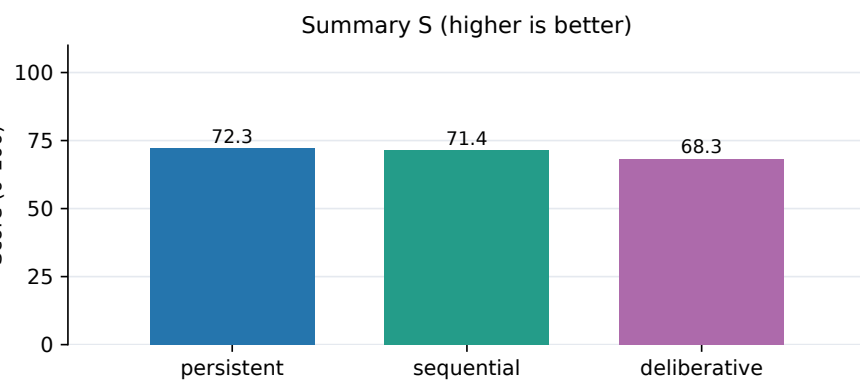
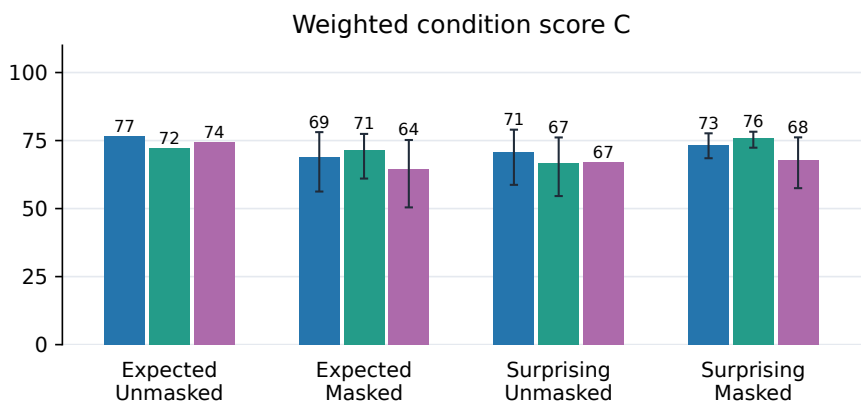
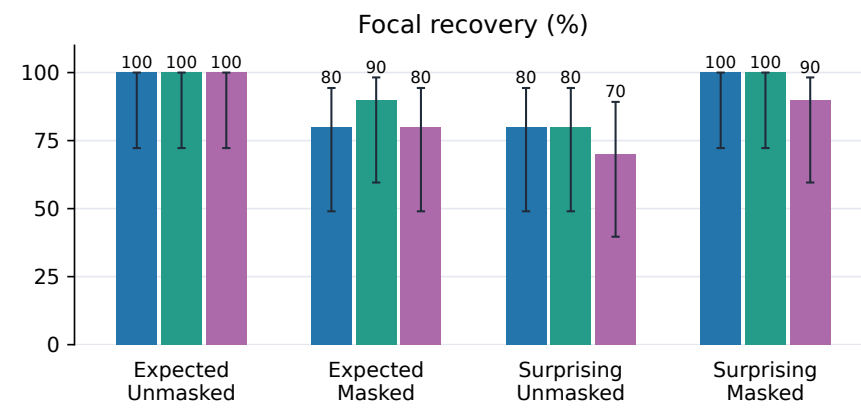
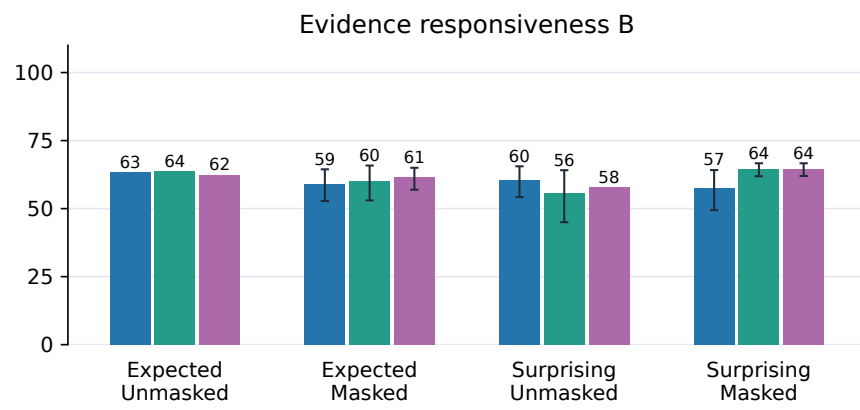
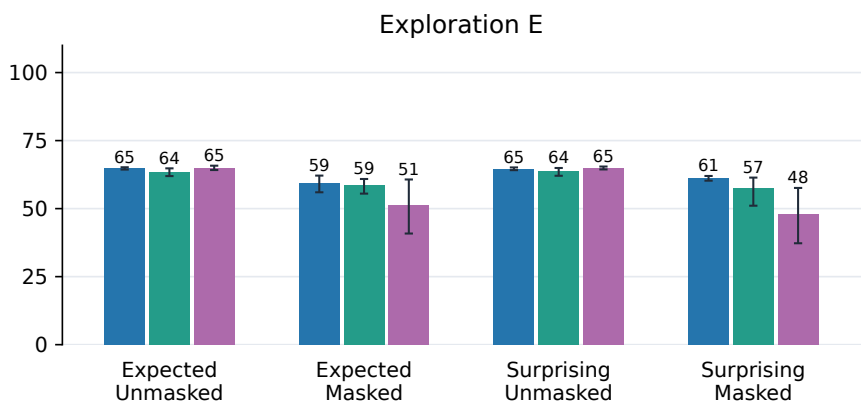
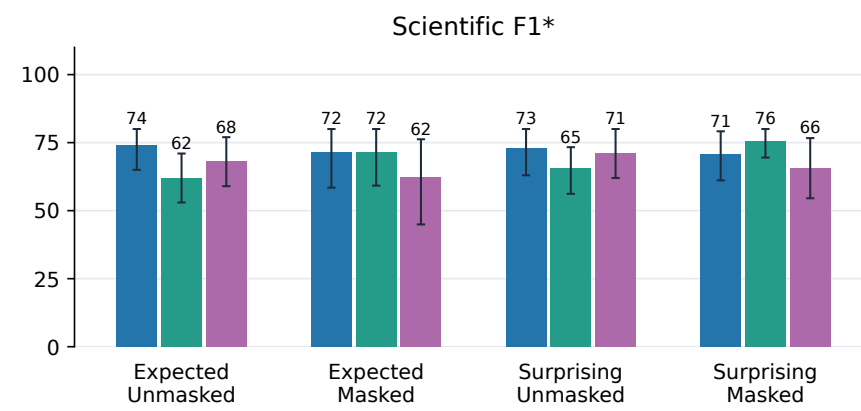
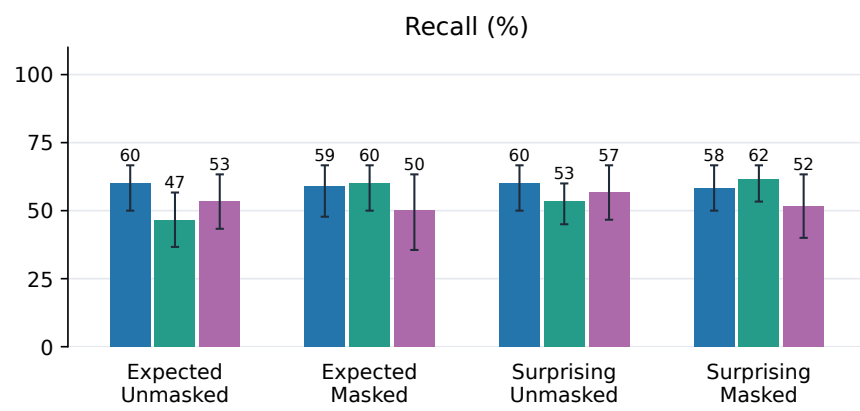
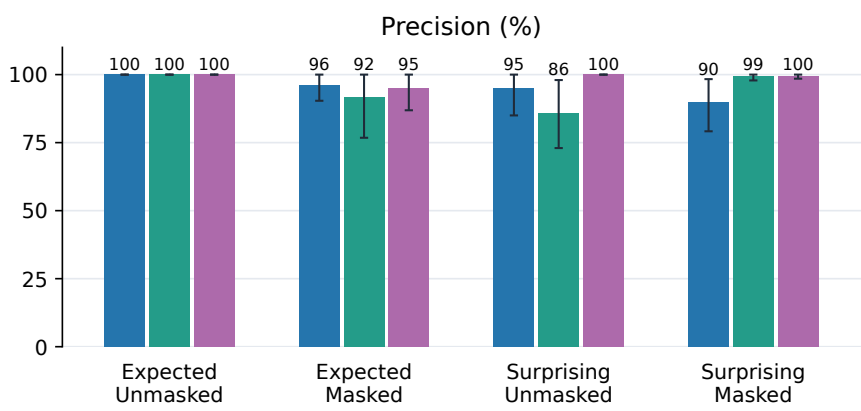


Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · S: equal condition weights; I: signed percentage points  
95% CIs: bootstrap; Wilson for focal recovery; Wilson-based MOVER for I. Fixed dataset pair. NA = unavailable; missing CIs indicate insufficient runs/evidence.

# Qwen 3.8 27B · Custom runner

Workflow and condition comparison · medium reasoning

persistent sequential deliberative

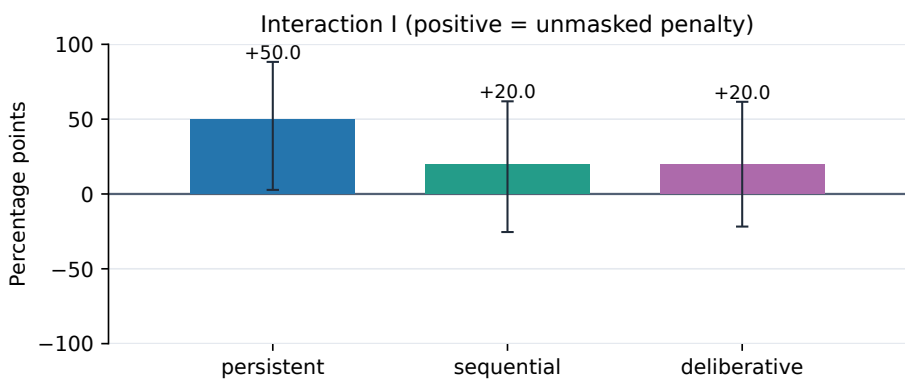
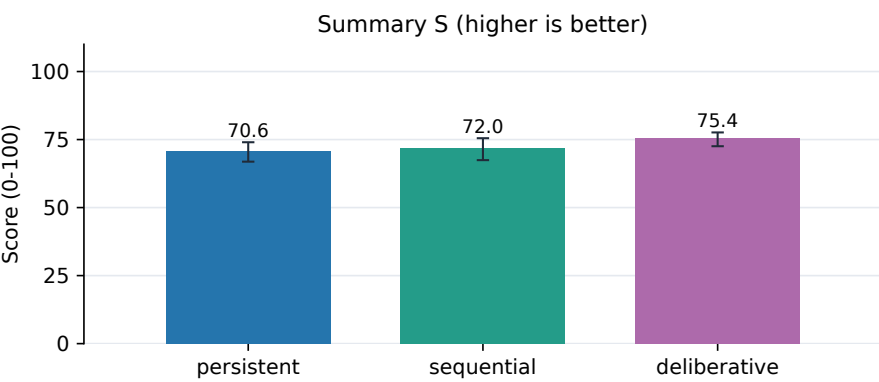
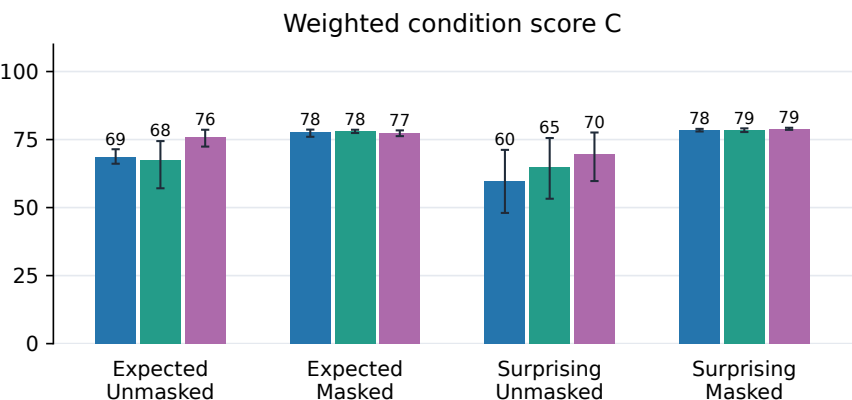
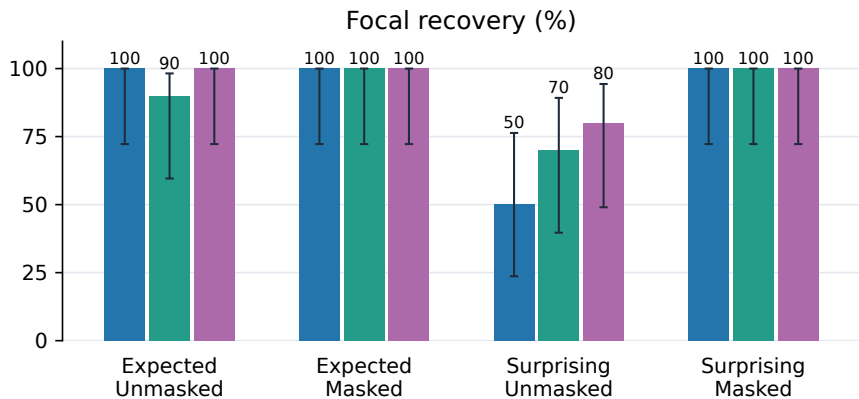
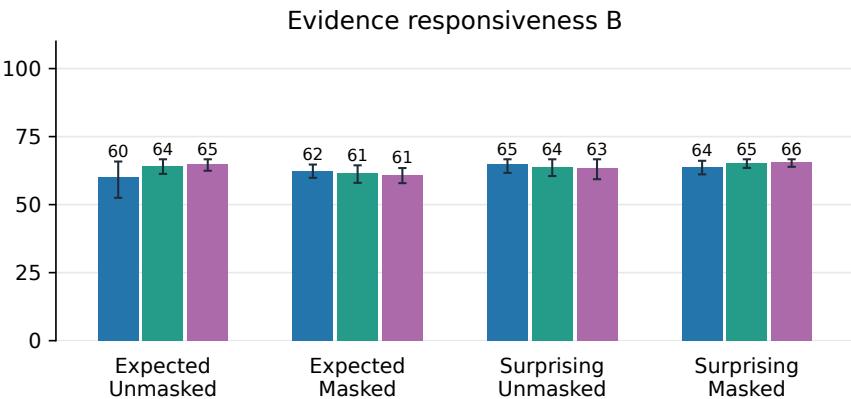
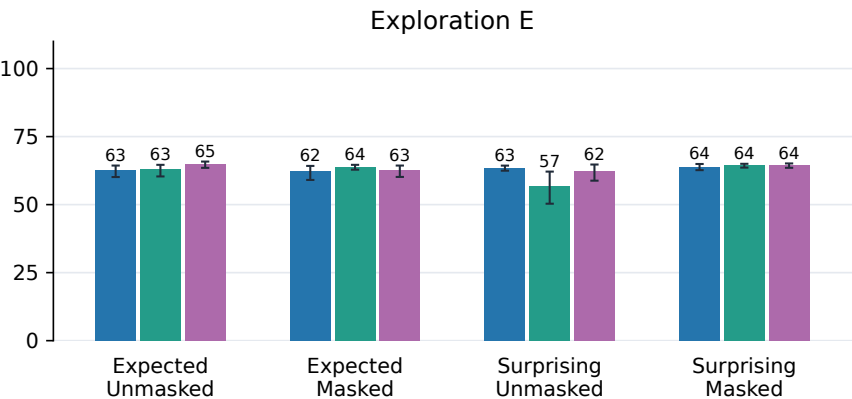
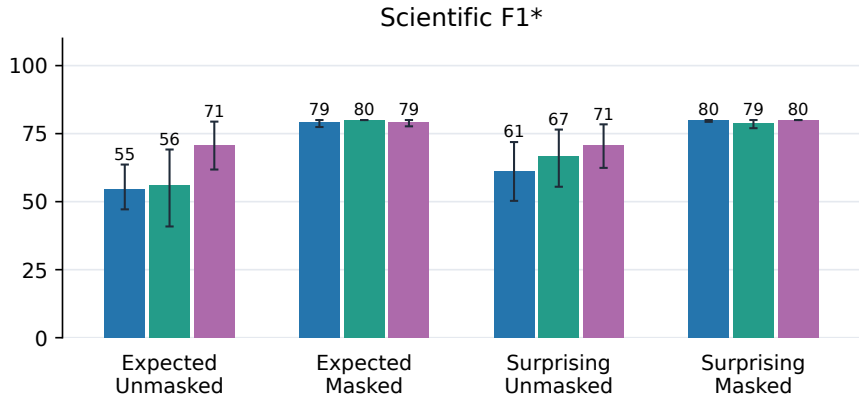
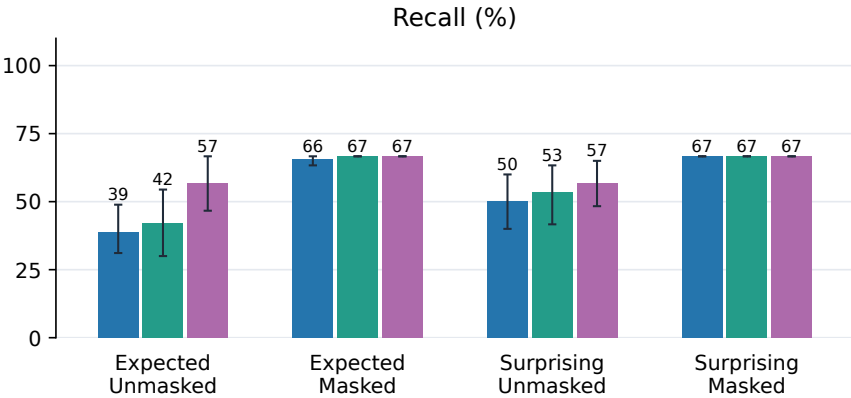
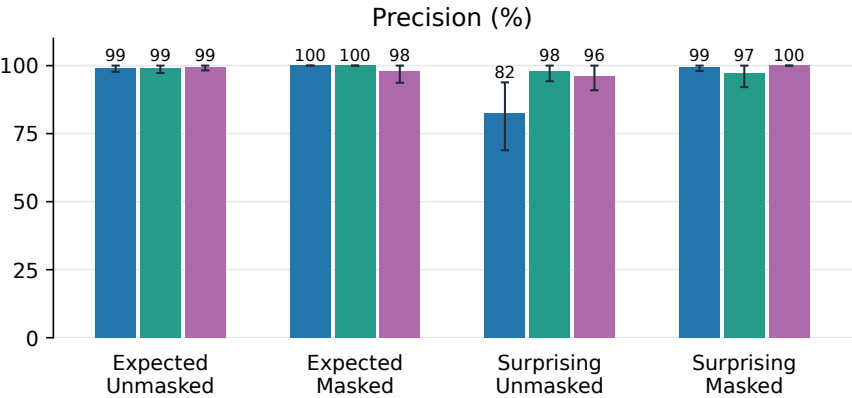


Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · S: equal condition weights; I: signed percentage points  
95% CIs: bootstrap; Wilson for focal recovery; Wilson-based MOVER for I. Fixed dataset pair. NA = unavailable; missing CIs indicate insufficient runs/evidence.

# Gemma 4 31B · Custom runner

Workflow and condition comparison · medium reasoning

persistent sequential deliberative



Snapshot 2026-09-12T20:27 UTC · Ended runs, including runs with errors · S: equal condition weights; I: signed percentage points  
95% CIs: bootstrap; Wilson for focal recovery; Wilson-based MOVER for I. Fixed dataset pair. NA = unavailable; missing CIs indicate insufficient runs/evidence.

# Source files and model identifiers

Frozen task definitions, saved result snapshot, and reproducible outputs

Dataset pair: **es-v2-nsclc\_clinical-42000**  
Biomni export generated: 2026-09-11T03:16:36.523825+00:00  
The report uses the saved result snapshot; rebuilding a PDF alone does not refresh ongoing experiments. Source hashes are recorded in clinical\_context.json and provenance.json.

data/expected\_surprising\_ledger/full\_runs/20260908\_clinical10pct\_workflows/input\_data/private/es-v2-nsclc\_clinical-42000/pair.json

data/expected\_surprising\_ledger/full\_runs/20260908\_clinical10pct\_workflows/input\_data/private/es-v2-nsclc\_clinical-42000/assignment.json

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data/expected\_surprising\_ledger/full\_runs/20260910\_clinical10pct\_masked\_workflows/input\_data/private/es-v2-nsclc\_clinical-42000/masking.json

data/expected\_surprising\_ledger/full\_runs/20260908\_clinical10pct\_workflows/input\_data/public/0dfe6f786ad7dd9d/data\_dictionary.json

| Display name  | Recorded model identifier           |
|---------------|-------------------------------------|
| Astra         | gpt-6-astra                         |
| Sol           | gpt-5.6-sol                         |
| Terra         | gpt-5.6-terra                       |
| Luna          | gpt-5.6-luna                        |
| Claude Opus 5 | claude-opus-5                       |
| Qwen 3.8 27B  | Inferact/Qwen3.8-27B-NVFP4          |
| Gemma 4 31B   | RedHatAI/Gemma-4-31B-IT-FP8-Dynamic |

Normal rerun: python3 scripts/expected\_surprising/build\_presentation\_results.py  
Saved-snapshot PDF rebuild: add --report-only. Redraw figures and rebuild the PDF: add --figures-only.  
Additional methods, source selection, aliases, metric availability, and version provenance: README.md, run\_metrics.csv, condition\_metrics.csv, and clinical\_context.json.